Reflectance Maps
for Non-Lambertian 3D Reconstruction

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Acknowledgements

First, I wish to thank my advisor, Professor Ko Nishino, for his continuous guidance since I was an undergraduate student. He invited me to this interesting research and always provided insightful advice. Without his help, my research projects including this thesis would not have been possible.

I would also like to thank co-authors of my projects. I would like to express my gratitude to Professor Shohei Nobuhara at Kyoto Institute of Technology. He provided continuous and practical advice not limited to research topics when he was at Kyoto University. I am also deeply grateful to Professor Vincent Lepetit at École des Ponts ParisTech. His interesting and helpful comments supported me a lot in making progress in my research. I would also appreciate Yuto Enyo, for his support, especially in image capture and data curation. Without his help, a real object dataset in this work would not have been possible.

I would also thank members of my thesis committee, Professor Shin'ya Nishida and Professor Tatsuya Kawahara, for taking the time to review my thesis and giving valuable feedback.

This work is also supported by the other members of Kyoto University Computer Vision Laboratory. I am grateful to a lot of advice and comments from Assistant Professor Marc Aurel Kastner, Shu Nakamura, Genki Kinoshita, as well as many other members.

I am also grateful to Japan Science and Technology Agency (JST) and Kyoto University Division of Graduate Studies for providing me the financial support during my graduate school life.

Finally, I would like to thank my parents for everything they have done for me.

Abstract

Image-based 3D reconstruction, reconstruction of object 3D shapes from images, is a fundamental problem in computer vision and can serve as a means for machines (*e.g.*, robots) to understand the real world. For passive 3D reconstruction, *i.e.*, reconstruction from images taken in unknown natural illumination, existing methods mainly relied on textures and Lambertian (diffuse) reflectance. Textures help us to establish correspondences across views, which enable reconstruction of relative camera poses. Once camera poses are obtained, by leveraging pixel-wise, view-independent features such as Lambertian shading, we can establish dense pixel-wise correspondences and recover surface depths by triangulation.

Real-world objects are, however, often textureless and non-Lambertian (*e.g.*, reflective). Reconstruction of textureless, non-Lambertian surfaces is still a challenging problem unless we assume dense sampling of input views and knowledge of camera poses, which are usually unavailable in real applications. The appearance of non-Lambertian surfaces is complex and view-dependent under natural illumination, which prevents us from establishing correspondences across views. Due to complex relationship between surface geometry and surface appearance and the ambiguity between geometry and the other physical quantities (reflectance and illumination), geometry estimation by a nonlinear optimization based the radiometric image formation model is also challenging under unknown natural illumination.

In this dissertation, we introduce a novel multi-view geometry reconstruction method that realizes 3D reconstruction of textureless, non-Lambertian surfaces from a sparse set of multi-view images (*e.g.*, 10 images that cover the whole object areas or 2 images for reconstruction of regions visible from the viewpoints) taken under unknown natural illumination.

First, we address 3D reconstruction from posed multi-view images (images with known camera poses) of a textureless, non-Lambertian object whose surface reflectance and surrounding illumination are known. Even with the information of the camera poses, the surface reflectance, and the surrounding illumination, geometry reconstruction is still challenging due to view-dependent, complex relationship between surface geometry and surface appearance. We overcome this by introducing *nLMVS-Net*, a deep multi-view geometry estimation network that first estimates surface normals from the appearance for each view and then exploits the normals as view-independent features for establishing cross-view correspondences. For plausible single-view surface normal estimation, we derive non-Lambertian shape-from-shading sub-network that explicitly computes camera-view reflectance maps from the reflectance and the illumination and leverages them in a coarse-to-fine manner. The camera-view reflectance maps are view-dependent mappings from surface normals to the surface radiances thus provide rich information for the surface normal estimation. We also integrate our method with a neural reflectance estimation method so

that we can apply nLMVS-Net to images of objects whose surrounding illumination environment is known but the surface reflectance is unknown.

Second, we tackle 3D reconstruction of a textureless, non-Lambertian object from only posed multi-view images, *i.e.*, without the knowledge of the surface reflectance and the surrounding illumination. A straightforward way to tackle this is formulating the problem as a joint estimation of geometry, reflectance, and illumination. It is, however, extremely challenging due to the fundamental ambiguity between reflectance and illumination in frequency and color. Different from such a naive joint estimation, we derive *DeepShaRM*, a deep estimation framework that does not decompose the surface appearance into reflectance and illumination and solves the problem as a joint estimation of geometry and reflectance maps. We exploit two deep neural networks for reflectance map recovery from the object shape and vice versa, and alternate between geometry reconstruction and reflectance map estimation. By bypassing the challenging joint estimation of reflectance and illumination and also leveraging the two deep neural networks, *i.e.*, by leveraging learned realistic prior on geometry and reflectance maps, we realize plausible geometry and reflectance map recovery from only posed multi-view images.

Finally, we address camera pose recovery from the object appearance for 3D reconstruction from unposed images (images without knowledge of camera poses). This is extremely challenging as, without multi-view observations with known camera poses, we cannot recover plausible surface normals for correspondence matching across views. In single-view estimation, there is a fundamental ambiguity between geometry and a reflectance map which is known as the bas-relief ambiguity. We resolve this by leveraging *Reflection Correspondences*, correspondences of the surrounding environment reflected on the surface, i.e., ones in reflectance maps. This novel type of correspondences enables us to recover plausible camera poses even from erroneous per-view estimates of surface normals and reflectance maps. By leveraging reflection correspondences and alternating between estimation of geometry, reflectance maps, and camera poses, we realize non-Lambertian 3D reconstruction even from two-view unposed images.

We also created nLMVS-Synth and nLMVS-Real, synthetic and real datasets of multi-view images of textureless, non-Lambertian objects for training and evaluation of the proposed methods. These datasets are unprecedented in the number of instances and can serve as a useful platform for not only geometry reconstruction but also other inverse rendering research.

Experimental analysis on newly created synthetic and real-world datasets shows the effectiveness of each of the proposed methods. Our method achieves dense surface geometry reconstruction of textureless, non-Lambertian surfaces from a sparse set (2 or more) of unposed multi-view images without requiring any additional inputs, which remains extremely challenging to existing geometry reconstruction methods. We believe our method can serve as practical means for understanding objects in the wild.

List of Publications

This dissertation consists of the following four publications.

International Conferences

- Kohei Yamashita, Yuto Enyo, Shohei Nobuhara, and Ko Nishino, "nLMVS-Net: Deep Non-Lambertian Multi-View Stereo", in Proc. of Winter Conference on Applications of Computer Vision (WACV), 2023. [1] [2]

- Kohei Yamashita, Shohei Nobuhara, and Ko Nishino, "DeepShaRM: Multi-View Shape and Reflectance Map Recovery Under Unknown Lighting", in Proc. of International Conference on 3D Vision (3DV), 2024. [3] [2]

- Kohei Yamashita, Vincent Lepetit, and Ko Nishino, "Correspondences of the Third Kind: Camera Pose Estimation from Object Reflection", submitted to IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024 (under review).

Journal Articles

- Kohei Yamashita, Shohei Nobuhara, and Ko Nishino, "DeepShaRM: Multi-View Shape and Reflectance Map Recovery Under Unknown Lighting", submitted to IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI) (under review).

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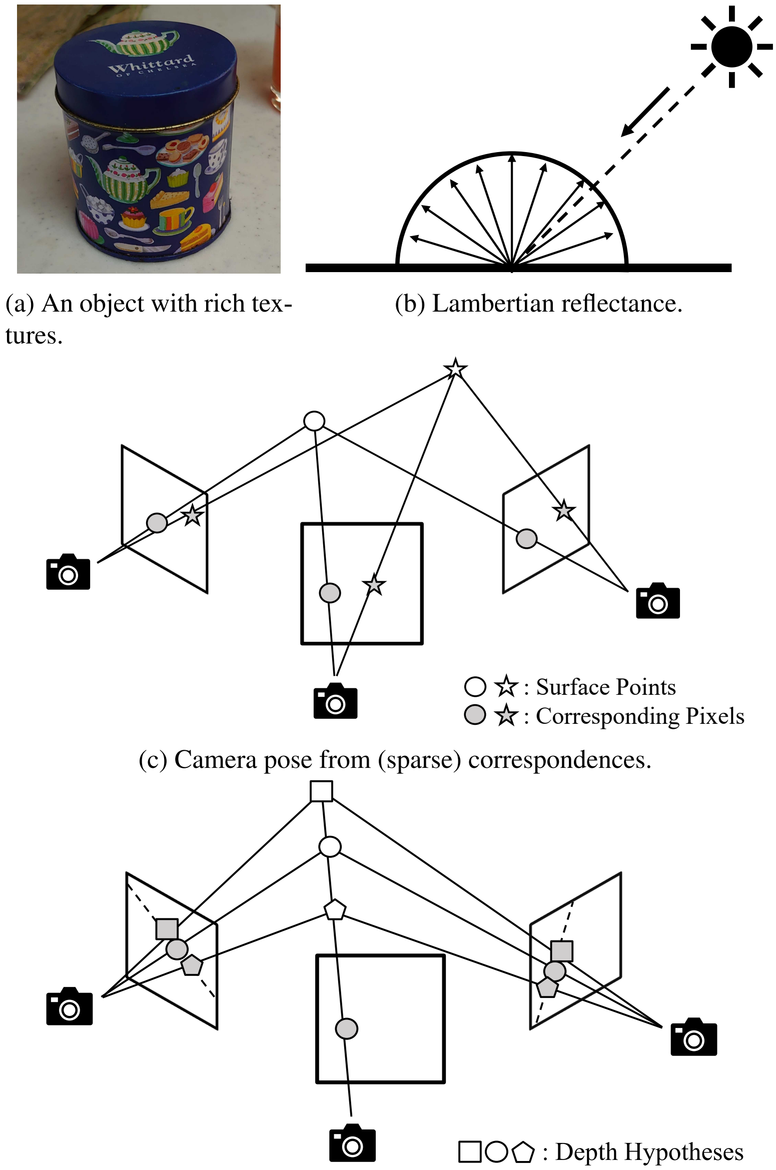
Chapter 1

Introduction

Image-based three-dimensional geometry reconstruction, *i.e.*, reconstruction of object 3D shapes from images, has been a central problem in computer vision. It finds many applications in different fields such as robotics, XR, and graphics. For instance, if robots could automatically recover 3D shapes of real objects around them, they can exploit the information for motion planning, *e.g.*, they can determine how to grasp objects. Geometry recovered by 3D reconstruction methods would also be helpful for predicting other object properties such as material and weight. For real world applications, passive methods that require only a small number (*e.g.*, 10) of images taken in the wild is especially important. In other words, we should recover the object 3D shape from only a small number of images taken in unknown natural illumination without exploiting any additional information such as knowledge of camera poses and the surrounding illumination.

For passive 3D reconstruction, existing methods mainly relied on textures on the surfaces (Fig. 1.1a) and shading caused by Lambertian reflectance (Fig. 1.1b). Textures, *i.e.*, view-independent, salient features, enable us to match pixels across views that correspond to the same surface point. Using the correspondences, as depicted in Fig. 1.1c, we can simultaneously obtain camera poses and 3D locations of the (sparse) surface points that are consistent with multi-view geometry. Once camera poses are obtained, we can recover dense surface geometry by leveraging Lambertian shadings, *i.e.*, shadings of matte surfaces whose radiances (observed pixel values) are independent of viewing directions. Using camera poses, as depicted in Fig. 1.1d, for each pixel and each hypothesis of surface depth (3D location of the surface point), we can compute expected corresponding pixels in different views. We can determine the surface depth as one that minimizes discrepancies between radiances of Lambertian shadings observed in the expected corresponding pixels. Geometry reconstruction based on such pixel-wise, view-independent feature has been studied as multi-view stereo (MVS). Recent methods exploit deep neural networks for the evaluation of discrepancies and the inference of surface depths from them [70]. By this, MVS methods realized accurate and dense geometry recovery from a sparse set (*e.g.*, 3) of images if the surfaces are textured and Lambertian.

As shown in Fig. 1.2a, real world objects are, however, often textureless and usually non-Lambertian. Cars, tableware, and bronze statues are with specular reflections and often textureless (unpainted or uniformly painted). Establishing correspondences on textureless surfaces for the camera pose recovery is non-trivial. Also, non-Lambertian reflectance affects accuracy of the dense surface recovery. This becomes critical especially when the surfaces are reflective (*e.g.*, made of metal or



(a) An object with rich textures.

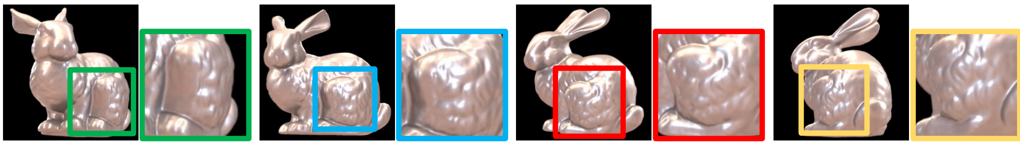
(b) Lambertian reflectance.

(c) Camera pose from (sparse) correspondences.

(d) Pixel-wise depth recovery.

Figure 1.1: Existing passive 3D reconstruction methods mainly relied on (a) textures and shading caused by (b) Lambertian reflectance whose radiance is independent of the viewing direction. (c) Textures, *i.e.*, view-independent, salient features, enable us to detect pixels in different views (gray circles and stars) that correspond to the same surface points (the white circle and the white star). Camera poses and 3D locations of the (sparse) surface points can be recovered from the correspondences. (d) Once we could recover camera poses, using radiances of the Lambertian shading as pixel-wise view-independent features, we can compute likelihoods of surface depth hypotheses for all pixels and all hypotheses in the images.

(a) Textureless, non-Lambertian objects in the real world.



(b) View-dependent appearance of a (synthetic) non-Lambertian object.

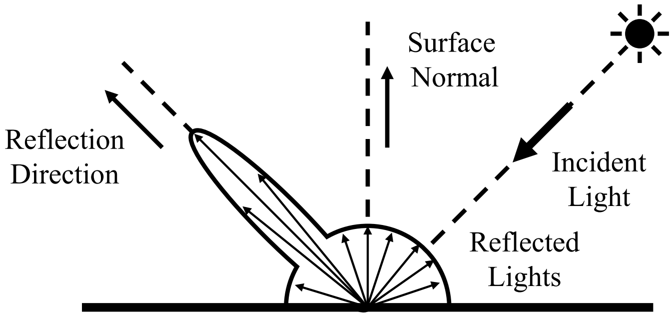
Figure 1.2: (a) Real world objects are, however, often textureless and non-Lambertian (*e.g.*, with specular highlights). (b) Appearance of a synthetic bunny with non-Lambertian reflectance. The appearance drastically changes across views due to the non-Lambertian reflectance (especially specular reflections). Such view-dependent appearance prevents us from establishing correspondences and extracting view-independent features. The images are synthesized using assets from existing databases [33, 14, 43].

plastic). As shown in Fig. 1.2b, appearance of such surfaces heavily depends on the viewing direction and the traditional methods completely fail.

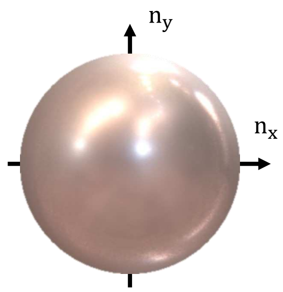
Some existing methods have tackled passive geometry reconstruction of textureless, non-Lambertian objects by inverting the radiometric image formation process (inverse rendering). They jointly optimize surface geometry, surface reflectance, surrounding illumination, and camera poses so that images synthesized from them based on the rendering equation [28] (a physics model of the image formation) become consistent with observed images. Carefully designed optimization frameworks and prior constraints on geometry, reflectance, and illumination have been proposed for the joint optimization of the unknown physical quantities. Full inverse rendering, *i.e.*, joint estimation of the all physical quantities, is, however, extremely challenging due to its large number of free parameters and ambiguity between the unknowns. Geometry estimate can easily fall into a local minima that is consistent with the current (wrong) estimates of reflectance, illumination, and camera poses. For these, existing inverse rendering methods rely on restrictive assumptions such as known illumination and an extremely large number (*e.g.*, 100) of input images.

1.1 Reflectance Maps for non-Lambertian 3D Reconstruction

In this dissertation, we focus on 3D reconstruction of a textureless, non-Lambertian object from a sparse set (*e.g.*, 10 for reconstruction of the whole object shape and 2 for reconstruction of regions visible from the input views) of multi-view images taken under unknown complex natural illumination. Again, due to lack of textures and Lambertian shading, *i.e.*, view-independent visual features, both camera pose estimation and dense surface recovery with known camera poses are challenging. We resolve these by introducing three novel methods (two for the dense 3D object



(a) Reflection on non-Lambertian surfaces.



(b) A reflectance map.

Figure 1.3: (a) Surface reflection on non-Lambertian surfaces. Incident lights from the surrounding illumination environment are reflected to different directions with different intensities. In particular, reflective surfaces (*e.g.*, metallic surfaces) reflect the light strongly in the reflection direction determined by the incident and surface normal orientations. (b) For an object with a certain reflectance property under a certain illumination environment, appearance of the object in an image (*i.e.*, for a single viewing direction) becomes a non-parametric function of the surface normal orientation, which is known as reflectance map [24].

shape recovery and one for the camera pose estimation). Our key idea is integrating the geometry recovery based on cross-view correspondences with the radiometric analysis., an analysis based on a physics-based model that models the relationship between radiances (pixel values) and physical quantities in the scene, *i.e.*, geometry, reflectance, and illumination (visualized in Fig. 1.3a). We show that, the radiometric analysis provides us additional information for the multi-view geometry recovery, and enables us to recover 3D shape of a textureless, non-Lambertian object with specular reflections even from two-view images without any additional knowledge about camera poses and the surrounding illumination.

1.1.1 Dense Surface Recovery from Posed Multi-View Images

In Chapter 5, we first consider dense geometry recovery under a setting where we know the surface reflectance, the surrounding illumination, and the camera poses. Given these, we can predict camera-view reflectance maps [24] of the object based on the radiometric image formation model. As depicted in Fig. 1.3b, camera view reflectance maps are view-dependent mappings from surface normals to surface radiances (pixel values) that represent the compound effects of the surface reflectance and the illumination environment. By comparing the predicted reflectance maps and the observed radiances, we can compute pixel-wise radiometric likelihoods of surface normals (how likely the surface is facing a specific orientation) which become additional constraints regarding surface geometry.

The geometry recovery is, however, non-trivial even with the radiometric likelihoods of surface normals. Under complex natural illumination, directional distributions of the radiometric likelihoods are noisy and have multiple peaks. A naive non-linear optimization of geometry based on the likelihoods can easily falls into a local minima (or results in overly smoothed surface). As depicted in Fig. 1.4, we overcome this difficulty by introducing *nLMVS-Net*, a deep multi-view stereo network

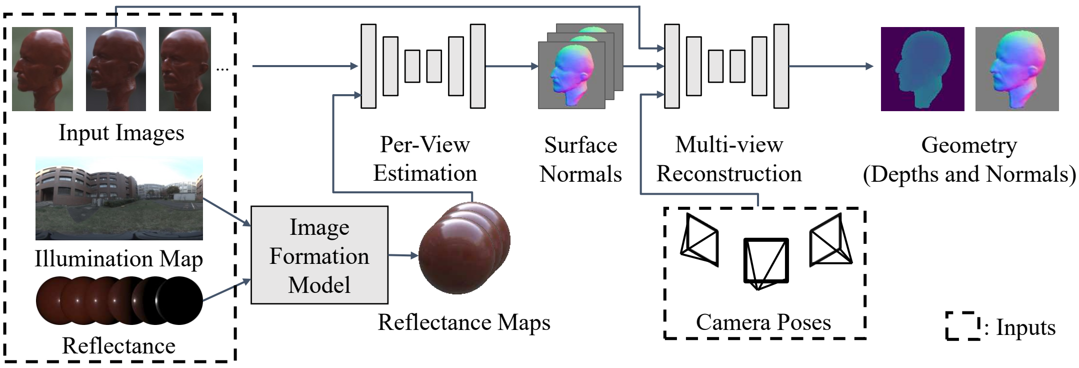
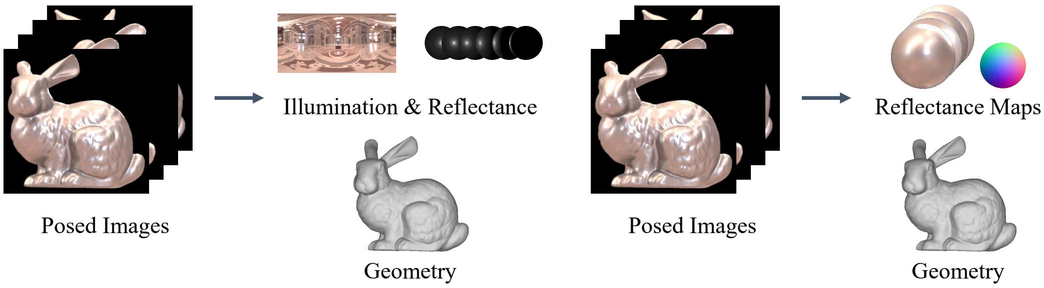


Figure 1.4: In Chapter 5, we first consider a setting where we know the surface reflectance, the surrounding illumination, and the camera poses. Even with these information, non-Lambertian 3D reconstruction is still challenging due to complex, view-dependent appearance of the object. We overcome this problem by introducing nLMVS-Net, a deep multi-view geometry estimation network that first estimates surface normals from the object appearance for each view and then leverages the surface normals as view-independent features for establishing correspondences across views.

that first estimates surface normals for each view and then exploits the estimated surface normals as view-independent features for dense correspondence detection. For each view, a deep shape-from-shading sub-network aggregates the information of the radiometric likelihoods in a coarse-to-fine manner and estimates surface normals as pixe-wise probability densities. A cost volume filtering sub-network (a deep network for dense surface recovery) then recovers depths and surface normals by exploiting the well-defined probability densities as input view-independent features for correspondence matching. We also integrate our deep geometry estimation method with a neural reflectance estimation [11] and jointly estimate surface geometry and surface reflectance. By these, we can in fact recover geometry of objects with unknown materials from posed multi-view images (multi-view images with known camera poses) taken under known but complex natural illumination.

Another challenge of dense geometry reconstruction from the object appearance is that the surface reflectance and surrounding illumination are also unknowns for real objects, *i.e.*, we cannot predict reflectance maps using them. A straightforward approach to tackle this problem is jointly estimating surface reflectance and surrounding illumination along with surface geometry similar to full inverse rendering. The joint estimation of reflectance and illumination, however, suffers from the fundamental ambiguity between the two unknowns in color and frequency. That is, appearance of red rough surfaces under white directional illumination is almost identical to one of smooth white surfaces under red blurry light.

As depicted in Fig. 1.5b, in Chapter 6, we address this problem by introducing *DeepShaRM*, a joint geometry and reflectance map estimation framework that requires only posed multi-view images as inputs. Our key idea here is bypassing solving the challenging joint estimation of reflectance and illumination but instead leveraging learned realistic prior on reflectance maps for direct estimation of reflectance maps. We realize this by introducing a deep reflectance map estimation sub-network that estimates a reflectance map for each view from an input image and a surface normal map. We train the sub-network with a large scale synthetic dataset of images



(a) Inverse rendering (b) DeepShaRM (Chapter 6)

Figure 1.5: In Chapter 6, we consider non-Lambertian 3D reconstruction from only posed multi-view images (images with known camera poses). (a) A naive approach to tackle this problem is inverse rendering, *i.e.*, joint estimation of geometry, reflectance, and illumination. This is, however, extremely challenging under natural illumination due to the fundamental ambiguity between the reflectance and the illlumination in color and frequency. (b) We bypass solving this problem by formulating the geometry reconstruction as joint estimation of geometry and reflectance maps.

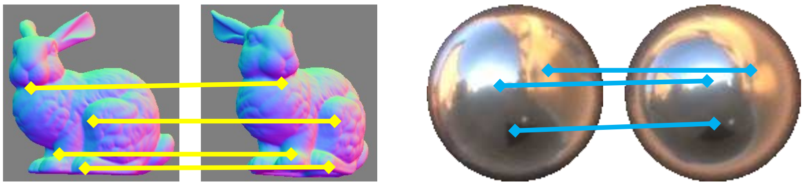
and reflectance maps so that it can implicitly learn realistic prior on reflectance maps such as statistical bias of their frequency characteristics. We alternate between the learning based reflectance map estimation and geometry recovery with the shape-from-shading sub-network, *i.e.*, we jointly estimate geometry and reflectance maps by leveraging realistic prior on them, and realize full 3D object shape recovery from only posed multi-view images.

1.1.2 Shape and Camera Pose Recovery from Unposed Images

We also address geometry reconstruction from unposed multi-view images, *i.e.*, without the knowledge of the camera poses. By applying the proposed joint geometry and reflectance map estimation framework to each image separately, we can recover surface normal and reflecance maps even for the unposed images. We can leverage the recovered surface normal maps for detecting pixel correspondences across views and the correspondences give us constraints for the camera pose estimation. Note that, although surface normals themselves are view-dependent in this setting due to rotation caused by the unknown camera poses, we can still extract view-independent features such as curvatures of the surfaces for the correspondence matching.

The single-view estimation of the surface normal and the reflectance maps, however, suffers from the ambiguity between them which is known as the bas-relief ambiguity [3]. There is a family of possible combinations of surface noromal and reflectance maps. For this, we cannot completely rely on correspondences detected from the surface normal maps and camera poses recovered from them can be erroneous. This becomes critical especially when we only have two views and the cameras are distant from the object (almost orthographic). In such setting, we cannot obtain a unique solution for the camera poses even if we could establish correct correspondences on the recovered surface normal maps.

As depicted in Fig. 1.6, in Chapter 7, we resolve this problem by exploiting a novel type of correspondences which we refer to as *Reflection Correspondences*. Reflection correspondences are correspondences that link pixels in two views whose



(a) Traditional correspondences (b) Reflection correspondences

Figure 1.6: In Chapter 7, we tackle non-Lambertian 3D reconstruction from unposed images, *i.e.*, images whose camera poses are unknown. Even if we recover surface normals for each view and establish (traditional) correspondences using them, accurate camera poses cannot be recovered due to the Bas-relief ambiguity [3], an ambiguity between geometry (surface normals) and a reflectance map in the single-view estimation. We resolve this by jointly estimating geometry and the relative camera poses with reflection correspondences, matches of the surrounding illumination environment observed in reflectance maps.

incident rays of specular reflections come from the same direction. We show that the detection of reflection correspondences can be realized as detection of corresponding surface normals in two-view reflectance maps, and that the reflection correspondences enable us to resolve the bas-relief ambiguity even if we have only two-view images. We derive a camera pose estimation algorithm that exploits correspondences in both normal and reflectance maps, and a joint estimation method that alternates between the camera pose estimation and geometry and reflectance map recovery (DeepShaRM). We show that we can in fact recover plausible surface geometry, reflectance maps, and camera poses from real-world two-view images.

1.1.3 Synthetic and Real Object Datasets

For plausible evaluation of the learning-based methods, large scale datasets that include objects with different shapes, reflectances and surrounding illumination environments are also important. Most of existing multi-view image datasets are, however, created for training and evaluation of multi-view stereo methods thus most of the images in the datasets are ones of textured or Lambertian surfaces. There is a few dataset [49] that contains multi-view images of textureless, non-Lambertian objects. The number of instances are, however, limited (4 objects under 3 illumination environments) and insufficient for training and evaluation of learning-based methods.

In Chapter 4, we introduce two novel datasets, nLMVS-Synth and nLMVS-Real, for training and evaluation of the proposed methods. The training set of the nLMVS-Synth dataset consists of 26850 images of 2685 synthetic objects whose ground truth 3D shapes, surface reflectance properties (bidirectional reflectance distribution functions, BRDFs), and surrounding illumination environments are known. This dataset is sufficient for training deep neural networks in our methods. For thorough quantitative evaluation, we also created a test set of synthetic images that consists of 4320 images of 216 synthetic objects. The nLMVS-Real dataset consists of 2569 multi-view high dynamic range real-world images of 120 combinations of 5 shapes, 4 materials, and 6 illumination environments. We created the objects captured in the nLMVS-Real dataset by 3D printing, *i.e.*, we know the ground truth surface geometry. We carefully calibrated camera parameters and 3D locations of the objects thus we can use the ground truth geometry for quantitative evaluation. We also provide

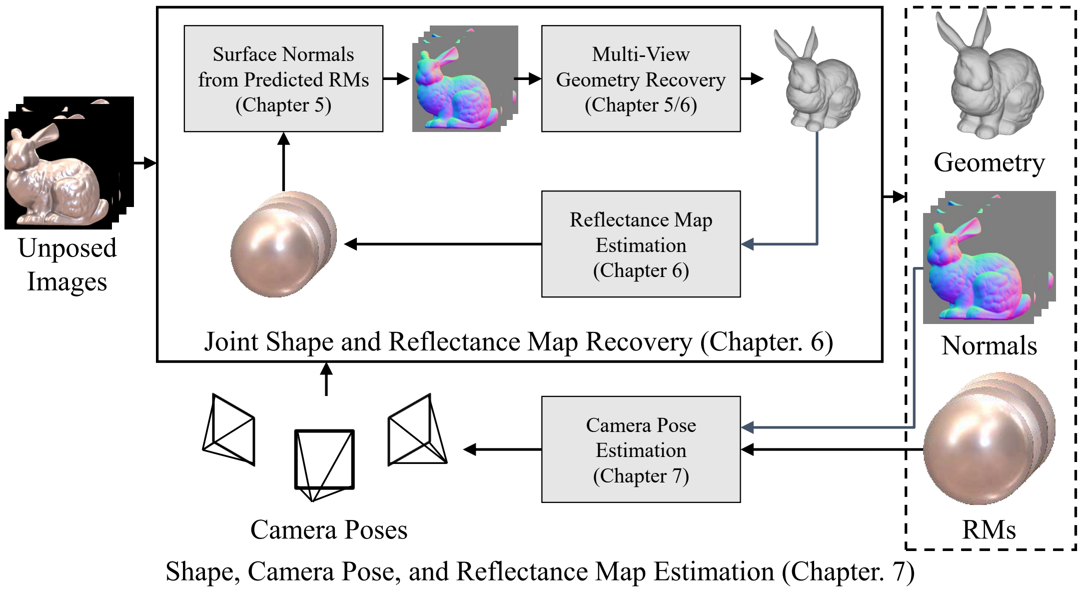


Figure 1.7: An overall outline of the proposed method(s). RMs refer to camera-view reflectance maps.

illumination maps captured by a 360 degree camera. This dataset is unprecedented in the number of instances and can serve as a useful platform for not only geometry reconstruction but also other inverse rendering research.

We conduct thorough experimental analysis on the nLMVS-Synth and the nLMVS-Real datasets including comparisons with existing methods and ablation studies about each of the proposed components. The results clearly demonstrate the effectiveness of each of the proposed methods. By integrating the three key methods, we realize geometry reconstruction of textureless, non-Lambertian surfaces from sparse (*e.g.*, 2) unposed multi-view images taken under static but unknown natural illumination, which remains extremely challenging to existing methods. We believe our novel methods and the newly created datasets opens a new avenue of applications of image-based 3D reconstruction in the wild.

1.2 Contributions

Figure 1.7 depicts an overall outline of the full method and how each key idea contribute to it. Towards 3D reconstruction of textureless, non-Lambertian surfaces from a sparse set (2 or more) of unposed multi-view images taken under unknown natural illumination, we make the following contributions.

In Chapter 4, we introduce nLMVS-Synth and nLMVS-Real datasets, synthetic and real image datasets for training and evaluation of the proposed methods.

In Chapter 5, we introduce a multi-view method that can recover dense surface geometry given posed multi-view images and reflectance maps computed from known reflectance and surrounding illumination. The method consists of a learning-based single-view surface normal estimation network that exploits the information of reflectance maps and a deep multi-view stereo network that leverages recovered surface normals as input view-independent features.

	Chapter 5	Chapter 6	Chapter 7
# of Views	5	2 - 10	2
Input Illum. Map	Needed	Not needed	Not needed
Input Camera Poses	Needed	Needed	Not needed
Geometry Estimation	For each view	Consolidated	Consolidated
Reflectance Estimation	✓		
w/o Specular Reflection	✓	✓	
Dynamic Illumination		✓	
Cameras	Distant	Distant	Distant
Reflectance	Homogeneous	Homogeneous	Homogeneous
Illumination	Distant	Distant	Distant

Table 1.1: Assumptions and experimental setup for each sub method.

In Chapter 6, we derive a joint shape and reflectance map estimation framework for geometry recovery from posed multi-view images taken under unknown natural lighting. For this, we introduce a deep reflectance map estimation sub-network that leverages learned realistic prior on reflectance maps. In this chapter, we also derive a method for geometry recovery from reflectance maps that consolidates the surface normals estimated by the shape-from-shading network and recovers the surface geometry as a single estimate that is consistent with all input views.

In Chapter 7, we propose a camera pose estimation method that exploits reflection correspondences (correspondences in reflectance maps) for joint shape, reflectance map, and camera pose estimation from unposed images.

Table 1.1 summarizes assumptions and experimental setup for each sub method. In general, the method in Chapter 7 is the least restrictive. However, the other two methods are superior in some aspects. For instance, DeepShaRM (Chapter 6) can deal with dynamic changes of the surrounding illumination as it recovers the camera view reflectance map for each view separately. Also, while the camera pose estimation method in Chapter 7 assumes that there are specular reflections on the object surfaces[1], the other two methods can be applied to objects without specular reflections.

1.3 Dissertation Overview

The rest of this dissertation consists of the following contents.

Chapter 2 reviews theoretical backgrounds of our method, *i.e.*, how the relationship between the object appearance and physical quantities in the scene is modeled. We first review the radiometric relationship between the object appearance (pixel values) and surface geometry, surface reflectance, and surrounding illumination. Based on the radiometric image formation model, we also explain how we can compute reflectance maps and the fundamental ambiguity in the full inverse rendering. We also explain how the projective geometry (the relationship between surface locations and

[1]Note that, without specular reflections, camera pose estimation from the appearance of a textureless object is extremely challenging even to traditional methods.

their locations in 2D image planes) can be leveraged in the geometry reconstruction based on cross-view correspondences.

Chapter 3 reviews existing image-based 3D reconstruction methods and explains the fundamental differences of our methods from them.

Chapter 4 provides details of the two novel datasets including assets used for the synthetic data creation, distributions of the viewpoints, and geometric calibration for the real-world dataset.

Chapter 5 introduces *nLMVS-Net*, a deep multi-view stereo method for 3D reconstruction from posed multi-view images of a textureless, non-Lambertian object whose surface reflectance and surrounding illumination are known. We experimentally validate that, by integrating the nLMVS-Net with a neural reflectance estimation [67], our method can in fact recover plausible surface geometry and also surface reflectance for textureless objects with unknown complex reflectances under known but complex natural illumination.

Chapter 6 derives *DeepShaRM*, a joint geometry and reflectance map estimation framework for reconstruction from posed multi-view images of a textureless, non-Lambertian object captured under unknown natural lighting. We experimentally validate that the joint estimation framework with two deep neural networks, *i.e.*, learned realistic priors on geometry and reflectance maps, is essential for plausible surface geometry reconstruction from the posed images taken under unknown natural illumination.

Chapter 7 proposes a joint shape, reflectance map, and camera pose estimation method that leverages *Reflection Corrspondences*, correspondences of the surrounding environment reflected on the surface (i.e., ones in reflectance maps). We theoretically and empirically validate that this novel type of correspondences is essential to resolve the ambiguity in the joint estimation when we only have two views. We also experimentally show that our joint estimation method with reflection correspondences successfully recovers plausible surface geometry, camera poses, and reflectance maps from synthetic and real-world two-view images.

Chapter 8 summarizes contributions of this dissertation and discusses possible future directions.

Chapter 2

Image Formation Process

In this chapter, we review how the object appearance in images is related to the object 3D shape and other physical quantities. Figure 2.1 depicts physical quantities related to the image formation process. The appearance of an object in an image is determined by the object 3D shape (surface location and surface normal), surface reflectance of the object, the surrounding illumination, and the camera pose.

We first review how the lights from the surrounding illumination are reflected on the object surface, which determines radiances (pixel values) observed in images. We also introduce camera-view reflectance maps which represents the radiometric relationship between surface normals and surface radiances. We then explain geometric relationship between 3D locations of surface points and its 2D locations in images.

2.1 Radiometric Image Formation

Let us first consider the relationship between surface geometry and surface radiances (pixel values) in terms of radiometry. We focus on geometry reconstruction of textureless, opaque objects whose surface reflectance property is homogeneous (independent of the surface location). For such objects, the relationship between the surface radiance and the physical quantities can be modeled by the rendering equation [28]

$$L_o(\boldsymbol{\omega}_\mathrm{o}) = \int L_i(\mathbf{x}, \boldsymbol{\omega}_\mathrm{i}) \rho({\boldsymbol{\omega}_\mathrm{i}}', {\boldsymbol{\omega}_\mathrm{o}}') \max(0, \boldsymbol{\omega}_\mathrm{i} \cdot \mathbf{n}) \mathrm{d}\boldsymbol{\omega}_\mathrm{i} \,, \tag{2.1}$$

where $\boldsymbol{\omega}_\mathrm{i}$, $\boldsymbol{\omega}_\mathrm{o}$, and $\mathbf{n}$ are incident, viewing, and surface normal directions, respectively. $L_i(\mathbf{x}, \boldsymbol{\omega}_\mathrm{i})$ is incident radiance for each incident direction $\boldsymbol{\omega}_\mathrm{i}$ and each surface location $\mathbf{x}$ which represents the surrounding illumination environment. $\rho({\boldsymbol{\omega}_\mathrm{i}}', {\boldsymbol{\omega}_\mathrm{o}}')$ is the bidirectional reflectance distribution function (BRDF) which represents the surface reflectance. ${\boldsymbol{\omega}_\mathrm{i}}'$ and ${\boldsymbol{\omega}_\mathrm{o}}'$ are incident and viewing directions in the local surface coordinate system

$${\boldsymbol{\omega}_\mathrm{i}}' \equiv \mathrm{R}_\mathbf{n} \boldsymbol{\omega}_\mathrm{i} \,, \tag{2.2}$$

$${\boldsymbol{\omega}_\mathrm{o}}' \equiv \mathrm{R}_\mathbf{n} \boldsymbol{\omega}_\mathrm{o} \,, \tag{2.3}$$

where $\mathrm{R}_\mathbf{n}$ is a transformation (rotation matrix) from the world coordinate system to the local surface coordinate system which is determined by the surface normal $\mathbf{n}$. In this dissertation, we assume that global illumination effects (shadows and interreflections) are negligible for most regions of the object surface. Under this assumption,

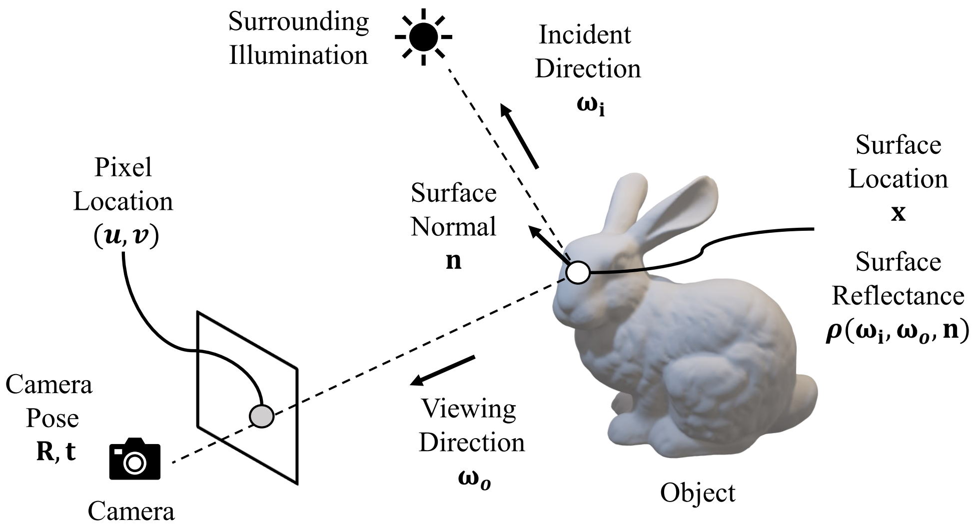


Figure 2.1: Appearance of an object in an image is determined by the object 3D shape (surface location and surface normal), surface reflectance of the object, the surrounding illumination, and the camera pose. The bunny model is from The Stanford 3D Scanning Repository [33].

the rendering equation can be simplified to

$$L_o(\boldsymbol{\omega}_\mathbf{o}) = \int L_i(\boldsymbol{\omega}_\mathbf{i})\rho({\boldsymbol{\omega}_\mathbf{i}}', {\boldsymbol{\omega}_\mathbf{o}}') \max(0, \boldsymbol{\omega}_\mathbf{i} \cdot \mathbf{n}) \mathrm{d}\boldsymbol{\omega}_\mathbf{i} \, . \tag{2.4}$$

When the surface is Lambertian, the BRDF becomes

$$\rho({\boldsymbol{\omega}_\mathbf{i}}', {\boldsymbol{\omega}_\mathbf{o}}') = \frac{c}{\pi} \, , \tag{2.5}$$

and the rendering equation is simplified to

$$L_o(\boldsymbol{\omega}_\mathbf{o}) = \frac{c}{\pi} \int L_i(\boldsymbol{\omega}_\mathbf{i}) \max(0, \boldsymbol{\omega}_\mathbf{i} \cdot \mathbf{n}) \mathrm{d}\boldsymbol{\omega}_\mathbf{i} \, . \tag{2.6}$$

where c is a constant determined by diffuse reflectance. As this equation is constant for the viewing direction, the appearance of Lambertian surfaces is view-independent regardless of the surrounding illumination.

The surface reflectance of real objects is, however, usually non-Lambertian and with specular reflections. As depicted in Fig. 2.2a, if we fix the incident direction $\boldsymbol{\omega}_\mathbf{i}$, the BRDF of a specular object is a function of $\boldsymbol{\omega}_\mathbf{o}$ that has a peak (specular lobe) in the reflection direction

$${\boldsymbol{\omega}_\mathbf{r}}'({\boldsymbol{\omega}_\mathbf{i}}', \mathbf{n}') \equiv -{\boldsymbol{\omega}_\mathbf{i}}' + 2 \left({\boldsymbol{\omega}_\mathbf{i}}' \cdot \mathbf{n}'\right) \mathbf{n}' \, , \tag{2.7}$$

where $\mathbf{n}' \equiv \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}^\mathrm{T}$ is the surface normal orientation in the local surface coordinate system. Note that concentration of the distribution of the specular lobe is determined by roughness of the surface. As depicted in Fig. 2.2b, this is equivalent to that, if we fix $\boldsymbol{\omega}_\mathbf{o}$, the BRDF is a function of $\boldsymbol{\omega}_\mathbf{i}$ that has a peak in ${\boldsymbol{\omega}_\mathbf{r}}'({\boldsymbol{\omega}_\mathbf{o}}', \mathbf{n}')$.

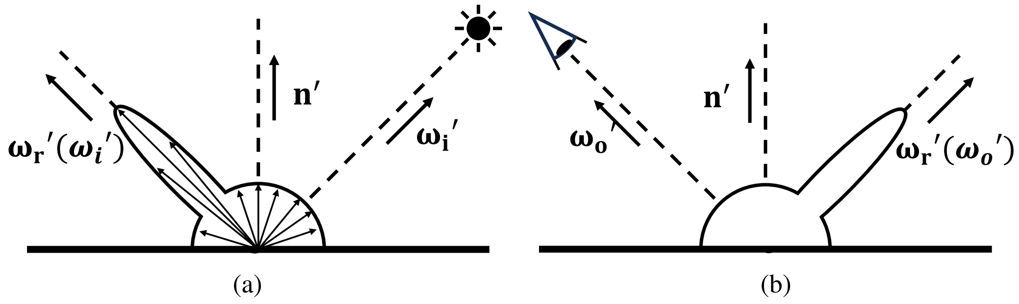


Figure 2.2: The surface reflectance of an opaque object is modeled by a bidirectional reflectance distribution function (BRDF) $\rho(\boldsymbol{\omega}_{\mathbf{i}}', \boldsymbol{\omega}_{\mathbf{o}}')$, a function whose value is determined by the incident and viewing directions in the local surface coordinate system (one aligned with the surface orientation). This figure depicts an example BRDF of a specular object for a certain incident direction (a) and a certain viewing direction (b).

Therefore, as depicted in Fig. 2.3a, we can regard the rendering equation as an approximated inner product of the illumination distribution $L_i(\boldsymbol{\omega}_{\mathbf{i}})$ and the reflectance distribution $\rho(\boldsymbol{\omega}_{\mathbf{i}}', \boldsymbol{\omega}_{\mathbf{o}}')$ that depends on the viewing and the surface normal orientations.

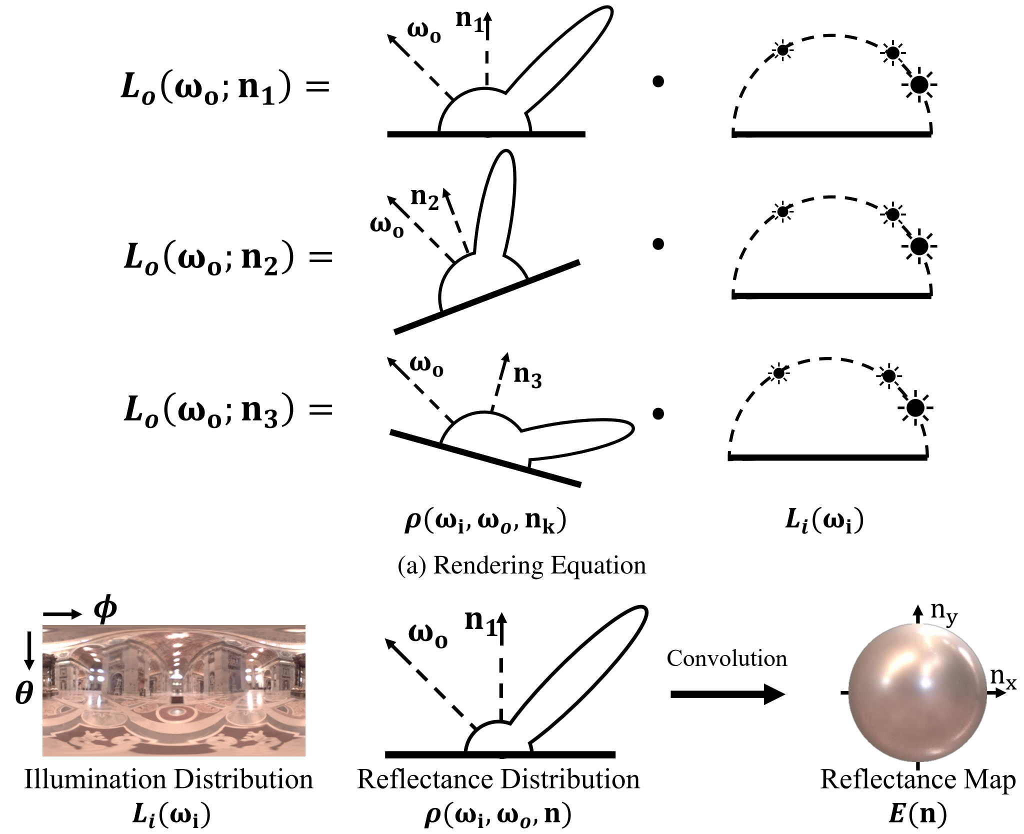
2.1.1 Camera-View Reflectance Maps

In this dissertation, we also assume that the viewing direction $\boldsymbol{\omega}_{\mathbf{o}}$ is almost constant for each view. Note that this assumption holds true as long as the distance between the object and the camera is much larger than the size of the object. Under this assumption, the surface radiance becomes a view-dependent function of the surface normal orientation

$$E_k(\mathbf{n}) \equiv \int L_i(\boldsymbol{\omega}_{\mathbf{i}})\rho(\boldsymbol{\omega}_{\mathbf{i}}', \mathbf{v}_{\mathbf{k}}') \max(0, \boldsymbol{\omega}_{\mathbf{i}} \cdot \mathbf{n}) \mathrm{d}\boldsymbol{\omega}_{\mathbf{i}} \,, \tag{2.8}$$

where $\mathbf{v}_{\mathbf{k}}'$ is the viewing direction of the k-th view in the local surface coordinate system. For reflective surfaces, as depicted in Fig. 2.3b, we can regard this as convolution of the illumination distribution $L_i(\boldsymbol{\omega}_{\mathbf{i}})$ and the reflectance distribution (especially the specular lobe). These view-dependent mappings from surface normals to surface radiances are known as Reflectance Maps [24].

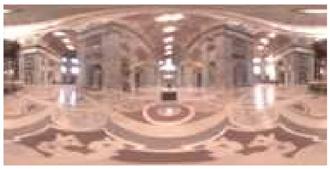
In our methods, we represent the object appearance by camera-view reflectance maps. There are three important properties of reflectance maps. First, as they explicitly represent the relationship between surface normals and surface radiances (apperance), they become clues for surface normal recovery from object appearance (shading and specular highlights). We explicitly leverage this in Chapter 5. Second, as reflectance maps are approximated convolutions of the surface reflectance and the surrounding illumination, there is a certain probability distribution of real-world reflectance maps caused by this formation process. For instance, as visualized in Fig. 2.4, frequency characteristics of reflectance maps are products of ones of reflectance and illumination. We train a reflectance map estimation network to learn such realistic prior on reflectance maps implicitly, and exploit it for joint estimation of geometry and reflectance maps (Chapter 6). Also, as the reflectance maps are something like a blurred version of the surrounding illumination map, we can detect



(a) Rendering Equation

(b) Reflectance Map Computation

Figure 2.3: (a) We can regard the observed radiances $L_o(\boldsymbol{\omega}_\mathbf{o})$ as the approximated inner product of the reflectance distribution $\rho(\boldsymbol{\omega}_\mathbf{i}', \boldsymbol{\omega}_\mathbf{o}')$ and the illumination distribution $L_i(\boldsymbol{\omega}_\mathbf{i})$. Note that the peak of the reflectance distribution is determined by the surface normal and the viewing direction. Note also that the illumination map is from the Light Probe Image Gallery [14]. (b) For each view, *i.e.*, if we fix the viewing direction $\boldsymbol{\omega}_\mathbf{o}$, the relationship between the surface radiances and the surface normals (*i.e.*, reflectance map) is an approximated convolution of the illumination distribution and the reflectance distribution. We visualize the illumination distribution as a panorama image whose x and y coordinates correspond to azimuth and zenith angles of the incident direction, respectively.



(a) Surrounding illumination

(b) Reflectance maps

Figure 2.4: Reflectance maps of objects with different reflectance properties (b) under the same illumination environment (a). As reflectance maps are approximated convolutions of the surface reflectance and the surrounding illumination (an illumination map from [14]), there is statistical bias in its frequency characteristics.

pixels in reflectance maps of different views that correspond to the same object (*e.g.*, a light source) in the surrounding environment. We leverage such correspondences regarding the surrounding environment for the joint estimation of object 3D shape, reflectance maps, and camera poses (Chapter 7).

2.1.2 Ambiguity between Reflectance and Illumination

The rendering equation shows that there is an ambiguity between reflectance and illumination in color (absolute reflectance). If a combination of $\rho(\boldsymbol{\omega}_{\mathrm{i}}', \boldsymbol{\omega}_{\mathrm{o}}')$ and $L_i(\boldsymbol{\omega}_{\mathrm{i}})$ is consistent with the observed radiance, another combination of $s\rho(\boldsymbol{\omega}_{\mathrm{i}}', \boldsymbol{\omega}_{\mathrm{o}}')$ and $\frac{1}{s}L_i(\boldsymbol{\omega}_{\mathrm{i}})$ is also consistent where s can be any positive value. There is also another ambiguity in frequency [53]. As depicted in Fig. 2.5, since the rendering equation is an approximated convolution of the surface reflectance and the surrounding illumination, it is extremely challenging to determine whether blurry appearance of an object is caused by the surface roughness or blurry illumination. For these, joint estimation of reflectance and illumination is extremely challenging even if we know the surface geometry [39, 11]. This is why we avoid the decomposition of reflectance maps into reflectance and illumination in the joint estimation of geometry and reflectance maps (Chapter 6).

2.2 Projective Geometry

There is also a relationship between a 3D location of a surface point $\mathbf{x}$ and its 2D locations (u_k, v_k) in images

$$d_k \begin{pmatrix} u_k \\ v_k \\ 1 \end{pmatrix} = \mathrm{K}_k \left(\mathrm{R}_k \mathbf{x} + \mathbf{t}_k \right) , \tag{2.9}$$

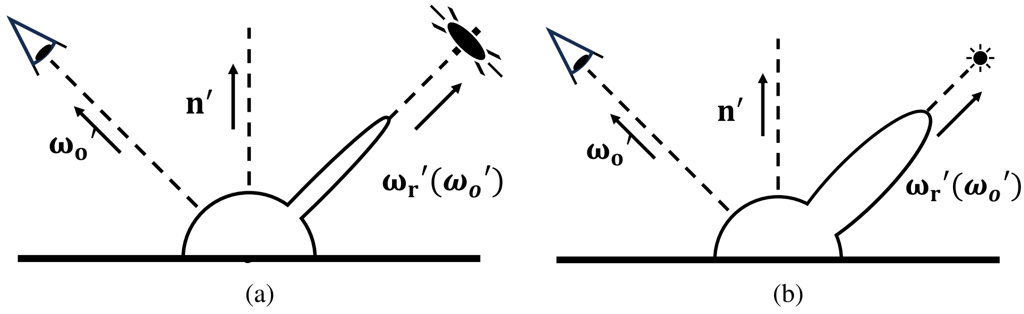


Figure 2.5: As the radiometric image formation is an approximated convolution of the reflectance and the illumination environment, it is extremely challenging to distinguish appearance of smooth surfaces under blurry illumination (a) from one of rough surfaces under directional illumination (b).

where d_k is a depth of the surface point in the k-th view and K_k is an intrinsic matrix of the k-th view which are determined by characteristics of the camera (*e.g.*, focal length). R_k and $\mathbf{t}_k$ are a rotation matrix and a translation vector of the k-th view which represent the 3D camera pose. By moving all terms except for $\mathbf{x}$ from the right hand side to the left, we have

$$\mathbf{x} = {\mathrm{R}_k}^{\mathrm{T}} \left[d_k \mathrm{K}_k^{-1} \begin{pmatrix} u_k \\ v_k \\ 1 \end{pmatrix} - \mathbf{t}_k \right] . \tag{2.10}$$

This is equivalent to

$$\mathbf{x} = \mathbf{c}_k + d_k \mathbf{l}_k , \tag{2.11}$$

where $\mathbf{c}_k$ is a 3D location of the camera

$$\mathbf{c}_k = -{\mathrm{R}_k}^{\mathrm{T}} \mathbf{t}_k , \tag{2.12}$$

and $\mathbf{l}_k$ is a looking direction of the camera

$$\mathbf{l}_k = {\mathrm{R}_k}^{\mathrm{T}} \mathrm{K}_k^{-1} \begin{pmatrix} u_k \\ v_k \\ 1 \end{pmatrix} . \tag{2.13}$$

Since the depth d_k can be any positive value, we cannot determine a 3D location of an observed surface point if we only have a single-view observation.

2.2.1 Camera Pose Estimation from Correspondences

Let us assume that we have N images captured from different views. As depicted in Fig. 1.1c, let us also assume that we can detect M correspondences of pixels across views that correspond to the same surface point. We can then solve for 3D locations $\mathbf{x}^i$ and depths d_k^i of the surface point, and the camera poses R_k and $\mathbf{t}_k$ by minimizing

$$\min_{\mathbf{x}^i, \{d_k^i\}, \{\mathrm{R}_k\}, \{\mathbf{t}_k\}} \sum_k \sum_i \left\| \mathbf{x}^i - \left(\mathbf{c}_k + d_k^i \mathbf{l}_k^i \right) \right\|^2 . \tag{2.14}$$

where i is the index of the correspondences. In practice, for instance, camera poses can be recovered for each view pair using 8 correspondences. Note that, the camera pose estimation with the pixel correspondences can becomes unstable when the cameras are distant from the object. Please see Chapter 7 for details.

2.2.2 Dense 3D Surface Recovery

Let us assume that we know camera poses and could extract pixel-wise features $\mathbf{F}_k(u, v)$ for a view pair that are independent of the viewing direction. Then, as depicted in Fig. 1.1d, for each pixel and each discretized hypothesis of the surface depth in the first view (u_1, v_1, d_1), we can compute how likely the depth hypothesis is to be correct by comparing the two-view feature maps

$$C(u_1, v_1, d_1) = M\left(\mathbf{F}_1(u_1, v_1), \mathbf{F}_2(u_2, v_2)\right) \ , \tag{2.15}$$

where u_2 and v_2 are a 2D location in the second view that we can compute from (u_1, v_1, d_1) using an equation

$$\begin{pmatrix} u_2 \\ v_2 \\ 1 \end{pmatrix} \propto \mathrm{K}_2 \left[\mathrm{R}_2 \left(\mathbf{c}_1 + d_1 \mathbf{l}_1\right) + \mathbf{t_2}\right] \ . \tag{2.16}$$

that we can derive from Eqs. (2.9) and (2.11). $M(\cdot, \cdot)$ is a metric function that evaluates the discrepancy between the two feature vectors. The 3D volume $C(u_1, v_1, d_1)$ is known as *Cost Volume* which is often leveraged as an intermediate feature for multi-view 3D reconstruction. As the cost volume is a 3D regular grid of the costs, we can directly feed it into a 3D convolutional neural network. If we have more than two views, we can still construct a cost volume by using a metric function that can take features from multiple views as inputs or simply summing up costs computed for each view pair.

Chapter 3

Related Work

In this chapter, we review relevant image-based geometry reconstruction methods. We also explain how our methods are fundamentally different from the existing methods. Table 3.1 summarizes differences between existing methods and ours. Reconstruction of textureless, non-Lambertian surfaces still remains challenging unless we have an extremely large number (*e.g.*, 100) of input images.

3.1 Multi-View Stereo (MVS)

Multi-View Stereo (MVS) methods recover surface depths from posed multi-view images based on correspondence detection across views and triangulation. Traditional methods [58, 16] exploited manually designed feature extraction methods and metrics for correspondence detection across views. They also proposed spatial aggregation of the information of correspondences to obtain more plausible and complete results [25, 16]. Hosni *et al.* [25], for instance, achieved this by constructing a cost volume and then filtering it using a bilateral filter. Recent works significantly improved the reconstruction accuracy and completeness by leveraging deep neural networks for both the feature extraction and the spatial aggregation (cost volume filtering) [70, 26]. Yao *et al.* [70] introduced an end-to-end learning-based multi-view stereo method. They jointly trained a 2D convolutional neural network for the feature extraction and a 3D convolutional neural network for the cost volume filtering in a supervised manner. Recursive [71] and coarse-to-fine [69, 19, 12] architectures for the cost volume filtering network were proposed to improve computational efficiency and resolution of the outputs. These methods are, however, fundamentally limited to textured and Lambertian surfaces as it is extremely challenging to extract view-independent features from appearance of non-Lambertian (*e.g.*, reflective) objects which drastically changes across views. Also, as they rely on triangulation, their outputs are limited to surface depths, *i.e.*, these methods cannot recover surface details (surface normals).

Kusupati *et al.* [31] introduced a deep MVS method that jointly estimates surface depths and surface normals. They improved the reconstruction accuracy by imposing a consistency loss between the depth and surface normal estimates. Their surface normal estimation, however, relies on features directly extracted from images. As the relationship between the appearance and surface normals are extremely complex for non-Lambertian objects, such a naive surface normal estimation method cannot generalize to textureless, non-Lambertian objects.

	# of Views	Textureless Non-Lambertian	Unknown Illum.
MVSNet [70]	3		✓
CVP-MVS [69]	3		✓
RC-MVS [8]	3		✓
IDR [73]	50	✓	✓
NeuS [64]	100	✓	✓
Oxholm & Nishino [50]	20	✓	
PhySG [76]	100	✓	✓
NDRMC [21]	100	✓	✓
NeRO [38]	100	✓	✓
Ours (Chapter 5)	5	✓	
Ours (Chapter 6)	2-10	✓	✓

(a) Dense surface recovery from posed images.

	# of Views	Textureless Non-Lambertian	Unknown Illum.
COLMAP [57]	50		✓
SAMURAI [5]	100	✓	✓
Ours (Chapter 7)	2	✓	✓

(b) Shape and camera pose recovery.

Table 3.1: Existing 3D reconstruction methods that handle images captured under natural illumination. Reconstruction of textureless, non-Lambertian surfaces still remains challenging unless we have an extremely large number (*e.g.*, 100) of input images.

Chang *et al*. [8] introduced a view synthesis loss for learning-based MVS that can implicitly deal with appearance of non-Lambertian objects. Their method, however, still relies on a photometric loss that assumes view-independent surface radiance (*i.e.*, Lambertian reflectance). They cannot handle objects whose appearance largely deviates from the Lambertian assumption.

3.2 Inverse Rendering

Inverse rendering methods recover the surface geometry by inverting the radiometric image formation process, *i.e.*, jointly estimating surface geometry, surface reflectance, and surrounding illumination so that images rendered from them become consistent with observed images [23, 27, 22, 2, 50]. Johnson and Adelson [27] recovered surface normals of Lambertian surfaces from a single image taken under know natural illumination by approximating Lambertian shading by a quadratic function of the surface normal. Oxholm and Nishino [50] jointly recovered surface normals and surface reflectance from a single image of a textureless, non-Lambertian object taken under known but natural illumination. They alternated between updating the surface normal estimates and the reflectance estimate by leveraging manually designed prior constraints on surface geometry and reflectance. These single-view methods are, however, fundamentally limited to reconstruction of surface normals and absolute depths cannot be recovered. They also make restrictive assumptions (*e.g.*, known illumination) to make the challenging single-view inverse problem tractable.

Some inverse rendering methods achieved reconstruction of absolute depths by optimizing a single geometry estimate using multi-view observations [50, 46, 65, 76, 13, 4, 21]. Oxholm and Nishino [50] extended their method for estimation from posed multi-view images taken under known but natural illumination by using a 3D mesh model as proxy geometry and updated it along with a reflectance estimate. Zhang *et al.* [76] jointly optimized surface geometry represented by a neural signed distance field (SDF), surface reflectance represented by another neural network, and surrounding illumination approximated by spherical Gaussians using posed multi-view images taken under unknown natural illumination. These methods, however, struggle to deal with objects with complex appearance (*e.g.*, reflective objects) as the relationship between the appearance and the surface geometry is extremely complex for such objects and the iterative nonlinear optimization based on it can easily falls into a local minima. To avoid this, they require an extremely large number (100) of input images especially when the illumination is unknown. They also simplify the rendering process or computation of their objective functions to reduce computational burden as evaluation of the rendering equation under complex natural illumination is computationally extremely expensive.

Reflectance Maps for Inverse Rendering As reflectance maps are known from 70s [24], some existing multi-view methods also leveraged reflectance maps for geometry reconstruction [50]. The most important difference between these methods and our nLMVS-Net (Chapter 5) is that, while inverse rendering methods directly leverage reflectance maps for nonlinear optimization of geometry, our method first filter them for each view using a deep shape-from-shading sub-network. By aggregating the observation likelihoods on the image plane and also leveraging learned realistic prior on surface geometry, the network can convert the complex, non-parametric observation likelihoods into well-defined, pixel-wise probability densities of surface normals which is much more tractable compared to the observation likelihoods. We show that the filtered surface normal estimates are in fact useful for geometry estimation based on multi-view stereo (nLMVS-Net) and multi-view SDF optimization (DeepShaRM). In Chapter 6, we also derive an estimation framework that can jointly recover surface geometry and reflectance maps from only posed multi-view images, which enable 3D reconstruction in the wild, *i.e.*, under unknown lighting.

3.3 Neural Image Synthesis

Instead of estimating physical quantities in the scene, some recent methods model the object appearance by a neural network and optimize its weights along with a neural geometry representation using posed multi-view images [48, 45, 73, 64]. Niemeyer *et al.* [48] optimized two MLPs that represent fields of occupancy values (geometry) and object colors by differentiable surface rendering with posed multi-view images. Mildenhall *et al.* (NeRF) [45] modeled the object appearance by an MLP that takes in not only the surface location but also the viewing direction so that it can learn view-dependent appearance of non-Lambertian objects. They also introduced a differentiable volume rendering framework which can be applied to images with cluttered backgrounds. Yariv *et al.* [73] exploited a neural appearance representation that also takes in a surface normal as an input. By this, the MLP can learn the appearance as a universal function that is almost independent of the surface location, *i.e.*, reflectance maps. Liu *et al.* [38] integrated inverse rendering with neural image synthesis so that the neural appearance representation can learn complex appearance of objects (*e.g.*, global illumination effects) that cannot be modeled by a simplified radiometric image formation model. Thanks to the carefully designed parameterization and inductive bias of the MLPs, these methods can recover surface geometry even for non-Lambertian objects if we have a large number (*e.g.*, 100) of input images. These methods, however, fail to recover plausible surface geometry on sparse inputs as they train their neural representation for each scene separately. A few methods exploit training data and handle sparse inputs by conditioning their neural representations with input images [40, 54]. Directly estimating geometry and appearance from images is, however, extremely challenging especially when the appearance of the object is complex, *i.e.*, the object is non-Lambertian.

3.4 Camera Pose Recovery

Structure-from-Motion Camera pose estimation from images has also been studied. Structure-from-motion (SfM) methods [78, 57] have established practical means for this task which only require multi-view images taken under unknown natural illumination as inputs. SfM methods first detect pixels in different views that correspond to the same surface point by leveraging view-dependent, salient image features. They then jointly solve for camera poses and 3D locations of the detected (sparse) surface points based on multi-view geometry. Image descriptors that are invariant to rotation and scale (*e.g.*, SIFT [42]) are often used for the view-invariant feature extraction. Also, for robust estimation, SfM methods often detect outlier (wrong) correspondences using RANSAC [15]. SfM methods, however, fail on textureless, non-Lambertian objects as their appearance is view-dependent and the detection of the corresponding pixels from their appearance is extremely challenging. Certainly, even if the target object has no texture, it is possible to recover camera poses from textures in backgrounds. For stable estimation, it, however, requires a dense-view (*e.g.*, 50 views) sampling of images which have high visual overlaps.

Neural Image Synthsis with Camera Pose Optimization A few recent neural image synthesis methods handle multi-view images whose camera poses are inaccurate or unknown by jointly optimizing camera pose estimates along with their neural geometry and appearance representations [73, 5]. These methods, however, still require good initial estimates of camera poses and dense view (*e.g.*, 100 view) sampling of input views to resolve the ambiguity between the unknown parameters.

Pose Estimation from Specular Reflections A few methods have tackled camera (or object) pose estimation from appearance of specular objects [34, 56, 9, 47]. Lagger *et al.* [34], for instance, introduced an optimization method that refines camera pose estimates using images and a 3D CAD model of a specular object. They recover camera-view environment maps from the inputs and updated the camera poses so that discrepancies between the environment maps are minimized. These methods, however, assume known geometry [34, 56, 9, 47], known illumination [9], or a simple lighting model (*i.e.*, directional or point light source) thus we cannot use them for image-based 3D reconstruction under unknown natural illumination.

In contrast, our joint shape and camera pose estimation method with reflection correspondences takes only images of a textureless, non-Lambertian object taken under arbitrary unknown natural illumination as inputs and does not require any background textures.

Chapter 4

nLMVS-Dataset

Different kinds of multi-view image datasets [1, 30, 72, 59] have been proposed for benchmarking of multi-view methods. Most of them, however, (implicitly) focus on evaluation of geometry reconstruction methods on textured and Lambertian surfaces. For this, we cannot use them for training and evaluation of our method. Oxholm and Nishino [50] created a multi-view image dataset that consists of images of textureless, non-Lambertian objects. Their dataset is, however, limited in terms of the number of instances (4 objects under 3 illumination environments). We also found that their dataset contains flaws about the image capture. That is, the images in their dataset are often saturated and geometric calibration of the camera poses are poor.

In this chapter, we introduce two novel multi-view image datasets which are essential for training and evaluation of non-Lambertian 3D reconstruction methods. First, we create nLMVS-Synth dataset, a large-scale synthetic image dataset which we can use for both training and thorough evaluation of our methods. We also create nLMVS-Real dataset, a real image dataset with ground truth 3D shapes and illumination maps which we can use for both qualitative and quantitative evaluations.

4.1 nLMVS-Synth

Figure 4.1 shows some example images from the nLMVS-Synth dataset. To make our methods robust to deviations from reflectance maps, we rendered images with shadows and interreflections based on Eq. (2.1). For this dataset creation, we used existing databases of synthetic shapes [66], measured BRDFs [43], and captured illumination maps [74, 17]. The shape, material, and illumination databases contains 5000 shapes, 100 BRDFs, 2685 illumination maps (452 from PolyHaven [74] and 2233 from Laval Indoor HDR datast [17]). As the number of BRDF data is relatively small, we augmented them using the conditional iBRDF model [11] as a generative model. We trained the conditional iBRDF model using 94 BRDFs from the material database [43]. Note that, we left 6 BRDFs for a test set described below. We then sampled 2685 BRDFs from a latent space learned by the conditional iBRDF model and used them along with 2685 synthetic shapes and 2685 illumination maps to render images of 2685 synthetic objects. For each object, we rendered images of randomly sampled ten-views. Therefore, the training set of the nLMVS-Synth dataset consists of 26850 images in total.

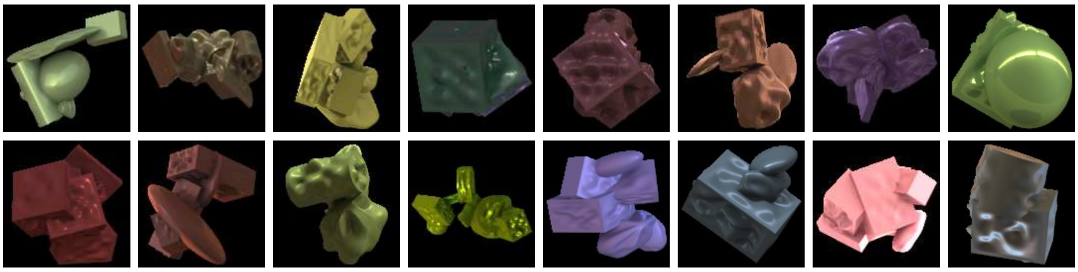


Figure 4.1: Example images in the training set of the nLMVS-Synth dataset. The training set consists of 26850 images of 2685 synthetic shapes.

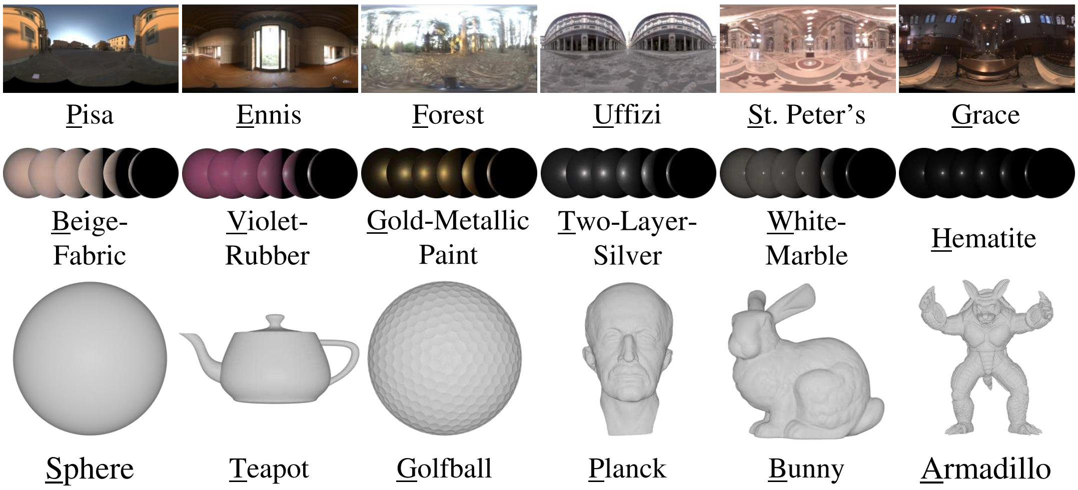


Figure 4.2: The test set of the nLMVS-Synth dataset consists of 4320 images of 216 different combinations of 6 illumination maps, 6 BRDFs, and 6 shapes. © 2023 IEEE [67]

Test Set For thorough qualitative and quantitative evaluation our methods, we also rendered a separate set of synthetic images. As showm in Fig. 4.2, we used 6 illumination maps [14, 32], 6 BRDFs [43], and 6 shapes [6, 33, 55] for the creation of the test set. We rendered images for all 216 combinations of the shapes, the materials, and the illumination environments. For each combination, we sampled 20 views on the horizontal line at equal intervals. For each sampled view, we added perturbations to the camera locations.

4.2 nLMVS-Real

We also create nLMVS-Real, a multi-view real image dataset for evaluation of the proposed methods. Figure 4.3 shows example images from the nLMVS-Real dataset. In addition to multi-view images, we provide calibrated camera poses and ground truth 3D mesh models for quantitative evaluations.

Target Non-Lambertian Subjects The bottom row of Fig. 4.4 shows examples of the target non-Lambertian subjects. We created them by replicating existing 5 different 3D mesh models ("Bunny" from the Stanford Computer Graphics Laboratory [33], "Planck" from the Suggestive Contour Gallery [55], "Horse" and "Shell" from the Multiview Objects Under Natural Illumination database [49], and "Sphere")

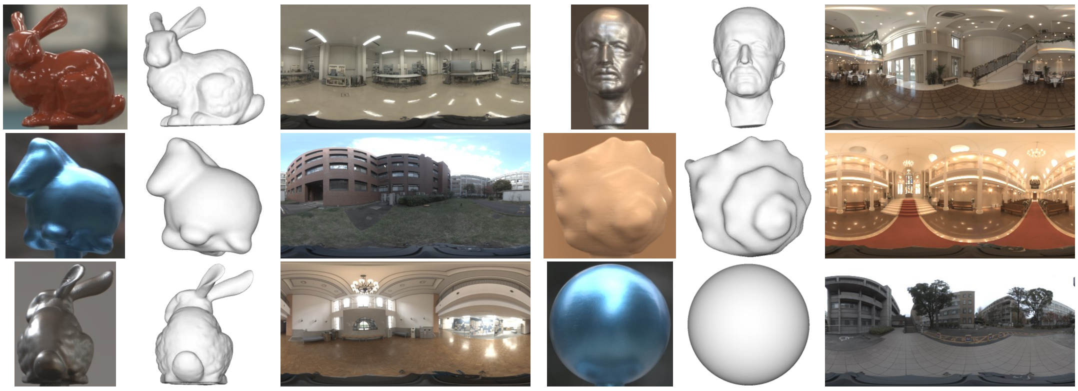


Figure 4.3: Example images from the nLMVS-Real dataset. For quantitative evaluation, we provide calibrated camera poses and aligned 3D mesh models (the second column for each object) for each object. We also provide illumination maps (the third column) captured by a 360 degree camera (RICOH THETA Z1).

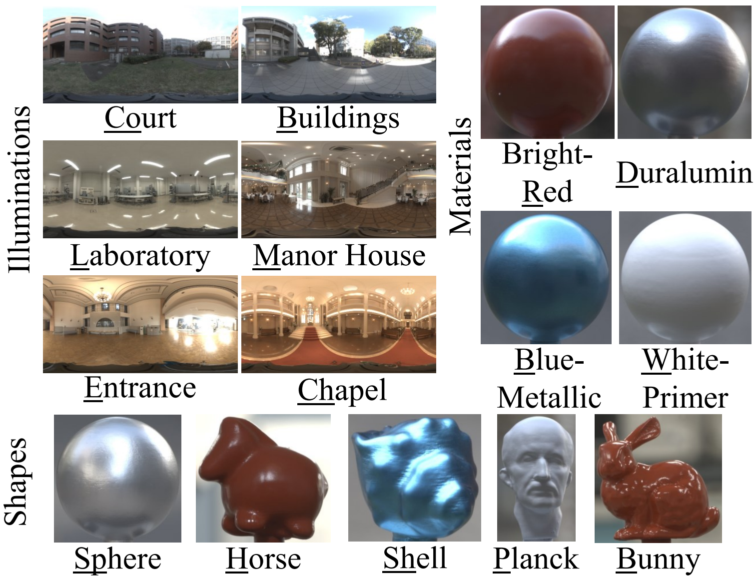


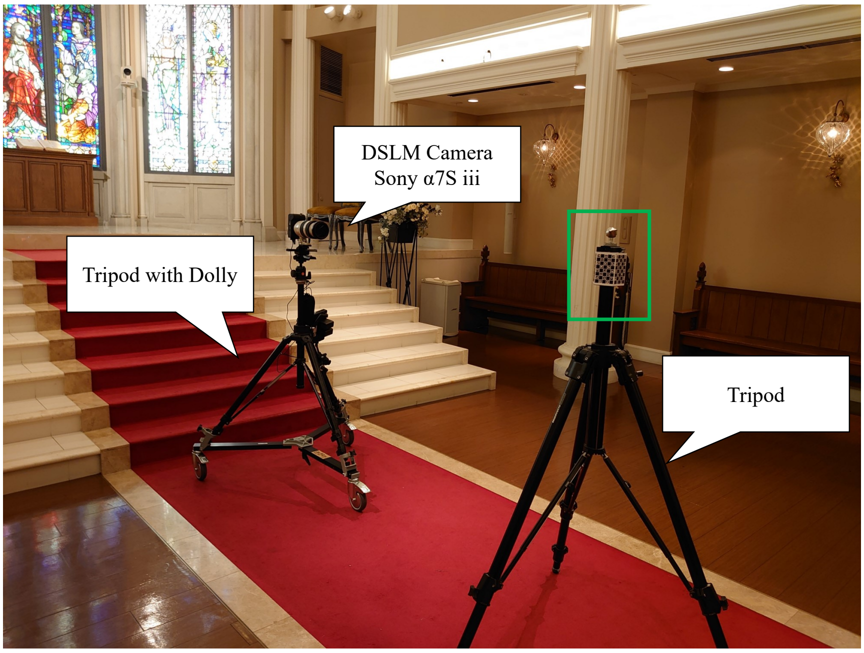
Figure 4.4: The nLMVS-Real dataset consists of 2569 multi-view images of 120 different combinations of 5 shapes, 4 materials, and 6 illumination environments. © 2023 IEEE [67]

using a 3D printer. We painted the 5 shapes with four paints ("Bright-Red", "Duralumin", "Blue-Metallic", and "White-Primer" in the right of Fig. 4.4).

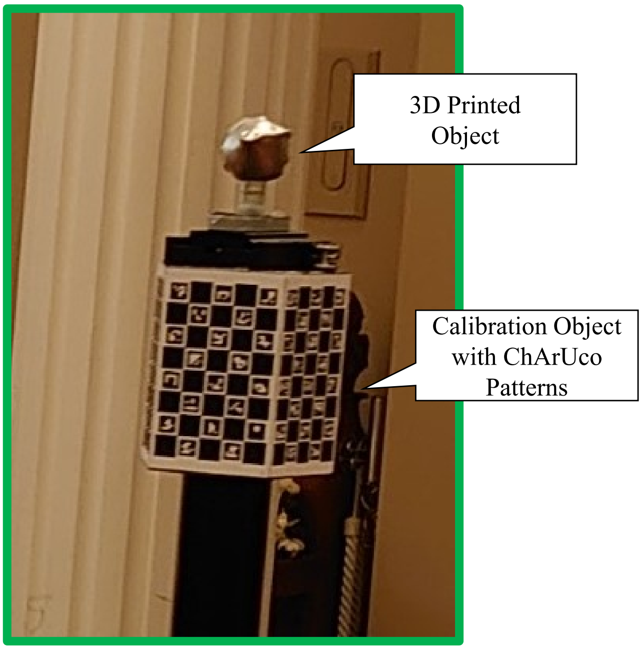
Surrounding Illumination Environments As shown in the top left of Fig. 4.4, we captured images of the target non-Lambertian subjects under 6 different illumination environments ("Court", "Buildings", "Laboratory", "Manor House", "Entrance", and "Chapel"). In total, we captured multi-view images for 120 combinations of 5 shapes, 4 paints, and 6 illumination environments.

4.2.1 Image Capture

Figure 4.5 shows our image capturing setup. We located a target subject on the top of a tripod so that the target object does not move during the image capture. A

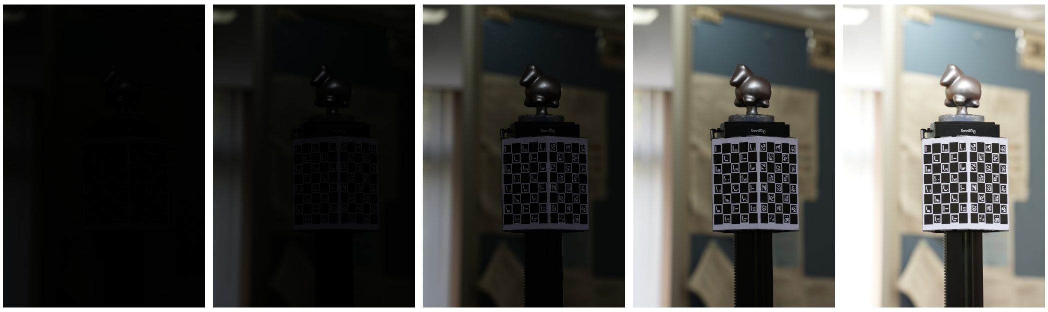


(a)

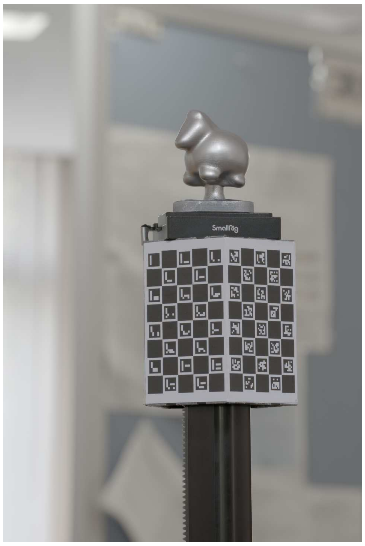


(b)

Figure 4.5: Image capturing setup. We located the target subjects on the top of a tripod. A calibration object with ChArUco patterns are also attached to the tripod for geometric calibration. We captured multi-view images of the subjects using a DSLM camera (Sony α7S iii) located on a tripod with dolly.



(a) SDR images with different exposure times



(b) Recovered HDR image

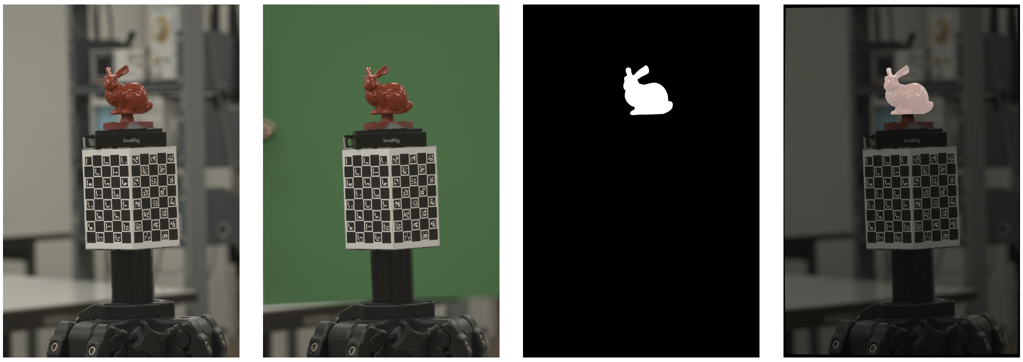
Figure 4.6: For each view, we captured standard dynamic range (SDR) images with 5 different exposure times and recover a high dynamic range (HDR) image from them. In this figure, we use the logarithmic function as a tone mapping function for the visualization of the HDR image.

calibration object with ChArUco patterns are also attached to the tripod for geometric calibration. We captured multi-view images of the objects using a DSLM camera (Sony α7S iii). We put the DSLM camera on a tripod with dolly so that the camera can move around the target subject during the image capture.

As irradiances of pixels in specular highlights are often much higher than those of other pixels, a single standard dynamic range (SDR) image cannot capture accurate irradiances in both regions. For this, as shown in Fig. 4.6, for each view, we captured SDR images with 5 different exposure times and recover a high dynamic range (HDR) image from them.

For each combination of a shape, a material, and an illumination environment, we captured roughly 20 HDR images captured from three different heights. In total, the nLMVS-Real dataset consists of 2569 images of the target subject.

Illumination Capture As shown in the right of Fig. 4.3, we also captured illumination maps of the surrounding illumination using a 360 degree camera (RICOH THETA Z1). For each combination of a shape, a material, and an illumination environment, before the image capture of the target subject, we put the 360 degree camera



(a) Captured image (b) Image with green background (c) Recovered mask (d) Calibration result

Figure 4.7: For calibration of 3D locations of the ground truth 3D mesh models, we exploit silhouette images of the subject (c) recovered from images with green background (b). In (d), we visualize the calibrated 3D location by overlaying the silhouette of the aligned 3D mesh model on the image, which demonstrates the accuracy of the calibration.

on the top of the (static) tripod and captured panorama images with 7 different exposure times and recovered a single HDR illumination map.

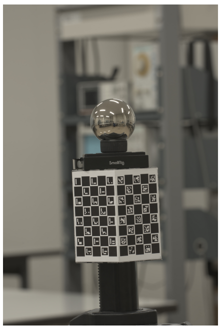
4.2.2 Geometric Calibration

For plausible evaluation of geometry reconstruction methods, accurate camera poses and 3D locations of the target subject are also essential. For this, we calibrated them using the calibration target with ChArUco patterns.

Intrinsic Calibration For each combination of a material and an illumination environment, we captured images of the ChArUco patterns from roughly 20 views and used them for calibration of intrinsic parameters of the camera (*i.e.*, focal length). We detected 2D-3D correspondences using the ChArUco patterns and used them as constraints for the calibration. For stable estimation, we used intrinsic parameters computed from metadata in the EXIF tags as an initial guess, and fixed the principal point (2D location of the optical axis in the image) during the calibration with the ChArUco patterns.

Camera Pose Calibration We also calibrated camera poses (extrinsic camera parameters) for each image of the target subject using ChArUco patterns observed in them. Similar to the intrinsic calibration, we detected 2D-3D correspondences using the ChArUco patterns and used them for optimization of the unknown parameters, *i.e.*, a rotation matrix and a translation vector which represent the camera pose relative to the calibration target.

Registration of The Ground Truth Geometry As the target subjects are replicated by a 3D printer, we can use the original 3D mesh models as the ground truth surface geometry. 3D locations of the ground truth geometry are, however, unknown and we cannot directly use them for quantitative evaluation. For this, we also calibrated the 3D locations of the target object. Figure 4.7 shows how we calibrate the



(a) An image of a mirror sphere

(b) Illumination map recovered from the images of the mirror sphere

(c) Illumination map captured by a 360 degree camera

(d) Alignment Result

Figure 4.8: We consolidate the coordinate system of the illumination map captured by a 360 degree camera (c) and the calibrated camera poses using images of a mirror sphere (a). Using the images of the mirror sphere and exploting ChArUco patterns in them, we can recover an illumination map aligned with the calibrated camera poses. We then consolidate its coordinate system and one of the illumination map captured by a 360 degree camera using point-to-point correspondences between the two illumination maps (d).

3D locations of the mesh models. We first initialize the 3D locations using approximated relative pose between the subject and the calibration target. We then detect silhouettes of objects using images with green background and use them to refine the 3D locations of ground truth 3D mesh models, *i.e.*, we optimize the 3D locations by minimizing discrepancies between the recovered silhouettes and ones of the 3D mesh models.

Alignment of The Illumination Maps We also aligned (rotated) the captured illumination maps so that the coordinate system of the illumination maps and one of the calibrated camera poses are consolidated. As shown in Fig. 4.8, we achieve this by capturing and exploiting images of a mirror sphere. From the images of the sphere, we can recover illumination maps by simply mapping pixels in them. We can also consolidate its coordinate system and one of the calibrated camera parameters using the camera pose for the image computed from the ChArUco patterns. We then align the illumination map captured by a 360 degree camera with one recovered from the images of the mirror sphere by detecting and leveraging point-to-point correspondences between them.

4.3 Summary

In this chapter, we introduced two novel datasets for training and evaluation of our geometry reconstruction methods for textureless, non-Lambertian objects. The nLMVS-Synth dataset consists of an extremely large number of images (26850 for training and 4320 for evaluation) which are sufficient for training of deep neural networks and thorough evaluations. Thanks to the carefully designed calibration method, the nLMVS-Real dataset provides not only multi-view high dynamic range (HDR) images but also calibrated camera poses, ground truth 3D shapes, and aligned illumination maps. We believe these novel datasets can serve as a new platform for not only geometry reconstruction methods but also other inverse rendering tasks (*e.g.*, joint reflectance and illumination estimation).

Chapter 5

nLMVS-Net: Deep non-Lambertian Multi-View Stereo

Let us assume, for now, that we know the camera poses, surface reflectance of the object, and the surrounding illumination environment. Even if we know them, *i.e.*, even if we could compute reflectance maps, geometry reconstruction of textureless, non-Lambertian objects is still challenging. The relationship between surface normals and surface appearance (surface radiances) is extremely complex, which prevents us from leveraging radiometric cues for 3D reconstruction. That is, if the reflectance maps are given, we can compute pixel-wise likelihoods of surface normals by comparing pixel values in the input images and ones in reflectance maps. As visualized in Fig. 5.3, the observation likelihoods are, however, complex and surface normal recovery from them for the multi-view geometry reconstruction is non-trivial. Some existing multi-view inverse rendering methods [50, 76] have tried to leverage the information of reflectance maps by formulating the inverse problem as a nonlinear optimization of the surface geometry based on the rendering consistency. Such a naive optimization with the complex relationship between surface radiance and surface normals, however, easily falls into a local minima.

In this chapter, we introduce a novel multi-view stereo method that can recover surface depths, surface normals of a textureless, non-Lambertian object from a sparse set (5) of posed multi-view images captured under known but complex natural illumination and corresponding reflectance maps. To overcome the difficulties of non-Lambertian 3D reconstruction under complex natural illumination, as depicted in Fig. 5.3, we introduce a deep shape-from-shading network that filters non-parametric, complex radiometric likelihoods of surface normals computed from reflectance maps. By aggregating the information radiometric likelihoods on each image plane (each view) and also leveraging learned realistic prior on surface geometry, the network can estimate surface normals as well-defined pixel-wise probability densities. Once we obtain such well-defined probability densities of surface normals, we can leverage them as view-independent features for correspondence detection and recover the surface geometry using a learning-based multi-view stereo framework (cost volume filtering). We also integrate our method with a neural reflectance estimation method [11] and, as shown in Fig. 5.1, demonstrate that we can in fact exploit our method for geometry recovery from images taken under known but complex natural illumination.

The filtering of the pixel-wise radiometric likelihoods is, however, non-trivial as they are non-parametric and four dimensional. We cannot process them using

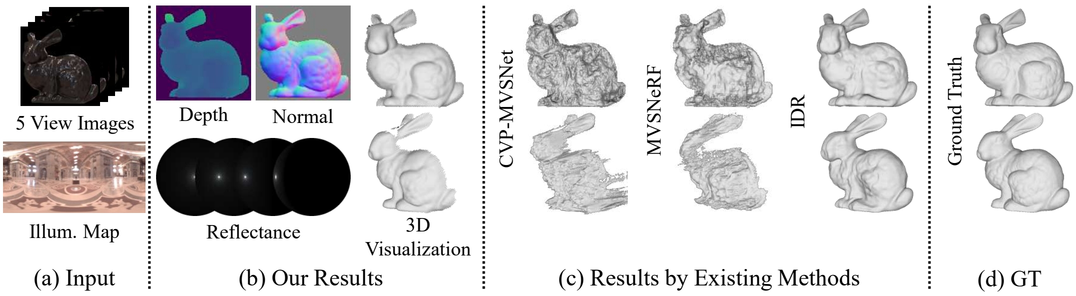


Figure 5.1: In this chapter, we introduce a novel deep multi-view stereo method that can recover depths and surface normals from posed multi-view images, surface reflectance, and a illumination map. By integrating it with a neural reflectance map estimation method, we also show that we can recover geometry and and reflectance (b) of a textureless, non-Lambertian object from a sparse set (5) of posed multi-view images taken under known but complex natural illumination (a). Surface geometry recovered by our method is consistent with the ground truth (d), while existing methods struggle to handle the complex appearance of the non-Lambertian object under natural illumination (c). © 2023 IEEE [67]

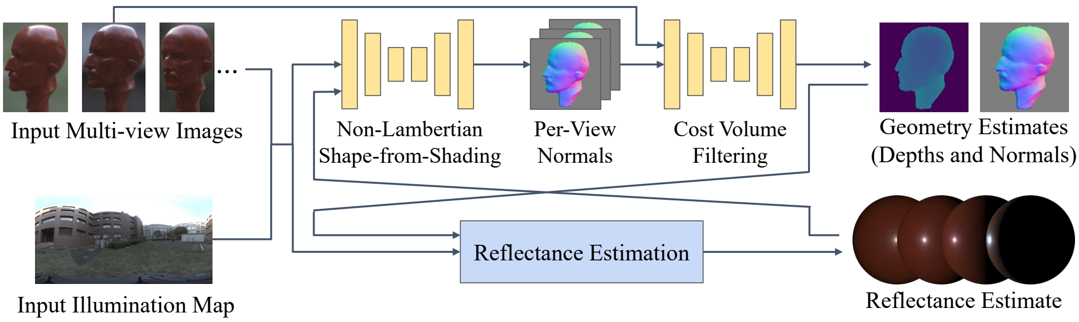


Figure 5.2: A method overview of nLMVS-Net. nLMVS-Net consists of a non-Lambertian shape-from-shading network (Sec. 5.2), a cost volume filtering network (Sec. 5.3), and a joint estimation of surface geoemtry and surface reflectance with a neural reflectance estimation method (Sec. 5.4). © 2023 IEEE [67]

a standard convolutional neural network as a naive convolution of 4D data is computationally extremely expensive. For this, we derive an architecture that samples radiometric likelihoods of surface normals in a coarse-to-fine manner. We also introduce a cost volume filtering network with novel feature extraction layers that extract view-independent features from the estimated probability densities which are also non-parametric.

5.1 Method Overview

Figure 5.2 depicts an overview of our nLMVS-Net. In Sec. 5.2, we explain how we compute the radiometric likelihoods of surface normals and how we filter them using the deep shape-from-shading network. In Sec. 5.3, we introduce a cost volume filtering network that exploits probability densities of surface normals estimated by the shape-from-shading network. In Sec. 5.4, we describe how we integrate our method with a neural relfectance estimation method for a joint estimation of surface geometry and surface reflectance.

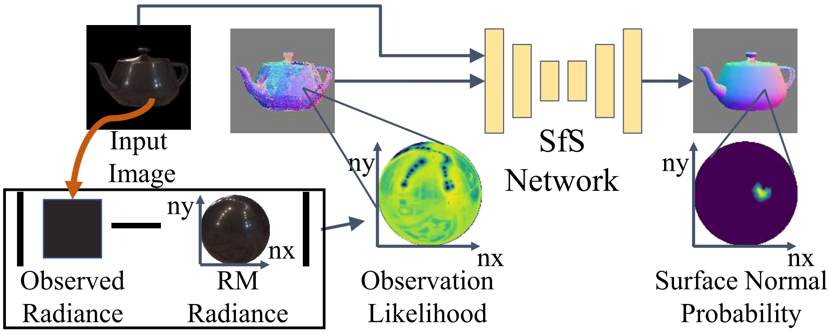


Figure 5.3: For the single-view non-Lambertian shape-from-shading, we first compute pixel-wise observation likelihoods of surface normals by comparing observed radiences and ones in a reflectance map (RM) computed from the input illumination map and a current estimate of surface reflectance. We then filter the complex observation likelihoods using a novel deep shape-from-shading (SfS) network.

5.2 Non-Lambertian Shape-from-Shading Network

At first, let us assume that we know the reflectance (the BRDF) of the object. Note that, in Sec. 5.4, we explain how to extend our method for a joint estimation of surface geometry and surface reflectance. As mentioned before, we also assume that we know the surrounding illumination, *i.e.*, an illumination map $L_i(\boldsymbol{\omega}_\mathbf{i})$ that represents the relationship between the incident direction and the incident radiance. Given the BRDF and the illumination map, we can (numerically) compute the reflectance map using Eq. (2.8) and then compute observation likelihoods of surface normals for each pixel by comparing the observed radiance $I_\mathbf{m}$ and the computed reflectance map $E(\mathbf{n})$

$$p(I_\mathbf{m}|\mathbf{n}) = \prod_{c=1}^{3} f(\log I_\mathbf{m}^{(c)}; \log E^{(c)}(\mathbf{n}), b)\,, \tag{5.1}$$

where $f(x; \mu, b)$ is the Laplace distribution and c is index of color channels. We optimize the parameter b during training of the shape-from-shading network. Figure 5.3 shows an example of the observation likelihoods for a pixel. They encode rich information about surface normals which we can extract from the inputs and the current reflectance estimate.

The observation likelihoods are, however, extremely complex and cannot be directly used as features for multi-view stereo. Evaluating similarities between the non-parametric observation likelihoods for all possible combinations of pixels is non-trivial and costly. For this, before using the observation likelihoods for multi-view stereo, for each view, we filter them using a deep shape-from-shading (SfS) network. As depicted in Fig. 5.3, the input of the SfS network are an input image and corresponding pixel-wise observation likelihoods. The network aggregates the information of the inputs on the image plane and outputs refined surface normal estimates as pixel-wise probability densities. Using the nLMVS-Synth dataset (*i.e.*, synthetic images with ground truth reflectace and normal maps), We train the network in a supervised manner. By this, the network can also learn to leverage learned realistic prior on surface geometry.

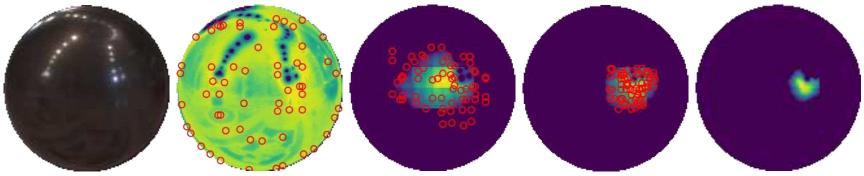


Figure 5.4: To process the four dimensional pixel-wise observation likelihoods (a 2D non-parametric map for each pixel), we derive a recursive estimation architecture that samples the observation likelihoods in a coarse-to-fine manner. © 2023 IEEE [67]

A technical challenge here is that the observation likelihoods are high dimensional (four dimensional) and cannot be processed by standard convolutional layers. In practice, we set the input image size as 128×128 and compute the reflectance map as an image of a sphere whose size is also 128×128. Thus the size of the observation likelihoods become 128^4, which we cannot process by a naive 4D convolutional neural network due to computational burden. We overcome this problem by introducing a recursive estimation architecture that exploits the observation likelihoods in a coarse-to-fine manner. Figure 5.4 visualizes how we leverage the observation likelihoods in the SfS network. First, for each pixel, we divide the map of observation likelihoods into a 8×8 grid. For each grid, we find the orientation $\mathbf{n}_j$ that maximizes the observation likelihood $p(I_{\mathbf{m}}|\mathbf{n}_j)$ by a brute-force-search. Inspired by PointNet [51], for each sampled orientation, we first extract a 64 dimensional feature vector from the surface normal $\mathbf{n}_j$ and corresponding observation likelihood $p(I_{\mathbf{m}}|\mathbf{n}_j)$, and then fuse the sample-wise features using max-pooling. We feed the fused pixel-wise feature vectors along with the input image to a 2D convolutional UNet and obtain pixel-wise filtered feature vector $\mathbf{a}$. We decode the filtered feature $\mathbf{a}$ into a refined probability densities of surface normals $\hat{p}(\mathbf{n})$ using a decoder MLP $g(\mathbf{n})$

$$\hat{p}(\mathbf{n}) = p(I_{\mathbf{m}}|\mathbf{n})g(\mathbf{b};\mathbf{a})\,. \tag{5.2}$$

Once we obtain the refined probability densities, we sample the observation likelihoods according the refined densities instead of the complex observation likelihoods. Specifically, in the subsequent iteration of the recursive estimation, we double the resolution of the grid for the brute-force search and, for each grid, we find the orientation that maximizes the probability densities. In the subsequent iterations, we sort the sampled orientations according to their probability densities, and use only the first 64 samples as inputs to the SfS network.

We train the shape-from-shading network with the nLMVS-Synth dataset in a supervised manner. In training, we exploit an image and a reflectance map computed from a ground truth BRDF and an illumination map as inputs. We impose the cross entropy loss on the corresponding outputs

$$L_{\mathrm{SfS}} = -\sum_j M(\mathbf{n}_j) \log\left(\frac{\hat{p}(\mathbf{n_j})}{\sum_{j'}\hat{p}(\mathbf{n_{j'}})}\right)\,, \tag{5.3}$$

where $\{\mathbf{n}_j\}$ is the set of sampled surface normals and M is a binary mask whose value is 1 iff the ground truth and the sampled surface normals are in the same grid. We impose this loss on the output of each iteration of the coarse-to-fine estimation.

As shown in Fig. 5.4, the filtering by the shape-from-shading network and the coarse-to-fine estimation significantly reduce the ambiguity in the observation likelihoods, and recovers a well-defined probability density for each pixel.

5.3 Cost Volume Filtering Network

Once we obtain the well-defined probability densities of surface normals, we can exploit them as view-independent features for multi-view stereo. Note that, although the probability densities of surface normals are estimated in each local camera coordinate system, we can consolidate the coordinate system using known camera poses. As depicted in Fig. 5.5, we achieve this multi-view stereo with the estimated surface normal probability densities using a novel cost volume filtering network. Inputs to the network are posed multi-view images and corresponding probability densities of surface normals. The network first extracts per-pixel features from the inputs and then uses them to construct a feature cost volume for one of the input views. The network further filters the feature cost volume using a 3D convolutional UNet and outputs surface geometry as depth and surface normal estimates in the reference view.

As the estimates of the shape-from-shading network are also pixel-wise probability densities, we cannot use a standard convolutional layers for feature extraction. As depicted in Fig. 5.5a, we derive a feature extraction layer specialized for the probability densities. Using the surface normal orientations sampled in the last iteration of the recursive estimation of the shape-from-shading network, we first construct pixel-wise sparse sets of surface normals and corresponding probability densities. As mentioned before, we consolidate the coordinate system of the surface normals using the known camera poses. We then convert the pixel-wise unstructured sets of sampled surface normals and probability densities into pixel-wise feature vectors. We process the unstructured set of probability densities using a feature extraction network similar to one for the shape-from-shading network. We further filter the pixel-wise features along with the input image to obtain pixel-wise feature vectors.

As depicted in Fig. 5.5b, once we obtain the pixel-wise features for each view, we can use them to construct and filter a cost volume. We first construct a latent cost volume for each pair of the reference view and one of the other views similar to Eq. (2.15). Instead of comparing features sampled from different views based on a metric function $M(\cdot, \cdot)$, here we construct a latent cost volume by simply concatenating the sampled features and filter it using a shallow 3D convolutional neural network. We then fuse the per-view latent cost volumes by simply taking the average of them and further filter it using a 3D convolutional UNet. The UNet outputs the estimated geometry as 3D volumes of depth probabilities $\hat{p}(d; \mathbf{m})$ and surface normals $\hat{\mathbf{n}}(d, \mathbf{m})$, which we can convert to depth and normal maps by computing the expected values

$$\hat{d}(\mathbf{m}) = \sum_{d} \hat{p}(d; \mathbf{m}) d \,, \tag{5.4}$$

$$\hat{\mathbf{n}}(\mathbf{m}) = \frac{\sum_d \hat{p}(d;\mathbf{m})\mathbf{n}(\hat{d,\mathbf{m}})}{\left\|\sum_d \hat{p}(d;\mathbf{m})\mathbf{n}(\hat{d,\mathbf{m}})\right\|} . \tag{5.5}$$

We train the cost volume network separately from the training of the shape-from-shading network, *i.e.*, we first train the shape-from-shading network and then fix its weights during the training of the cost volume filtering network. Using multi-view images and corresponding ground truth depth and normal maps in the nLMVS-Synth dataset, we impose depth and normal supervisions

$$L_d = \sum_{\mathbf{m}} \left\| \hat{d}(\mathbf{m} - d_{\mathrm{gt}}(\mathbf{m})) \right\|_1 , \tag{5.6}$$

$$L_{\mathbf{n}} = \sum_{\mathbf{m}} \arccos\left(\hat{\mathbf{n}}(\mathbf{m}) \cdot \mathbf{n}_{\mathrm{gt}}(\mathbf{m})\right) , \tag{5.7}$$

where $d_{\mathrm{gt}}(\mathbf{m})$ and $\mathbf{n}_{\mathrm{gt}}(\mathbf{m})$ are the ground-truth depth and surface normal. We also ensure the consistency between the depth and surface normal estimates by imposing a consistency loss

$$L_{\mathrm{dn}} = \sum_{\mathbf{m}} \arccos\left(\hat{\mathbf{n}}(\mathbf{m}) \cdot \bar{\mathbf{n}}(\mathbf{m})\right) , \tag{5.8}$$

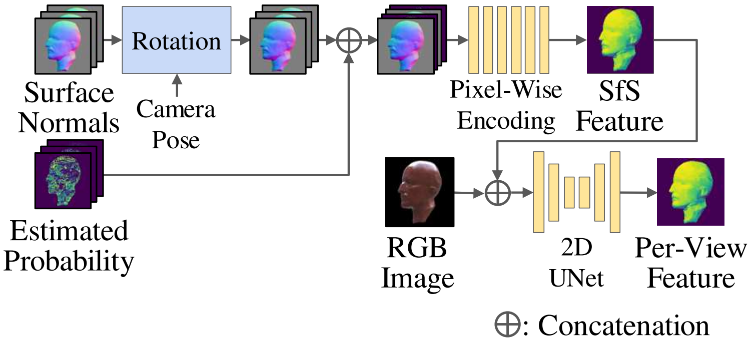
where $\bar{\mathbf{n}}(\mathbf{m})$ is the surface normal computed from the estimated depths $\hat{d}(\mathbf{m})$, *i.e.*, we compute cross product of tangent vectors on the surfaces. Note that, we estimate the surface normals as separate quantities as the fidelity of the depth estimates is limited by the resolution of the cost volume ($128 \times 128 \times 128$ in practice) and normals computed from them can be erroneous especially when the surfaces are not C0 and C1 continuous.

5.4 Joint Shape and Reflectance Estimation

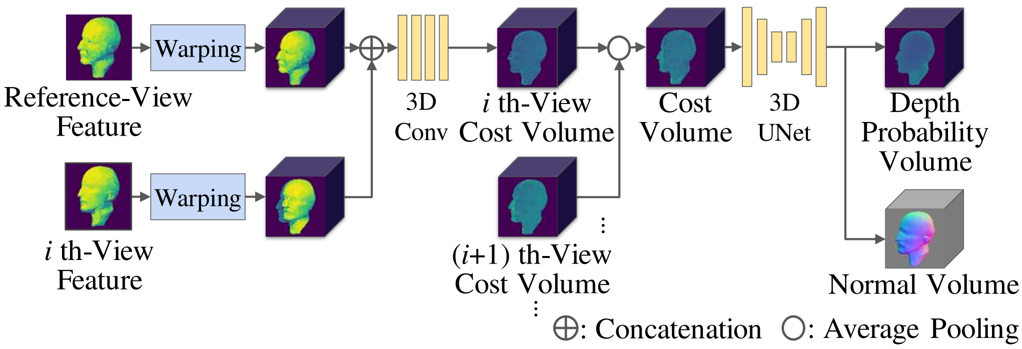
Even if we do not know the reflectance property of the object, we can still recover the surface geometry by alternating between estimating surface geometry using the proposed networks and updating a reflectance estimate using a neural reflectance estimation method. In the reflectance estimation, we represent the BRDF with the conditional invertible neural BRDF model (conditional iBRDF model) [11] and optimize its latent code. We realize the optimization by minimizing discrepancies between the input images and ones rendered from the current depth, surface normal, and BRDF estimates.

5.5 Whole 3D Shape Recovery

Our nLMVS-Net can recover per-pixel depths and surface normals given an image and its neighboring views (typically 4 neighboring views). If we have multi-view images that captures the entire regions of the object surfaces, we can recover the whole object 3D shape by integrating the per-view estimates of depths and surface normals. We achieve this by converting the depth and surface normal maps into a set of oriented points and then applying the Poisson surface reconstruction [29] to



(a) Feature extraction for each view



(b) Cost volume construction and filtering

Figure 5.5: Once we obtain well-defined, pixel-wise probability densities of surface normals, we can leverage them as view-independent features for multi-view stereo. We achieve this with a novel cost volume filtering network with a feature extraction layer for the probability densities. © 2023 IEEE [67]

them. As demonstrated in Sec. 5.6.4, we can exploit the recovered whole 3D surface geometry for novel-view synthesis and relighting.

5.6 Experiments

We focus our experiments on answering the main questions below. Can the shape-from-shading network recover surface normals as well-defined probability densities? Does the joint estimation of surface geometry and reflectance improves the estimation accuracy? How does our method work compared to MVS [69, 8], inverse rendering [50, 76], and neural image synthesis methods [73, 10]? How does each component of our method contribute to the estimation accuracy? Can our method generalize to real-world data? To answer these questions, we validate the following points.

- The probability densities recovered by the shape-from-shading network is qualitatively and quantitatively plausible.
- Our joint estimation framework is essential for geometry reconstruction of textureless, non-Lambertian surfaces.
- Our geometry reconstruction results on a sparse set (5) of multi-view images are more accurate than those of existing methods.
- Ablation studies show the effectiveness of each component in the nLMVS-Net.
- Our nLMVS-Net can work even with real-world images.

5.6.1 Accuracy of the Shape-from-Shading Network

We first evaluate the accuracy of the proposed shape-from-shading network on the nLMVS-Synth dataset. In this evaluation, we used the ground truth reflectance maps as inputs to the SfS network. For quantitative evaluation, we computed the mean and the median errors between the ground truth and the expected directions of the estimated probability densities. We also computed the rate of pixels whose estimation errors are lower than 10 degrees. Table 5.1a shows the accuracy for each shape. We also compare our results with those of existing methods on blobby shapes. The results show that the accuracy of our method, *e.g.*, our median error on the complex "Bunny" shape is more accurate than one of Oxholm on Nishino [50] on simple blobby shapes. In Tab. 5.1b, we report the accuracy of the shape-from-shading network for each iteration in the coarse-to-fine estimation. In the table, we also report errors of the expected surface normal orientation computed from the observation likelihoods. The results clearly show the effectiveness of the shape-from-shading network and the coarse-to-fine estimation.

In Tab. 5.1c, we also evaluate how the grid size of for the sampling of the observation likelihoods affects the estimation accuracy. In this table, we also report corresponding memory footprint required for training ("Mem"). The results show that the higher resolution results in better accuracy. Note that we couldn't increase the grid size beyond 8×8 due to limitation of GPU memory.

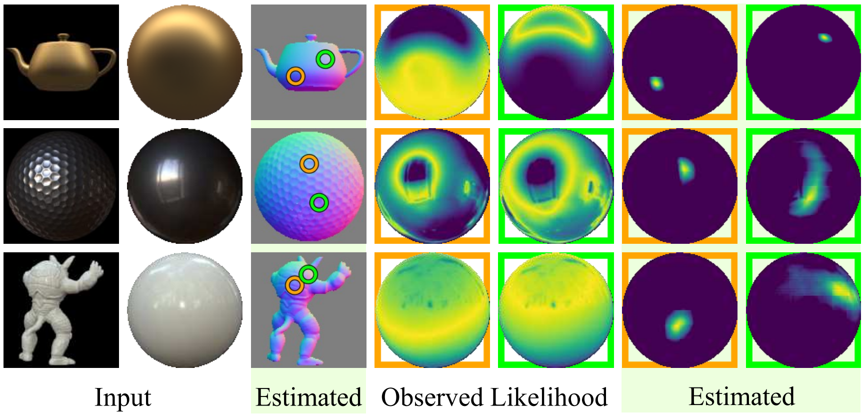


Figure 5.6: Results of the shape-from-shading network. Surface normals are successfully recovered as well-defined pixel-wise probability densities. © 2023 IEEE [67]

Figure 5.6 shows the expected surface normal orientations computed from the estimated probability densities ("Estimated"). For each object, we also visualize the observation likelihoods and the estimated probability densities for two pixels that correspond to circles in the "Estimated" results. The results show that our shape-from-shading network successfully reduces the ambiguity in the observation likelihoods for objects with different shapes and materials.

5.6.2 Joint Shape and Reflectance Estimation

We then evaluate the accuracy of the joint shape and reflectace estimation method in the nLMVS-Synth dataset. We first evaluated our method on five view images from the nLMVS-Synth dataset. We compared our results with those of MVS [8, 69] and neural image synthesis [10, 73] methods that also handle sparse inputs. For quantitative evaluation, we computed mean absolute errors of the estimated depths and surface normals. Note that, when evaluating the accuracy of the depth estimates, we normalized its scale so that the diagonal of the ground truth 3D shape becomes 1.

Figures 5.7 to 5.9 and Tab. 5.2 show qualitative and quantitative results. Our method successfully recovers surface geometry and reflectance consistent with the ground truth, while the existing methods struggle to deal with the objects with complex non-Lambertian reflectance. The quantitative results clearly show the accuracy of our method.

Effectiveness of the surface normal probability densities We evaluate the effectiveness of the use of the surface normal probability densities estimated by the shape-from-shading network. We achieve this by comparing our method with our method "w/o SfS" (without shape-from-shading) that constructs a cost volume only from multi-view image features. Figure 5.10 and Tab. 5.2 show qualitative and quantitative results. The results show the effectiveness of the use of the surface normal probability densities.

Effectiveness of the alternating estimation In Fig. 5.11, we report mean depth and surface normal estimation errors for each iteration of the alternating estimation.

	Material	Shape	$\leq$ 10 deg (↑)	Median (↓)
Ours	non-Lambertian	Sphere	99.7%	3.1 deg
		Teapot	91.3%	3.7 deg
		Golfball	84.7%	7.3 deg
		Planck	85.8%	7.2 deg
		Bunny	75.6%	12.4 deg
		Armadillo	40.7%	24.2 deg
JA [27]	Lambertian	blobs	90%	N/A
ON [50]	non-Lambertian	blobs	N/A	15 deg

(a) Accuracy for each shape

	$\leq$ 10 deg (↑)	Median (↓)	Mean (↓)
Likelihood	56.6%	49.4 deg	19.8 deg
1st	75.9%	26.9 deg	9.4 deg
2nd	80.3%	26.5 deg	8.4 deg
Final (Ours)	**83.4%**	**24.2 deg**	**7.7 deg**

(b) Accuracy for each stage

Grid Size	Mem.	$\leq$ 10 deg (↑)	Median (↓)	Mean (↓)
2×2	6.6 GB	74.2%	31.5 deg	10.0 deg
4×4	16.6 GB	79.7%	41.3 deg	8.9 deg
8×8 (ours)	43.5 GB	**83.4%**	**24.2 deg**	**7.7 deg**

(c) Effectiveness of the grid size

Table 5.1: Accuracy of the shape-from-shading network. Please see the text for details and discussions.

	Depth (↓)	Normal (↓)
RC-MVS [8]	5.76 %	-
MVSNeRF [10]	5.59 %	-
CVP-MVS [69]	4.16 %	-
IDR [73]	1.11%	11.0 deg
w/o SfS	1.12 %	11.2 deg
Ours	**0.94 %**	**9.8 deg**

Table 5.2: Mean depth and surface normal estimation errors on five-view images in the nLMVS-Synth dataset.

	Depth (↓)	Normal (↓)
w/o Depth Supervision	27.74 %	7.1 deg
w/o Normal Supervision	0.49 %	9.0 deg
w/o Consistency Loss	0.35 %	**6.3 deg**
Ours	**0.34 %**	**6.3 deg**

Table 5.3: Ablation study on the loss functions for the training of the cost volume filtering network.

The results clearly show the effectiveness of the joint shape and reflectance estimation.

Effectiveness of the loss functions for training of the cost volume filtering network In Tab. 5.3, we evaluate how each of the proposed loss functions for training of the cost volume filtering network affects the estimation accuracy by ablation study. In this evaluation, we used the ground truth reflectance maps as inputs. The results show the effectiveness of each of the loss functions.

Effectiveness of the probabilistic representation of shape-from-shading results In Tab. 5.4, we evaluate the effectiveness of representing the single-view shape-from-shading results by probability densities by comparing our results with those by our method "w/o PDF" (without probability density function) whose cost volume filtering network exploits the expected surface normal orientations of the estimated probability densities instead of the probability densities themselves. In this evaluation, we used the ground truth reflectance maps as inputs. Table 5.4 shows quantitative results. While the use of probability densities improves the estimation accuracy, the improvement is subtle. This is because the probability densities estimated for each view is already well-defined and, in many cases, the expected orientations are sufficient for representing the density distributions.

Accuracy for each shape and each material In Tab. 5.5, we report the quantitative estimation errors for each shape and each material. In Tab. 5.5c, we report the log-space root-mean-square errors of the estimated BRDFs. From the results, we can notice that the "Beige-Fabric" and the "White-Marble" materials are relatively challenging to our method. The "Beige-Fabric" material is challenging as, due to its diffuse reflectance, there are few radiometric cues in the appearance. Although

	Depth Error (↓)		Normal Error (↓)	
	Ours	w/o PDF	Ours	w/o PDF
Sphere	0.29%	0.29%	1.9 deg	1.8 deg
Teapot	0.38%	0.38%	4.3 deg	4.3 deg
Golfball	0.21%	0.20%	5.1 deg	5.2 deg
Planck	0.25%	0.26%	4.9 deg	5.0 deg
Bunny	0.36%	0.38%	6.7 deg	6.9 deg
Armadillo	0.55%	0.61%	14.6 deg	15.4 deg
Average	**0.34%**	0.35%	**6.3 deg**	6.4 deg

Table 5.4: Ablation study on the probabilistic surface normal representation. Please see the text for discussions.

we can observe specular highlights in the appearance of objects with the "White-Marble" material, as the color of its specular highlights and diffuse shadings are both white, it is difficult to correctly distinguish the two kinds of the radiometric cues. From Tab. 5.5, we can also notice that the "Golfball" object is also challenging to our method. The shape is challenging as its local geometry is repetitive and we cannot easily detect correct correspondences.

The number of input views We also evaluate how the number of input views affects the estimation accuracy. For 6 objects from the nLMVS-Synth dataset, we changed the number of input views from 3 to 9 and evaluated the depth, surface normal, and reflectance estimation accuracy. Table 5.6 shows quantitative results. Our results are plausible even with three-view images.

5.6.3 Whole 3D Shape Recovery

We then evaluated the accuracy of the whole object 3D shape recovery method on ten-view images from the nLMVS-Synth dataset. For quantitative evaluation of the recovered mesh models, we computed root-mean-square of the distances from points on the reconstructed surface to the nearest points on the ground truth mesh. Table 5.7 and Fig. 5.12 show quantitative and qualitative results. Even though we recover the whole object 3D shape by simply applying the Poisson surface reconstruction [29], our results are quantitatively comparable to the state-of-the-art methods. Note also that, as we can see in Fig. 5.12, our results for objects with complex, non-Lambertian materials have fewer artifacts.

5.6.4 Evaluation on Real Data

We also evaluated our method on real-world images in the nLMVS-Real dataset. Figure 5.13 shows recovered depths, surface normals, and whole object 3D shapes. The results are consistent with the ground truth. Figure 5.14 shows recovered BRDFs for each shape and material under the "Court" environment. Although, the ground truth BRDFs cannot be easily measured, the results are consistent across different shapes, which demonstrates the accuracy of our method.

	Fabric	Rubber	Gold	Silver	Marble	Hematite	Average
Sphere	1.45	0.60	0.52	0.30	0.64	0.36	0.64
Teapot	1.15	1.03	0.52	0.56	1.23	0.47	0.83
Golfball	1.36	0.87	3.25	2.34	2.21	0.86	1.82
Planck	1.81	1.09	0.55	0.46	0.94	0.47	0.89
Bunny	1.15	0.94	0.59	0.45	0.82	0.43	0.73
Armadillo	1.02	0.88	0.49	0.56	0.88	0.40	0.71
Average	1.32	0.90	0.99	0.78	1.12	0.50	0.94

(a) Depth Errors (%) (↓)

Sphere	4.6	3.1	2.3	2.4	3.3	2.6	3.0
Teapot	8.4	8.0	5.2	5.6	8.2	5.1	6.7
Golfball	9.9	8.4	19.2	13.7	14.0	9.5	12.5
Planck	14.6	11.7	7.6	7.0	10.5	7.5	9.8
Bunny	15.8	14.1	10.7	8.6	12.9	8.7	11.8
Armadillo	17.7	16.3	12.7	13.1	16.4	12.0	14.7
Average	11.8	10.3	9.6	8.4	10.9	7.6	9.8

(b) Normal Errors (degree) (↓)

Sphere	4.09	2.43	1.61	0.25	0.11	0.31	1.47
Teapot	4.66	2.66	1.95	0.29	0.17	0.39	1.69
Golfball	4.51	2.79	3.75	0.99	0.17	0.45	2.11
Planck	4.32	2.63	2.05	0.37	0.13	0.47	1.66
Bunny	4.31	3.07	3.36	0.56	0.19	0.51	2.00
Armadillo	4.72	2.96	2.86	0.50	0.20	0.62	1.98
Average	4.43	2.76	2.60	0.49	0.16	0.46	1.82

(c) Reflectance Errors (log RMSE) (↓)

Table 5.5: Estimation accuracy for each shape and each material. Please see the text for discussions.

Table 5.8 shows mean depth and surface normal estimation errors for each shape and each material in the nLMVS-Real dataset. The mean depth error for the overall dataset is 2.52% and the mean surface normal error is 15 degrees. Although the results are relatively poor compared to the results on the nLMVS-Synth dataset (Tab. 5.5), they are still plausible even though the real data do not strictly satisfy the assumptions (*e.g.*, the captured illumination map can be inconsistent with the images due to dynamic changes of the illumination environment). This demonstrates the robustness of our method.

Comparison with Oxholm and Nishino [50] For a direct comparison with a multi-view method by Oxholm and Nishino [50], we run our method on their dataset [49] and compared our results with theirs. Figure 5.15 shows the recovered shapes of our method and Oxholm and Nishino [50] (ON). Our results have fewer artifacts and capture details of the object (*e.g.*, the ear of the horse). Figure 5.16 shows recovered BRDFs, the results are consistent across two different illumination environments.

Estimation from real-world five-view images In Fig. 5.17, we show our geometry estimation results on five-view real-world images from the nLMVS-Real dataset. In the figure, we also show the results by IDR [73]. Our method successfully recovers the surface geometry even from the sparse sets of real-world images.

Free Viewpoint Relighting As shown in Fig. 5.18, once we recover the whole object 3D shape and the surface reflectance (BRDF), we can exploit them to synthesize images from arbitrary viewpoints and under arbitrary illumination environments.

5.7 Summary

In this chapter, we introduced nLMVS-Net, a novel deep multi-view stereo method for textureless, non-Lambertian surfaces that recovers surface geometry from posed multi-view images of a textureless, non-Lambertian object whose surface reflectance and surrounding illumination are known. The shape-from-shading sub-network effectively filters the complex observation likelihoods of surface normals computed from reflectance maps in a coarse-to-fine manner. The refined surface normal probability densities serve as tractable view-independent cfeatures for learning-based cost volume filtering. By integrating with a neural reflectance estimation, our nLMVS-Net successfully handles both synthetic and real-world objects with unknown materials. We believe our nLMVS-Net establishes a basis towards passive non-Lambertian 3D reconstruction in the wild.

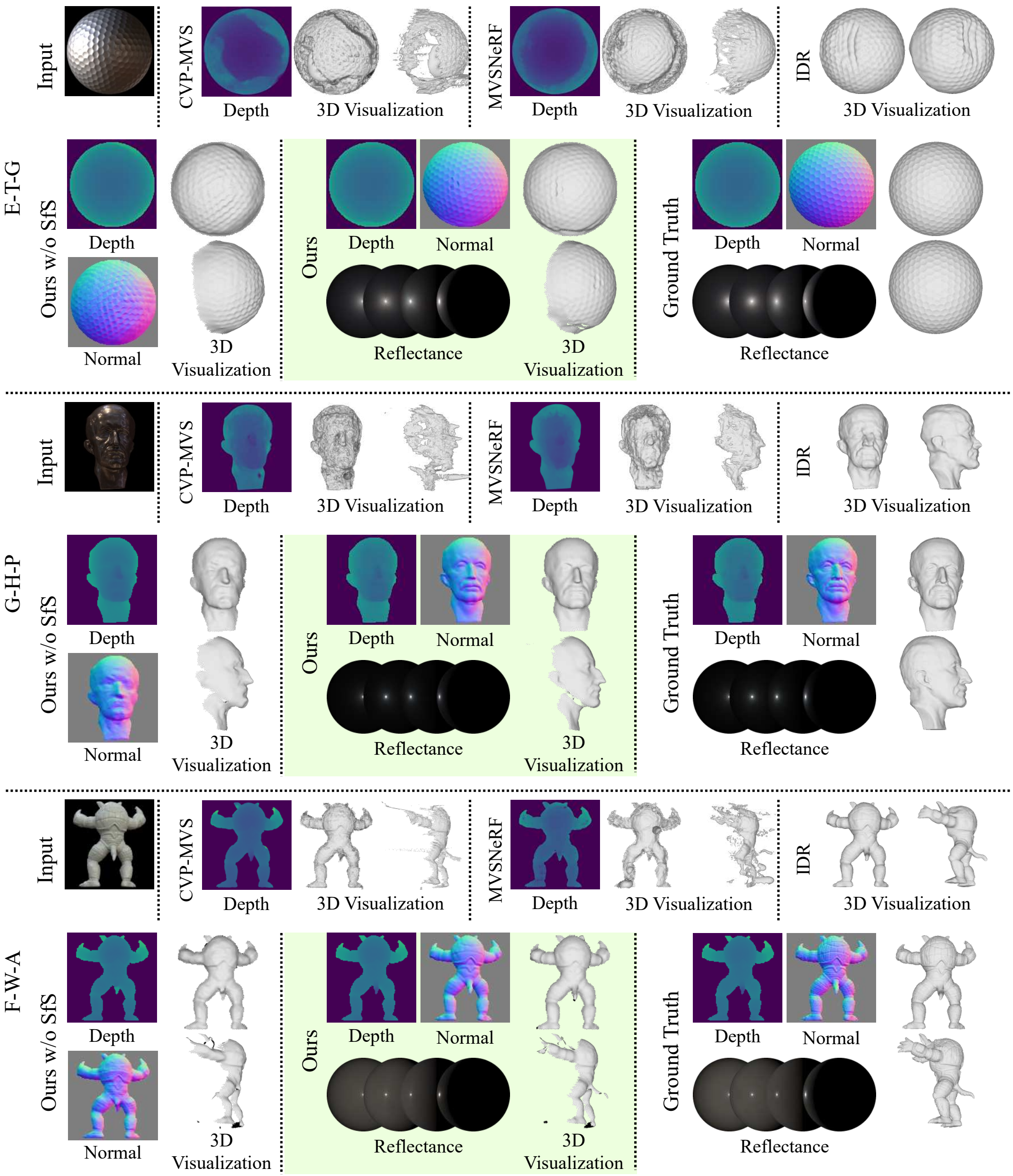


Figure 5.7: Recovered depths, surface normals, and reflectance from five-view images in the nLMVS-Synth dataset. The initials on the left correspond to names of shapes, materials, and illumination environments (Please see Fig. 4.2 for the names). © 2023 IEEE [67]

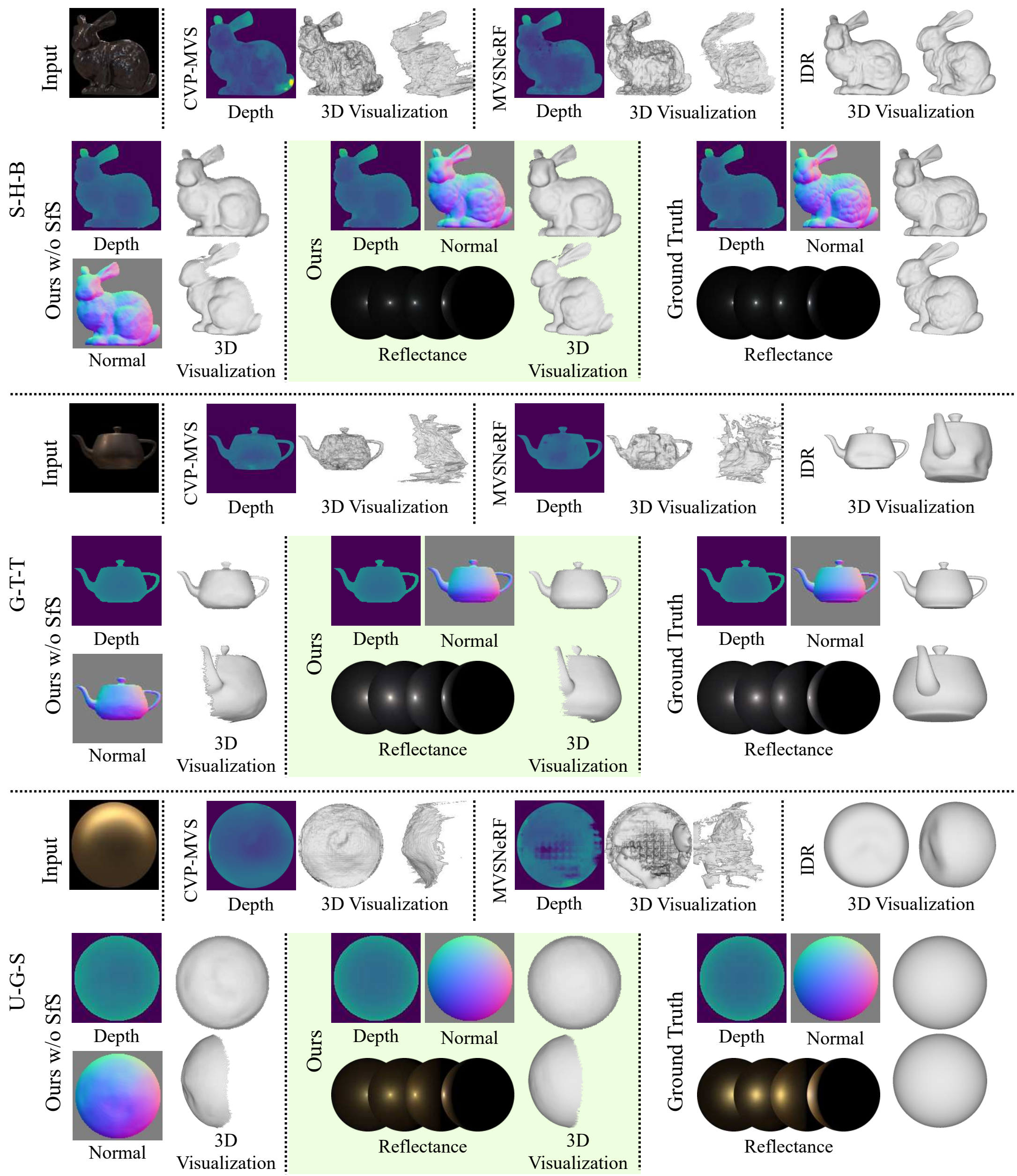


Figure 5.8: Recovered depths, surface normals, and reflectance from five-view images in the nLMVS-Synth dataset. The initials on the left correspond to names of shapes, materials, and illumination environments (Please see Fig. 4.2 for the names). © 2023 IEEE [67]

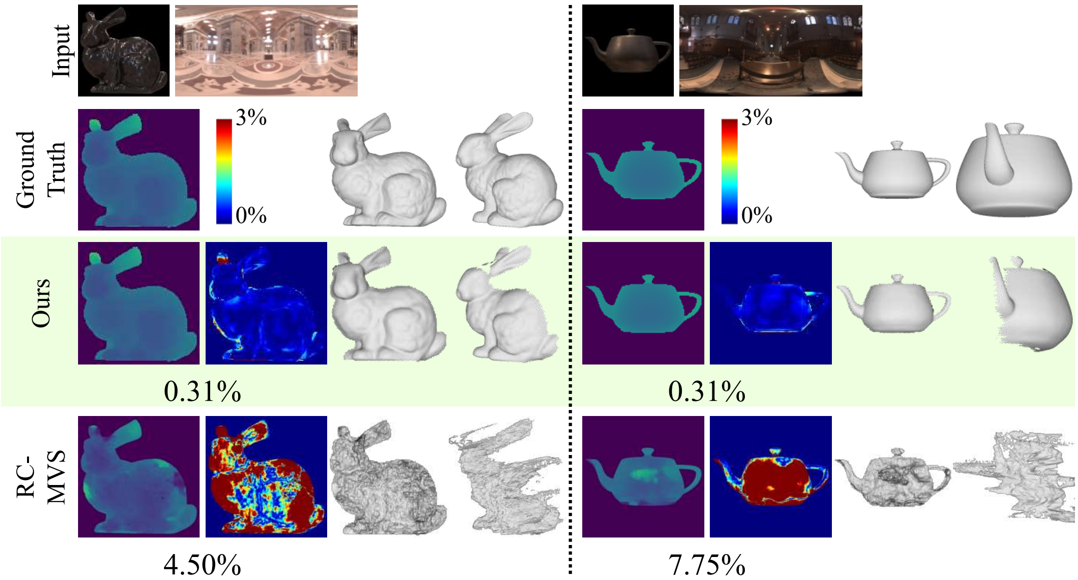


Figure 5.9: Comparison with Chang *et al*. [8] (RC-MVSNet). The color maps show the estimation errors. The numbers are mean estimation errors (lower is better). © 2023 IEEE [67]

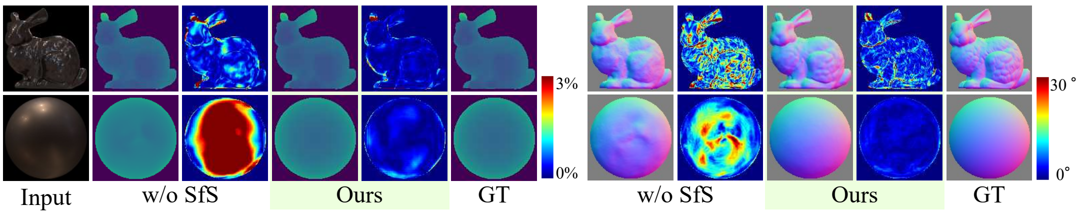


Figure 5.10: Visualization of estimation errors of our method and ours without the shape-from-shading. The color maps show the estimation errors. © 2023 IEEE [67]

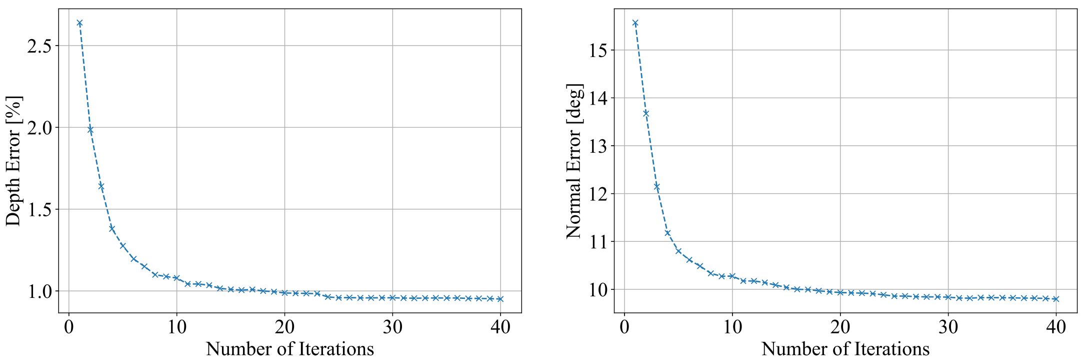


Figure 5.11: Mean depth and surface normal estimation errors for each iteration in the joint shape and reflectance estimation.

	# of Views			
	3	5	7	9
Ennis-Silver-Golfball	0.54	0.47	0.46	0.48
Grace-Hematite-Planck	0.44	0.37	0.33	0.32
Forest-Marble-Armadillow	0.31	0.28	0.25	0.26
St.Peters-Hematite-Bunny	0.37	0.31	0.32	0.31
Grace-Silver-Teapot	0.41	0.31	0.40	0.37
Uffizi-Gold-Sphere	0.95	0.59	0.67	0.93
Average	0.50	0.39	0.41	0.44

(a) Depth Errors (%) (↓)

	# of Views			
	3	5	7	9
Ennis-Silver-Golfball	6.7	6.7	6.8	6.8
Grace-Hematite-Planck	6.2	5.9	5.7	5.7
Forest-Marble-Armadillo	10.4	10.2	10.1	10.3
St.Peters-Hematite-Bunny	7.3	6.9	7.3	7.5
Grace-Silver-Teapot	4.9	4.2	4.1	4.2
Uffizi-Gold-Sphere	2.7	2.3	2.3	2.4
Average	6.4	6.0	6.1	6.2

(b) Normal Errors (degree) (↓)

	# of Views			
	3	5	7	9
Ennis-Silver-Golfball	0.19	0.15	0.21	0.14
Grace-Hamatite-Planck	0.42	0.43	0.45	0.44
Forest-Marble-Armadillo	0.10	0.10	0.10	0.10
St.Peters-Hematite-Bunny	0.47	0.44	0.46	0.40
Grace-Silver-Teapot	0.29	0.26	0.25	0.27
Uffizi-Gold-Sphere	1.77	1.72	1.73	1.77
Average	0.54	0.52	0.53	0.52

(c) Reflectance Errors (log RMSE) (↓)

Table 5.6: Relationship between the number of input views and estimation errors. Please see the text for discussions.

	Mesh (↓)
NeRS [75]	0.63 %
PhySG [76]	0.61 %
IDR [73]	**0.25 %**
Ours + PSR [29]	<u>0.38</u> %

Table 5.7: Accuracy of the whole 3D shape recovery on ten-view images from the nLMVS-Synth dataset. Please see the text for discussions.

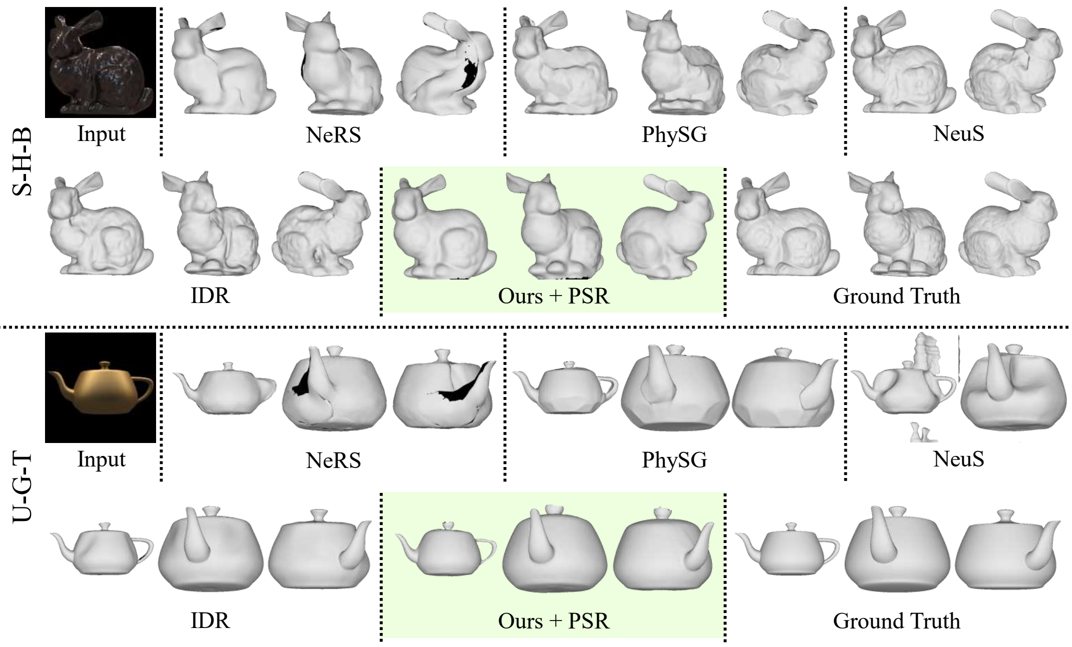


Figure 5.12: Recovered mesh models from 10 view images of the nLMVS-Synth dataset. © 2023 IEEE [67]

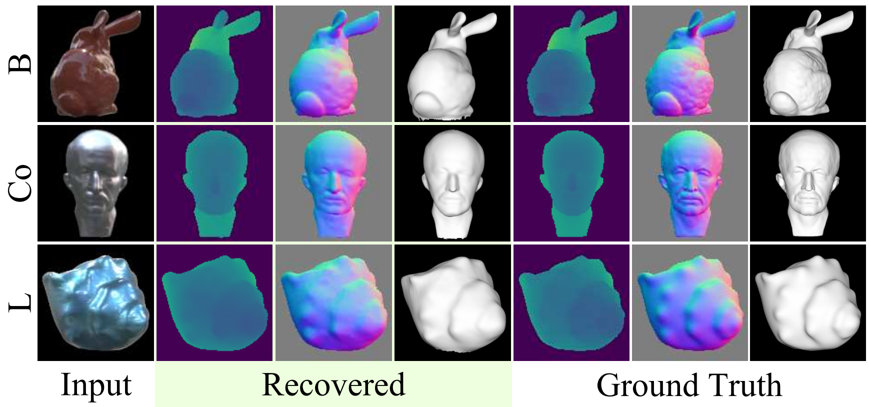


Figure 5.13: Depths, surface normals, and whole 3D shapes recovered from images and illumination maps in the nLMVS-Real dataset. © 2023 IEEE [67]

		Reflectances				
		Blue-Metallic	Bright-Red	Duralumin	White-Primer	Average
Shapes	Spere	1.67	2.88	4.20	5.65	3.60
	Bunny	1.27	1.74	2.86	4.27	2.53
	Horse	2.13	1.51	3.37	3.14	2.54
	Planck	1.55	1.33	1.80	2.81	1.87
	Shell	1.76	1.94	1.78	2.69	2.05
	Average	1.68	1.88	2.80	3.71	2.52

(a) Depth Errors (%) (↓)

		Reflectances				
		Blue-Metallic	Bright-Red	Duralumin	White-Primer	Average
Shapes	Spere	7.92	9.61	13.52	18.70	12.44
	Bunny	10.61	10.61	15.40	20.93	14.39
	Horse	14.24	10.53	18.59	19.24	15.65
	Planck	14.20	11.77	13.41	19.25	14.66
	Shell	17.82	17.68	18.31	22.40	19.05
	Average	12.96	12.04	15.85	20.10	15.24

(b) Normal Errors (degree) (↓)

Table 5.8: Depth and surface normal estimation accuracy on the nLMVS-Real dataset.

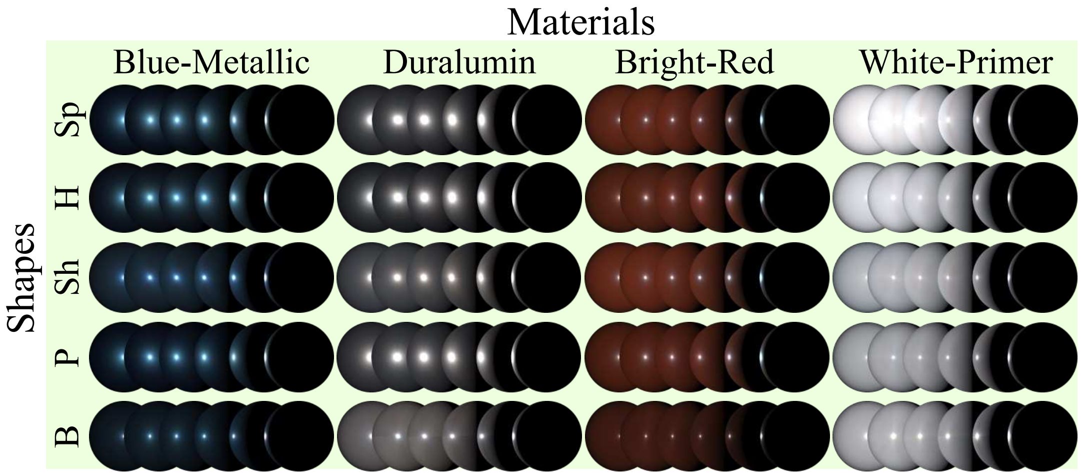


Figure 5.14: Recovered BRDFs on the nLMVS-Real dataset. The results are qualitatively consistent across different shapes.

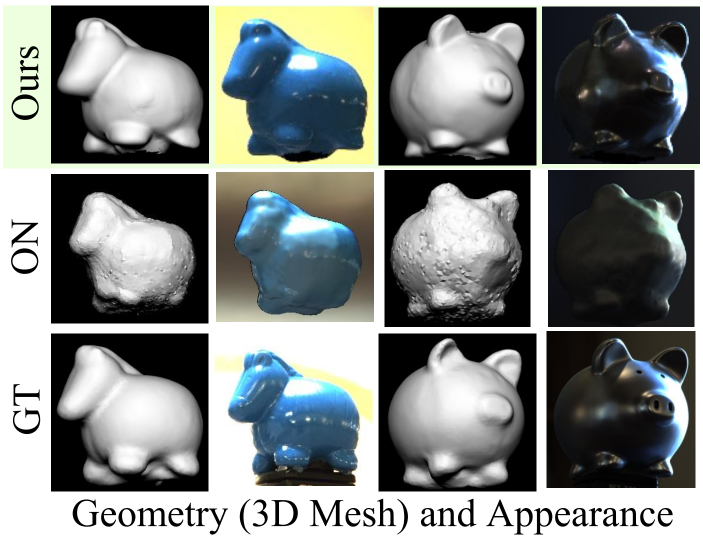


Figure 5.15: Comparison with Oxholm and Nishino [50] (ON) on the Multiview Objects Under Natural Illumination Database [49]. Our results have fewer artifacts and capture details of the object (*e.g.*, ear of the horse). © 2023 IEEE [67]

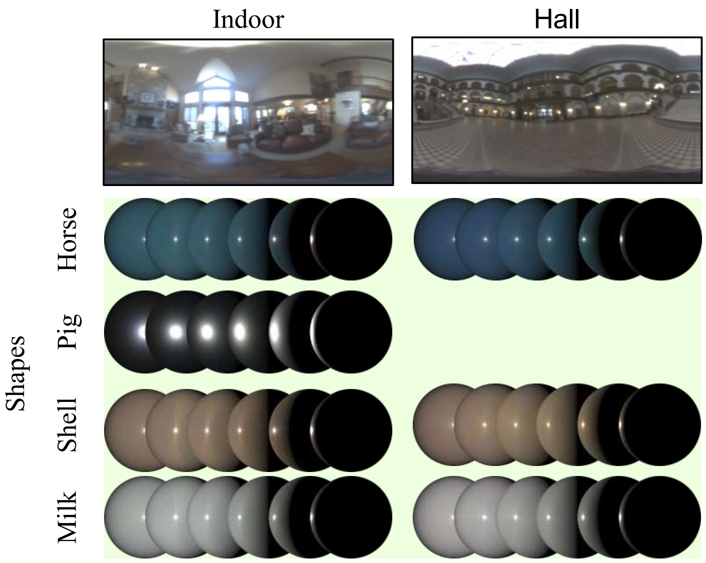


Figure 5.16: BRDFs recovered from images in the Multiview Objects Under Natural Illumination Database [49]. © 2023 IEEE [67]

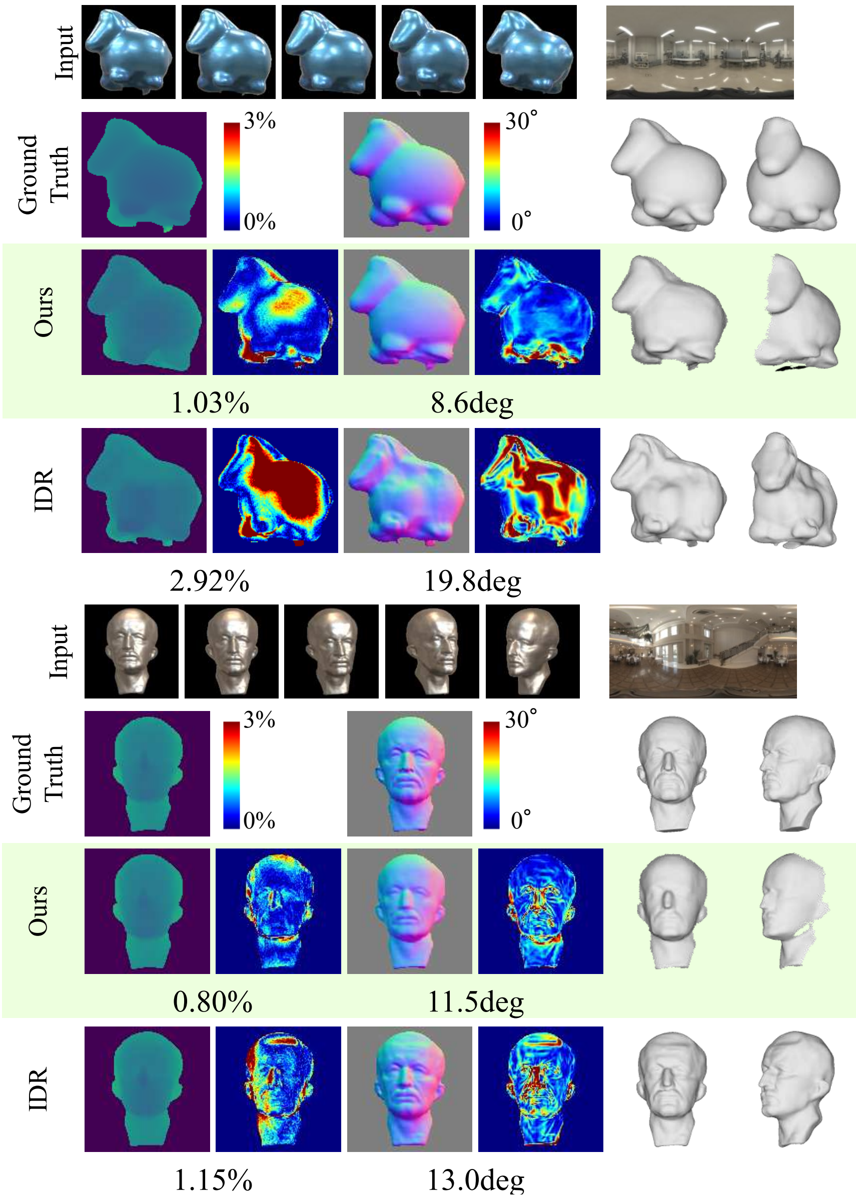


Figure 5.17: Geometry recovered from five-view imaegs in the nLMVS-Real dataset. The numbers are mean estimation errors (lower is better). © 2023 IEEE [67]

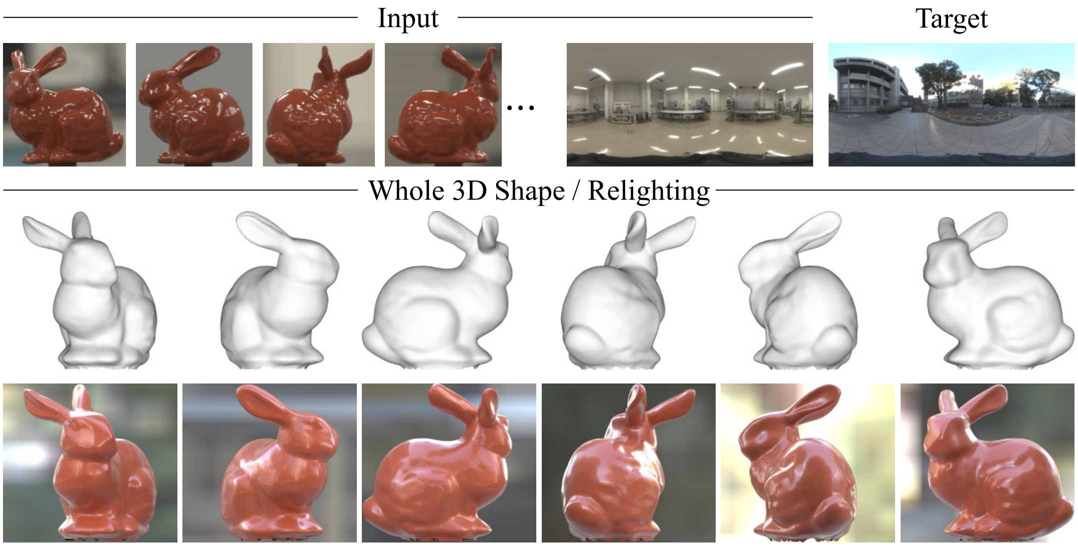


Figure 5.18: Using the recovered whole 3D object shape and surface reflectance, we can synthesize images of the object from arbitrary viewpoints and under arbitrary illumination environments.

Chapter 6

DeepShaRM: Multi-View Shape and Reflectance Map Recovery

In the last chapter, we introduced nLMVS-Net that realizes geometry reconstruction from posed multi-view images taken under known but complex natural illumination. nLMVS-Net showed that the learned per-view filtering of radiometric likelihoods significantly reduces the ambiguity of the single-view estimation and enabled plausible geometry reconstruction of textureless, non-Lambertian objects if we could compute reflectance maps from the input illumination map and a reflectance estimate. Non-Lambertian 3D reconstruction under unknown natural illumination is, however, still non-trivial even with the nLMVS-Net.

A few existing methods have tried to solve this problem as full inverse rendering, *i.e.*, joint estimation of surface geometry, surface reflectance, and surrounding illumination based on the rendering equation (Eq. (2.1)). It is, however, extremely challenging due to the fundamental ambiguity between the three unknown physical quantities. Even with known geometry, joint estimation of reflectance and illumination is still a challenging problem [39, 11]. There is the fundamental ambiguity between the two unknowns in color and frequency. It is difficult to distinguish appearance of a smooth white shiny surface under red blurry illumination from one of a red rough surface under a white sharp light.

Our key observation here is that, for geometry reconstruction, we do not need to jointly estimate surface reflectance and surrounding illumination, *i.e.*, do not need to deal with the ambiguity between them, as long as we can recover camera-view reflectance maps. Can we leverage this for geometry reconstruction from only posed multi-view images taken under unknown natural lighting?

As shown in Fig. 6.1, in this chapter, we introduce DeepShaRM, an estimation framework for geometry estimation from posed multi-view images captured under unknown natural illumination. Our key idea is bypassing solving the challenging joint estimation of reflectance and illumination but fully leveraging learned realistic prior on the camera-view reflectance maps for joint estimation of geometry and reflectance maps. As we do not decompose the camera-view reflectance maps, DeepShaRM does not suffer from difficulties caused by the fundamental ambiguity between reflectance and illumination. Although reflectance maps are entangled representations, we can still leverage realistic prior on reflectance maps for resolving the ambiguity between surface geometry and reflectance maps. That is, as the reflectance maps are approximated convolutions of reflectance and illumination (Sec. 2.1), there is a statistical bias of the frequency characteristics of reflectance maps that we can learn using a deep neural network.

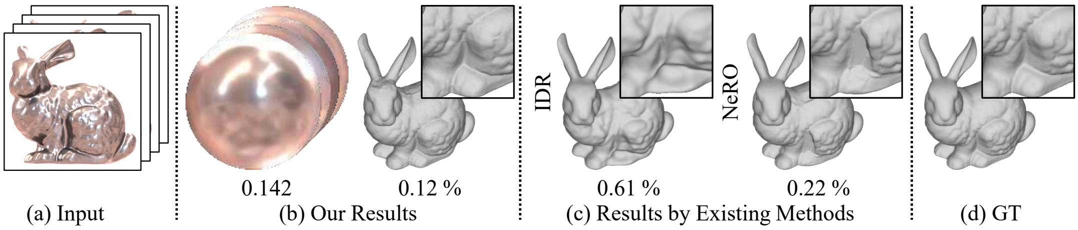


Figure 6.1: In this chapter, we introduce a method that can recover geometry and camera-view reflectance maps of a textureless, non-Lambertian object (b) from posed multi-view images captured under unknown natural illumination (a). While existing methods struggle to handle the reflective object (c), our method recovers plausible surface geometry consistent with the ground truth (d). The numbers are the root-mean-square errors for the 3D shapes and the log-scale mean absolute error for the reflectance maps. Each inset figure shows details around the leg.

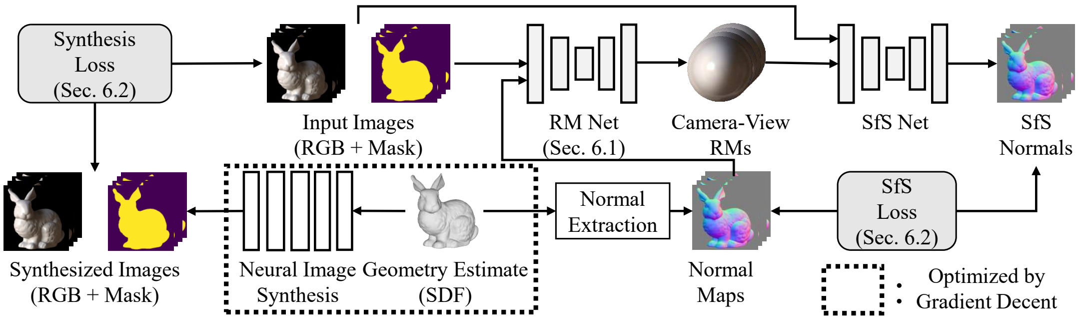


Figure 6.2: An overview of DeepShaRM. Given a initial coarse geometry recovered from silhouettes of the object, we alternate between estimating reflectance maps from the geometry estimate and updating the geometry estimate using the reflectance maps. We realize each of the reflectance map estimation and the geometry reconstruction with a deep neural network so that we can leverage learned realistic prior on geometry and reflectance maps for the joint estimation. Inspired by recent methods [73, 64], we also integrate our estimation framework with a geometry optimization with a neural image synthesis method. © 2024 IEEE [68]

Figure 6.2 shows an overview of DeepShaRM, we formulate this joint estimation of geometry and camera-view reflectance maps as alternating estimation with two deep neural networks. Given an initial coarse geometry recovered from silhouettes of the object (a visual hull) as an initial estimate, for each view, we first recover a camera-view reflectance map using an input images and a normal map rendered from the current geometry estimate. We achieve this with a deep reflectance map estimation network. Once reflectance maps are recovered, we exploit them as inputs to a deep shape-from-shading network. We update the geometry estimate using surface normals estimated by the shape-from-shading network and alternate these until convergence. Inspired by recent methods [73, 64], we also integrate our method with geometry optimization with a neural image synthesis method.

6.1 Reflectance Map Estimation Network

Given images and an initial coarse estimate of surface geometry, the first step of our method is estimating a camera-view reflectance map for each view. The reflectance map recovery is challenging as the observations are sparse. Even if we map pixel values using ground truth surface normals, we can recover only a sparse reflectance map. Therefore, strong realistic prior learned by the reflectance map estimation network is essential for not only the joint estimation but also interpolation of the sparse observations.

Figure 6.3 depicts the architecture of the reflectance map estimation network (RM network). Inspired by Georgoulis *et al.* [18], we explicitly map pixels in the input image onto the spherical domain using the input surface normals and then filter them using a 2D convolutional neural network. A naive mapping is, however, insufficient for handling objects with complex geometry. Some of the input pixels are inconsistent with the (ground-truth) reflectance map due to global illumination effects (shadows and interreflections) and sharp changes of surface normals within a single pixel. Such deviations from reflectance maps affects the estimation results. For this, here we introduce two novel ideas. First, we introduce an image feature extraction sub-network (Feature Net) before the mapping so that the network can learn to implicitly refine pixels that deviate from the reflectance map. Second, we introduce confidence score estimation sub-networks that learn to filter such inconsistent pixels. At first, we estimate pixel-wise confidence scores of pixel-wise observations from the inputs (Initial Confidence Net). We use the estimated scores to weight pixel-wise features when mapping them onto the spherical domain. Specifically, we derive a differentiable mapping from an image feature map $I'_{\mathbf{m}}$ to a reflectance map feature $R'(\mathbf{n})$ with the input surface normal $\mathbf{n}_{\mathbf{m}}$ and the estimated score $w_{\mathbf{m}}$

$$R'(\mathbf{n}) = \frac{\sum_{\mathbf{m}} w'_{\mathbf{m}} I'_{\mathbf{m}}}{\sum_{\mathbf{m}} w'_{\mathbf{m}}}\,, \tag{6.1}$$

where

$$w'_{\mathbf{m}} \equiv w_{\mathbf{m}} \max(\mathbf{n}_{\mathbf{m}} \cdot \boldsymbol{\omega}_{\mathbf{o}}, 0) \exp(s\mathbf{n}_{\mathbf{m}} \cdot \mathbf{n}')\,. \tag{6.2}$$

In addition to the estimated weight $w_{\mathbf{m}}$, we use weights determined by distances between the input normal $\mathbf{n}_{\mathbf{m}}$, the query normal $\mathbf{n}'$, and the viewing direction $\boldsymbol{\omega}_{\mathbf{o}}$. In training, we optimize the parameter s along with weights of the network. In practice, we use the angular fisheye projection to convert reflectance maps into 2D images. We compute Eq. (6.1) for each pixel in the projected reflectance map. A reflectance map estimate is decoded from the reflectance map feature using another 2D convolutional sub-network (RM decoder). Once we obtain the estimate of the camera-view reflectance map, we can detect deviations of pixels from the reflectance map estimate by comparing the input image and one rendered from the reflectance and the normal maps. We leverage this by alternating between estimating confidence scores and recovering the reflectance map. In the subsequent iteration of the alternating estimation, we use another confidence score estimation network (Confidence Net) that takes in not only the inputs but also an image rendered from the current reflectance map estimate and the input normal map.

Using the nLMVS-Synth dataset, we train the reflectance map estimation network in a supervised manner. We feed synthetic images and corresponding (ground-truth) normal maps to the network and obtain reflectance map estimates. We optimize weights of the network so that discrepancies between the reflectance map estimates and corresponding ground truth are minimized. Specifically, we impose a log-space

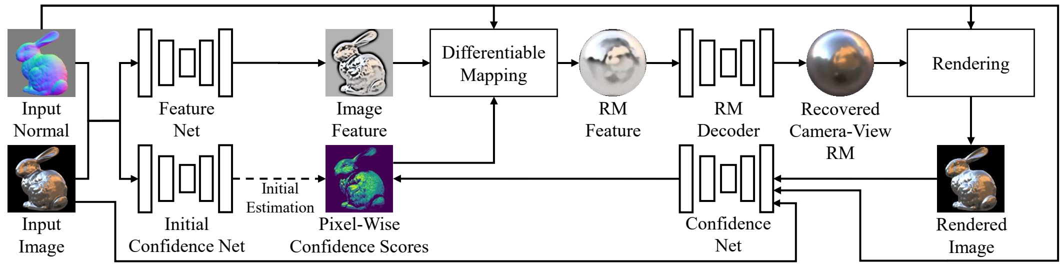


Figure 6.3: Architecture of our reflectance map estimation network. Inspured by Georgoulis *et al.* [18], we explicitly map pixels in the input image onto the spherical domain using the input surface normals. To deal with deviations of pixels from the reflectance map, we introduce two novel modules, image feature filtering before the mapping and pixel-wise confidence score estimation. © 2024 IEEE [68]

L1 loss

$$L_{\mathrm{L1}} = \sum_{\mathbf{n} \in N} \left\| \log \hat{E}(\mathbf{n}) - \log E(\mathbf{n}) \right\|_1 , \tag{6.3}$$

a log-space gradient loss

$$L_{\mathrm{G}} = \sum_{\mathbf{n} \in N} \left\| g\left(\log \hat{E}(\mathbf{n})\right) - g\left(\log E(\mathbf{n})\right) \right\|_1 , \tag{6.4}$$

and a perceptual loss

$$L_{\mathrm{P}} = \sum_{\mathbf{n} \in N} \left\| f\left(\log \hat{E}(\mathbf{n})\right) - f\left(\log E(\mathbf{n})\right) \right\|_1 , \tag{6.5}$$

where $\hat{E}(\mathbf{n})$ is the reflectance map estimate, $E(\mathbf{n})$ is the ground truth, and N is a set of surface normals that correspond to pixels in the reflectance maps. $g(\cdot)$ is an operation that computes gradients of the input image with respect to horizontal and vertical directions. $f(\cdot)$ is a feature extraction layer of a pretrained VGG16 model. Note that we can apply the gradient computation and the VGG16 feature extraction layer to the reflectance maps as we convert them into 2D images using the angular fisheye projection. We also compute an image reconstruction loss

$$L_{\mathrm{I}} = \sum_{\mathbf{m}} \alpha_{\mathbf{m}} \left\| \log \hat{E}(\mathbf{n_m}) - \log I_{\mathbf{m}} \right\|_1 , \tag{6.6}$$

where $I_{\mathbf{m}}$ and $\mathbf{n_m}$ are the input pixel value and the input surface normal of a pixel $\mathbf{m}$. We compute a pixel-wise weight $\alpha_{\mathbf{m}}$

$$\alpha_{\mathbf{m}} = \exp(-10 \left\| \log E(\mathbf{n_m}) - \log I_{\mathbf{m}} \right\|) , \tag{6.7}$$

so that we can impose this loss only on pixels consistent with the ground truth reflectance map $E(\mathbf{n})$. We use the weighted sum of these loss functions

$$L_{\mathrm{RM}} = w_{\mathrm{L1}} L_{\mathrm{L1}} + w_{\mathrm{G}} L_{\mathrm{G}} + w_{\mathrm{P}} L_{\mathrm{P}} + w_{\mathrm{I}} L_{\mathrm{I}} , \tag{6.8}$$

as the training loss for the RM network. In practice, we set the weights w_{L1}, w_{G}, w_{P},

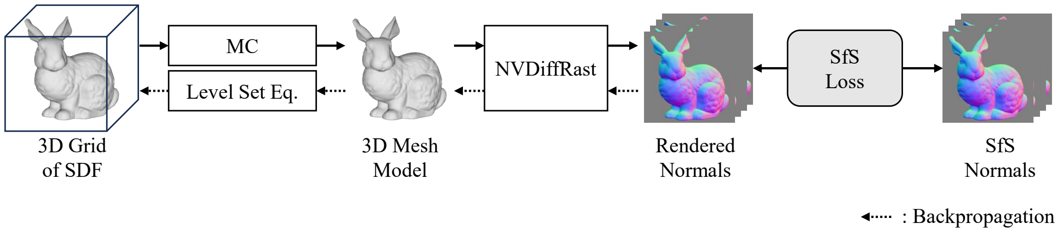


Figure 6.4: Inspired by Neural Implicit Evolution [44], given surface normals estimated by the shape-from-shading network, we update a unified geometry estimate (a 3D grid of a signed distance function (SDF)) using a level set method and NVDiffRast [35], a differentiable renderer for 3D mesh models. © 2024 IEEE [68]

and w_{I} to 1, 0.1, 0.001, and 1, respectively. As the RM network outputs a reflectance map estimate for each iteration of the alternating estimation, we compute this loss for the output of each iteration.

6.2 Joint Shape and Reflectance Map Estimation

Once we obtain estimates of camera-view reflectance maps, we can used them to update the geometry estimate. We can obtain pixel-wise surface normal estimates for each view by feeding a reflectance map estimate and an input images into a deep shape-from-shading network similar to one in the nLMVS-Net.

The shape-from-shading network, however, can recover only surface normals (*i.e.*, absolute depths cannot be recovered from a single-view estimate) which can be inconsistent across views. Although the cost volume filtering network inside the nLMVS-Net can aggregate information from one view and its neighboring views to recover surface depths, its estimates are still view-dependent and can be inconsistent across views. For this, we derive a geometry optimization framework that can update a unified surface geometry estimate using the per-view surface normal estimates. We represent the surface geometry by a 3D grid of a signed distance function (SDF) so that we can handle any topological change of the surface geometry during the optimization.

Figure 6.4 depicts how we update the 3D grid of a signed distance function (SDF) using the surface normals estimated by the shape-from-shading network (SfS normals). Inspired by Neural Implicit Evolution [44], we realize this with a level set method and NVDIffRast [35], a differentiable renderer for 3D mesh models. We first extract a 3D mesh model from the SDF grid using Marching Cubes [41]. We then render a normal map from the extracted mesh for each input view and compute mean absolute error between the rendered surface normal $\bar{\mathbf{n}}_{\mathbf{m}}$ and the SfS normal $\hat{\mathbf{n}}_{\mathbf{m}}$

$$L_{\mathrm{SfS}} = \arccos\left(\hat{\mathbf{n}}_{\mathbf{m}} \cdot \bar{\mathbf{n}}_{\mathbf{m}}\right) , \tag{6.9}$$

for each pixel $\mathbf{m}$. We refer to this as SfS loss. We compute gradients of the SfS loss with respect to vertex positions of the extracted mesh using NVDiffRast. We then compute gradients with respect to parameters of the SDF using the level set equation and use them to update the SDF parameters. Specifically, given the gradients of the loss function $\frac{\partial L}{\partial \mathbf{v}_j}$ with respect to vertex positions $\mathbf{v}_j$, we compute the gradients with respect to the SDF values $f_\theta(\mathbf{v_j})$

$$\frac{\partial L}{\partial f_\theta(\mathbf{v}_j)} = -\nabla f_\theta(\mathbf{v}_j) \cdot \frac{\partial L}{\partial \mathbf{v}_j}, \tag{6.10}$$

and then compute the gradients with respect to parameters of the SDF θ (values in the SDF grid)

$$\frac{\partial L}{\partial \theta} = \sum_j \frac{\partial f_\theta(\mathbf{v}_j)}{\partial \theta} \frac{\partial L}{\partial f_\theta(\mathbf{v}_j)}, \tag{6.11}$$

where j is index of the vertex. Note that we compute the SDF value $f_\theta(\mathbf{v}_j)$ from the parameter θ (the SDF grid) using cubic B-spline interpolation so that the SDF values used for the Marching cubes become smooth [62]. Note also that we compute the gradient $\frac{\partial f_\theta(\mathbf{v}_j)}{\partial \theta}$ by automatic differentiation. We feed the obtained gradient $\frac{\partial L}{\partial \theta}$ to the Adam optimizer to achieve the SDF optimization.

Image Synthesis Loss Using the ground truth segmentation masks (silhouette images), for each pixel $\mathbf{m}$, we also impose a mask loss

$$L_{\text{Mask}} = \|\hat{s}_{\mathbf{m}} - s_{\mathbf{m}}\|^2, \tag{6.12}$$

where $s_{\mathbf{m}}$ is a pixel value in the ground truth silhouette and $\hat{s}_{\mathbf{m}}$ is one in a mask rendered from the current geometry estimate. Note that, thanks to NVDiffRast [35], we can compute its gradient with respect to the SDF parameters approximately.

Inspured by neural image synthesis methods [45, 73, 64], we also synthesize RGB images using a neural representation of the appearance and jointly optimize its weights along with the SDF grid. Specifically, we approximate the object appearance by an MLP $\mathbf{c}(\mathbf{x}, \mathbf{v}, \mathbf{n})$ that takes in a surface location $\mathbf{x}$, a viewing direction $\mathbf{v}$, and a surface normal $\mathbf{n}$ and outputs a surface radiance. For each pixel $\mathbf{m}$, we impose an image synthesis loss (NeRF loss)

$$L_{\text{NeRF}} = D(\log(\mathbf{c}(\bar{\mathbf{x}}_{\mathbf{m}}, \mathbf{v}_{\mathbf{m}}, \bar{\mathbf{n}}_{\mathbf{m}})), \log(I_{\mathbf{m}})), \tag{6.13}$$

where ($\bar{\mathbf{x}}_{\mathbf{m}}$ and $\bar{\mathbf{n}}_{\mathbf{m}}$ are a surface location and a surface normal computed from the SDF grid, respectively, $\mathbf{v}_{\mathbf{m}}$ is the viewing direction computed from the given camera pose, and $D(\cdot,\cdot)$ is the smooth L1 loss. Although our method can work without this loss, we empirically found that this loss improves reconstruction of surface details. Note also that, we experimentally show that, even with this image synthesis loss, the proposed SfS loss is still essential for plausible surface geometry reconstruction especially when the number of input views is extremely small (*e.g.*, 2). We also impose a Laplacian regularization on the SDF grid so that artifacts in unobserved regions can vanish.

6.3 Estimation without Segmentation Masks

Even if we do not now the segmentation masks (silhouettes) for the object, we can still recover them using a well-established salient object detection method [52]. Using the estimated mask, we can initialize the geometry estimate and determine which pixels we use for the reflectance map estimation. The estimated masks are, however, sometimes erroneous and we cannot completely rely on them in the geometry optimization. If we use them to impose the mask loss, estimation errors in the masks strongly affect the geometry reconstruction result. For this, we derive a method that can recover the surface geometry and the reflectance maps without using the mask loss. Inspired by recent methods [77, 64, 38], we achieve this by modeling the backgrounds in the images with another neural network [60]. We synthesize images with backgrounds by using this network along with the MLP for the object appearance and impose the image synthesis loss on the rendered images. We jointly optimize weights of the two neural appearance representations and the SDF grid. To impose the SfS loss on all surfaces visible from the input views, we mask the input images to the SfS network using silhouettes of the current geometry estimate, *i.e.*, we make sure that the SfS network estimates surface normals for all pixels that are insider the silhouettes of the current geometry estimate. The joint optimization of two neural appearance representation is, however, challenging due to the ambiguity between them. For this, we derive an RM loss that ensures the consistency between the object appearance learned by the color MLP $\mathbf{c}(\mathbf{x}, \mathbf{v}, \mathbf{n})$ and the reflectance map estimate $\hat{R}(\mathbf{n})$

$$L_{\mathrm{RM}} = D\left(\log\left(\mathbf{c}(\bar{\mathbf{x}}_{\mathbf{m}}, \mathbf{v}_{\mathbf{m}}, \bar{\mathbf{n}}_{\mathbf{m}})\right), \log\left(\hat{R}(\bar{\mathbf{n}}_{\mathbf{m}})\right)\right) . \tag{6.14}$$

We mildly impose this loss (*i.e.*, with a small weight) so that the color MLP can learn the object appearance that deviates from the reflectance map estimate (*e.g.*, shadows and interreflections).

Another challenge here is that surface normals estimated by the SfS network are completely wrong for pixels that are outside the object region. If we impose the SfS loss on all pixels with the same weight, the wrong surface normal estimates affect the geometry estimation result. For this, we weight the SfS loss

$$L'_{\mathrm{SfS}} = w_{\mathrm{RM}} L_{\mathrm{SfS}} , \tag{6.15}$$

according to weights determined by the consistency between the surface normal estimated by the SfS network $\hat{\mathbf{n}}_{\mathbf{m}}$, the reflectance map estimate $\hat{R}(\mathbf{n})$, and the observed radiance $I_{\mathbf{m}}$

$$w_{\mathrm{RM}} = \exp\left(-\left\|\log\left(\hat{R}(\bar{\mathbf{n}}_{\mathbf{m}}) - \log\left(I_{\mathbf{m}}\right)\right)\right\|_1\right) . \tag{6.16}$$

6.4 Experiments

We focus our experiments on answering the questions below. Can the RM network recover accurate reflectance maps? Does the joint estimation framework improve geometry reconstruction accuracy? How does our method work compared to inverse rendering [76, 21, 38] and neural image synthesis [73, 64] methods? How does our method generalize to real data? How does each component of our method contribute to the estimation accuracy? To answer these questions, we validate the following points.

- Estimation results of RM network are quantitatively and qualitatively more accurate compared to baseline methods

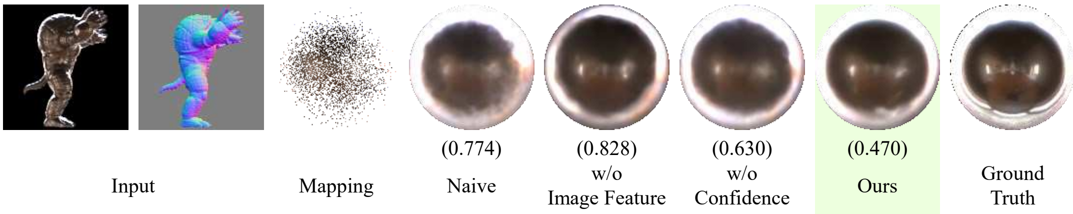


Figure 6.5: Example estimation results of the reflectance map estimation network. The numbers are log-scale mean absolute errors (lower is better). Please see the text for details. © 2024 IEEE [68]

	Log-MAE (↓)
Naive	0.162
w/o Image Feature	0.125
w/o Confidence	0.135
Ours	**0.121**

Table 6.1: Accuracy (Log-MAE) of recovered reflectance maps on the nLMVS-Synth dataset.

- Our joint estimation framework improves the accuracy of the recovered geometry.
- The recovered 3D shapes are more accurate than those of existing methods on both synthetic and real data.
- Each of the proposed components contribute to plausible geometry and reflectance map recovery.

Training We trained the RM and the SfS networks using synthetic images from the nLMVS-Synth dataset. We applied random rotation and zoom to the training images as data augmentation.

6.4.1 Accuracy of RM Network

We first evaluate the accuracy of the RM network using images and (ground truth) surface normal maps fron the test set of the nLMVS-Synth dataset. To evaluate the effectiveness of the proposed image-space feature extraction and the pixel-wise confidence estimation, we compared our method with its three variants, "w/o Confidence", "w/o Image Feature", and "Naive". "w/o Confidence" does not estimate pixel-wise confidence scores and uses a constant weight during the weighted mapping. "w/o Image Feature" does not exploit the image feature map and input pixel values are used as alternatives. "Naive" does not use either the image feature extraction or the confidence score estimation similar to Georgoulis *et al.* [18]. Figure 6.5 and Tab. 6.1 show qualitative and quantitative results. In Fig. 6.5, as a reference, we also show a sparse reflectance map obtained by simply mapping input pixels using the normal map. The results clearly show the effectiveness of the RM network.

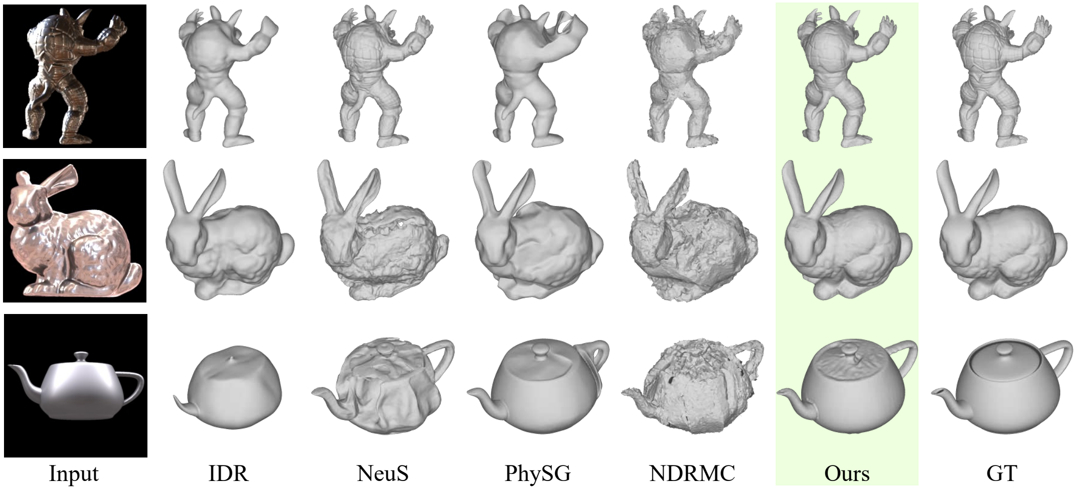


Figure 6.6: Object 3D shapes recovered from ten-view images (without backgrounds) in the nLMVS-Synth dataset. © 2024 IEEE [68]

6.4.2 Joint Shape and Reflectance Map Recovery

We evaluate the accuracy of the overall joint shape and reflectance map estimation method on the test set of the nLMVS-Synth dataset. We used ten-view images as inputs for each object. Here we use the ground truth masks to impose the mask loss so that we can evaluate the effectiveness of the core components of our method. We compared our method with its two variants, ours without the SfS loss ("w/o SfS loss") and ours without the image synthesis loss ("w/o NeRF"), to evaluate the effectiveness of each of the loss functions. For quantitative evaluation, we compute the root-mean-square (RMS) distance between the recovered and ground truth 3D shapes. We computed the RMS error using only surfaces visible from the input views. Figure 6.7 and Tab. 6.2b show qualitative and quantitative results. In Tab. 6.2b, we also show the accuracy of the initial geometry estimate (*i.e.*, visual hull). The results show that the proposed SfS loss is essential to recover the overall structure of the object especially when the surface is highly specular, while the image synthesis loss improves reconstruction of surface details. We also compared our method with recent inverse rendering [76, 21] and neural image synthesis [73, 64] methods that also take in masked images as inputs. Figure 6.6 and Tab. 6.2a show qualitative and quantitative results. Our method achieves the state-of-the-art accuracy of the nLMVS-Synth dataset.

Accuracy for Each Material Figure 6.8 shows geometry estimation accuracy for each material in the nLMVS-Synth dataset. DeepShaRM works well especially on reflective surfaces. This shows that DeepShaRM successfully leverage rich information of appearance caused by specular reflections. Note that, the results on less reflective surfaces including matte surfaces (ones with the "Fabric" material) are still plausible. This shows the robustness of DeepShaRM.

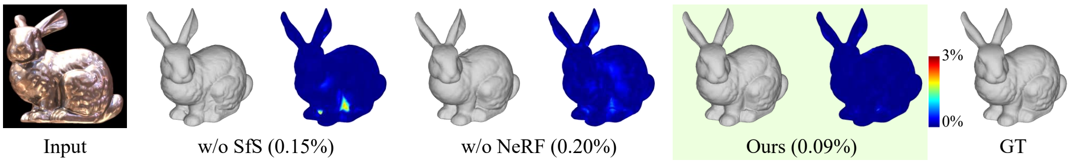


Figure 6.7: Effectiveness of each of the loss functions for the joint shape and reflectance map recovery. Color maps visualize distances from the recovered surface to the nearest point on the ground truth. © 2024 IEEE [68]

	RMS1 (↓)	RMS2 (↓)
IDR [73]	0.26 %	0.27 %
NeuS [64]	0.96 %	0.66 %
PhySG [76]	0.70 %	0.49 %
NDRMC [21]	0.64 %	0.44 %
Ours	**0.15 %**	**0.15 %**

(a) Comparison with existing methods.

	RMS1 (↓)	RMS2 (↓)
Initial	1.10 %	1.03 %
w/o SfS	0.16 %	0.15 %
w/o NeRF	0.21 %	0.20 %
Ours	**0.15 %**	**0.15 %**

(b) Ablation study.

Table 6.2: Geometry reconstruction accuracy (RMSE) on 10 view images in the nLMVS-Synth dataset. RMS1 is the root-mean-square error (RMSE) from the recovered surface to the nearest ground truth surface and RMS2 is RMSE from the ground truth surface to the nearest recovered surface.

Estimation from Very Sparse Inputs

We also test our method on very sparse sets of synthetic images. As shown in the first row of Fig. 6.9, we evaluated our method on five different settings: “5 views (outside-in)”, “4 views (outside-in)”, “5 views”, “3 views”, and “2 views”. In “5 views (outside-in)” and “4 views (outside-in)” settings, we sample views at equal intervals from the original 20 views in the nLMVS-Synth dataset. In “5 views”, “3 views”, and “2 views” settings, we use one view and its neighboring views as inputs.

The third row in Fig. 6.9 shows our results. In the second row, we also show the results of our method without the SfS loss. While the accuracy of our method without the SfS loss heavily depends on the number of input views, our method successfully recovers the object 3D shape even from two-view images. Figure 6.10 shows additional qualitative results on the two-view inputs. The results are qualitatively plausible for objects with different shapes and materials.

Changes of the Surrounding Illumination

As our method recovers a camera-view reflectance map for each view, we can recover surface geometry even from images taken under completely different natural illumination environments. Figure 6.11 shows example results on synthetic images. The results are qualitatively plausible even on the challenging inputs. This demonstrates an advantage of our method, *i.e.*, we can recover surface geometry even from real-world images taken under an illumination environment that dynamically changes across views.

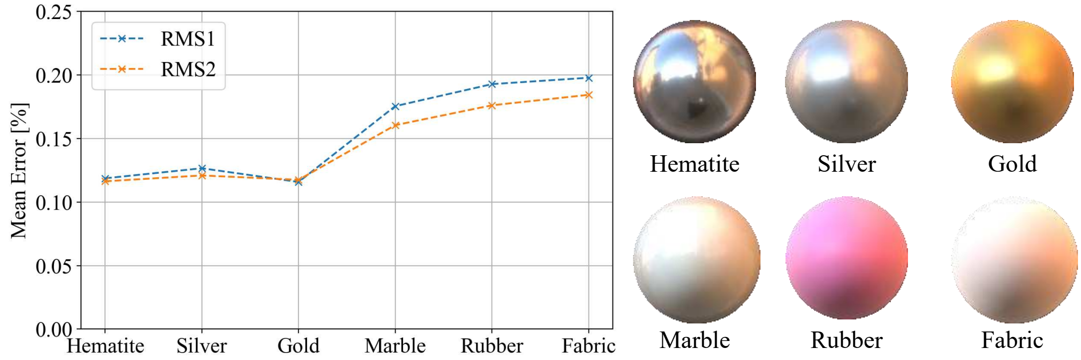


Figure 6.8: Geometry estimation accuracy for each material in the nLMVS-Synth dataset. DeepShaRM works well especially on reflective surfaces, while results on less reflective surfaces (including diffuse surfaces) are still plausible.

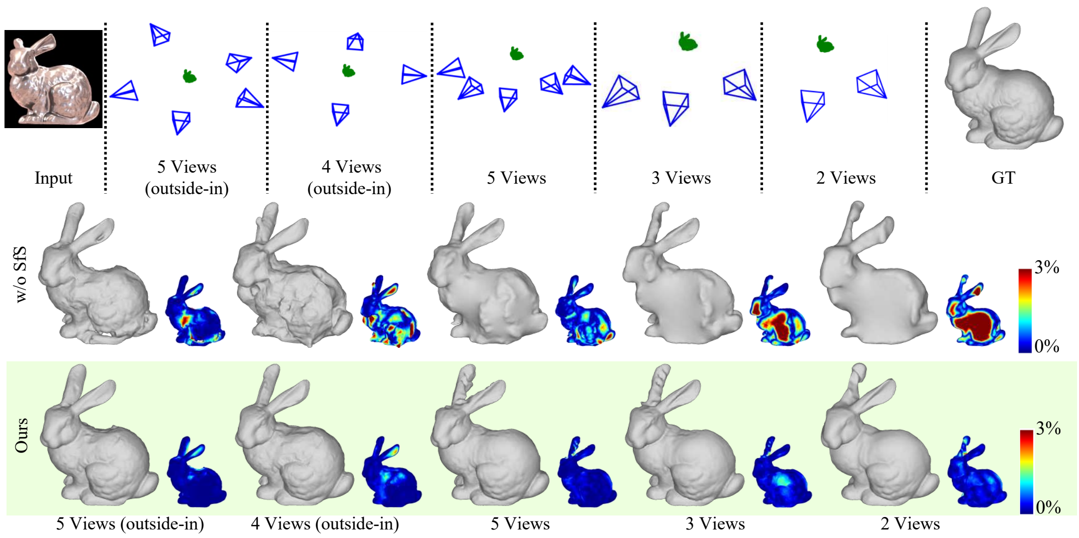


Figure 6.9: Relationship between the number of input views and the recovered geometry. The SfS loss is essential especially when the input views are extremely sparse.

6.4.3 Estimation from Images with Backgrounds

We then test our method on synthetic images with backgrounds. To evaluate the effectiveness of the jount estimation with the neural background representation, the RM loss, and the weighting of the SfS loss, we compare our (full) method with its five variants: "Ours (Naive)", "Baseline", "Baseline w/o SfS", "w/o RM loss", and "w/o Weighted SfS Loss". "Ours (Naive)" imposes the mask loss using masks estimated by a salient object detection method [52] instead of the joint optimization with the neural background representation. "w/o RM Loss" and "w/o Weighted SfS Loss" are ablation studies about the proposed components. "Baseline" does not exploit either the RM loss or the weighting of the SfS loss. "Baseline w/o SfS" is a naive neural image synthesis with only the image synthesis (NeRF) loss. We also compare our method with NeRO [38] that also recovers the object 3D shape from images with backgrounds.

Figure 6.12 shows qualitative results and corresponding quantitative errors. Our results are better than or at least comparable to the other methods. While the results of

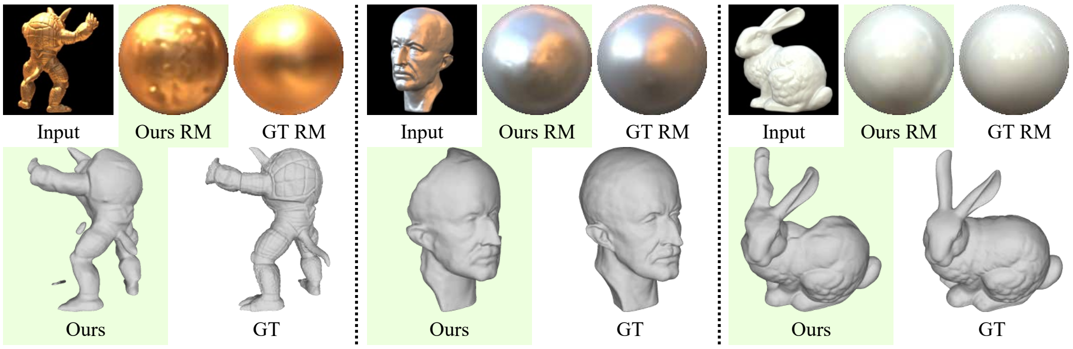


Figure 6.10: Geometry and reflectance maps recovered from two-view images in the nLMVS-Synth dataset. Our method can recover plausible geometry and reflectance maps even from two-view inputs.

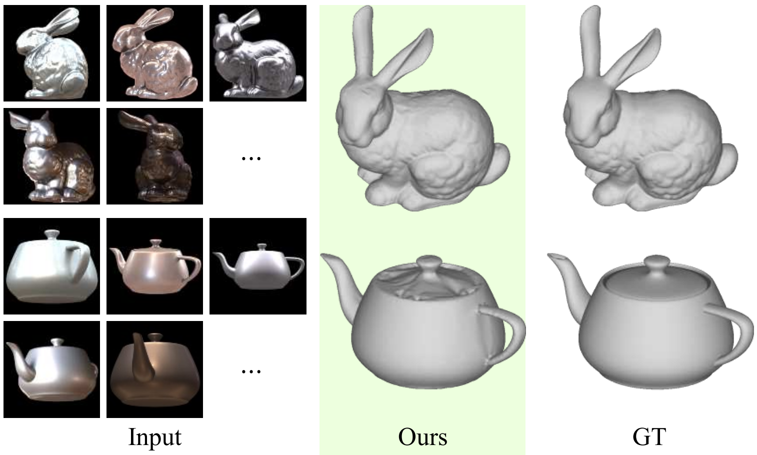


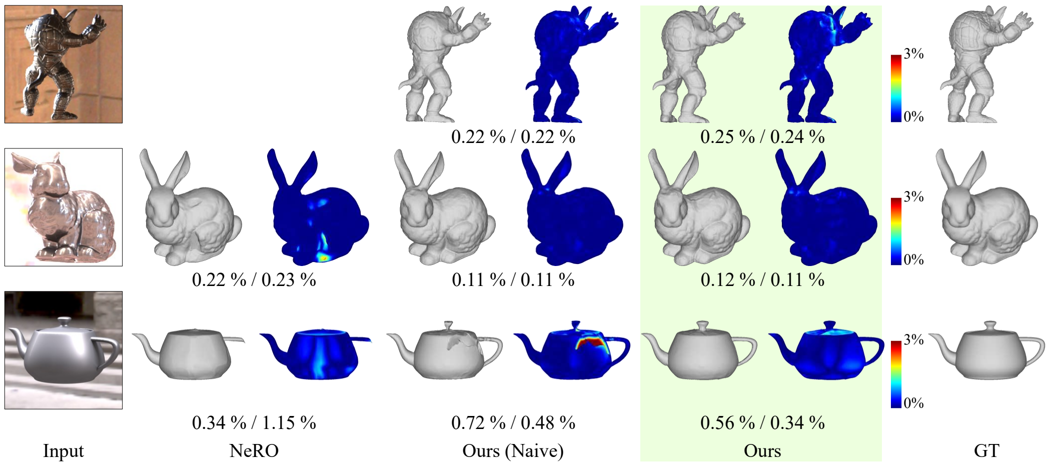
Figure 6.11: Geometry recovered from ten-view synthetic images under different illumination environments. As our method recovers a camera-view reflectance map for each view separately, it can handle changes of the surrounding illumination environments.

"Ours (Naive)" are plausible for most cases, as shown in the third row of Fig. 6.12a, when the segmentation masks estimated by the salient object detection method are erroneous, they strongly affect the geometry estimation results. The ablation study in Fig. 6.12b shows that both the RM loss and the weighting of the SfS loss are essential for accurate reconstruction.

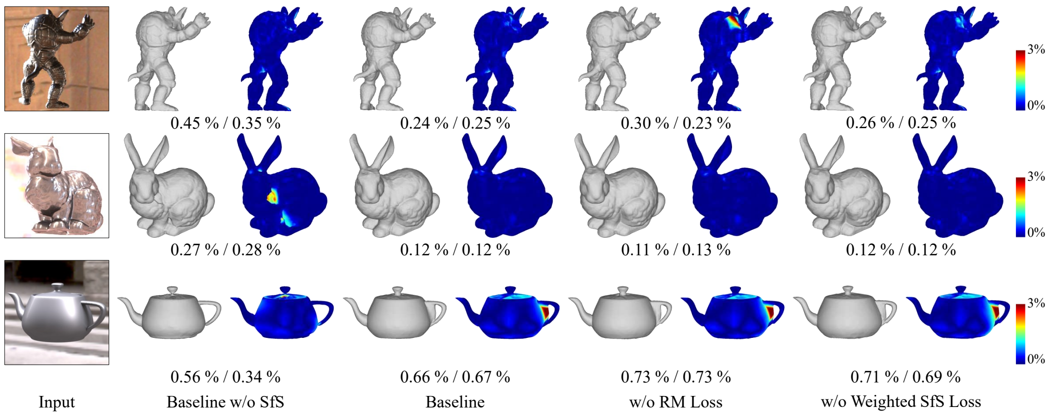
6.4.4 Evaluation on Real Data

We also evaluated our DeepShaRM on real-world images from the nLMVS-Real Dataset. We tested our method on roughly ten-view images with and without the ground truth segmentation masks. Figures 6.13 and 6.15 and Tab. 6.3 show qualitative and quantitative results of our method and those of existing methods. Our method successfully recovers accurate geometry on the reail-world images. This clearly demonstrates the robustness of our method.

Figure 6.14 shows recovered reflectance maps from the real-world images in the nLMVS-Real dataset. Although the ground truth reflectance maps are not available,



(a) Comparison with NeRO [38] and a naive extension of our method. NeRO [38] completely failed to recover the 3D shape of the object in the first row.



(b) Abation study.

Figure 6.12: Geometry recovered from ten-view synthetic images with backgrounds. The left and the right numbers are the RMS errors from the recovered surface to the ground truth and ones from the ground truth to the recovered surface. Please see the text for details and discussions.

the results are qualitatively consistent with corresponding reflectance maps of mirror objects created from ground truth illumination maps.

We also tested our method on a very sparse setup, *i.e.*, estimation from only five-view real-world images (one image and its four neighboring images). Figure 6.16 shows qualitative results. Our method can work even on these very sparse in-the-wild images. This further demonstrates the effectiveness of our method.

Limitations As we can see in Tab. 6.3, our results on the "Entrance-White-Horse" object are relatively poor. As the SfS network exploits radiometric cues (*i.e.*, shadings and specular highlights) for the surface normal recovery, surfaces whose radiances are almost identical regardless of the surface orientations are challenging to our method. Also, as we approximate the object appearance by camera-view reflectance maps, surfaces that deviate from the reflectance maps (*e.g.*, ones in shadows) are also challenging. Note, however, that the experimental results show that our method is robust to small deviations from the assumptions thanks to the learned realistic prior on

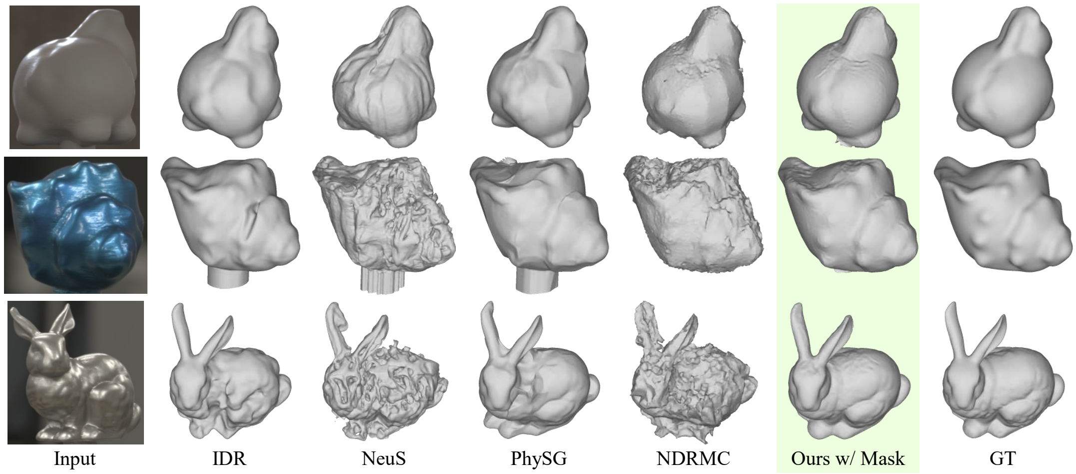


Figure 6.13: Geometry recovered from roughly ten-view images and ground truth masks in the nLMVS-Real Dataset. © 2024 IEEE [68]

geometry and reflectance maps. Note also that, in the reflectance map estimation, we explicitly handle such deviations by introducing the image feature extraction and the confidence score estimation.

6.5 Summary

In this chapter, we introduced DeepShaRM, a deep multi-view geometry and reflectance map estimation method for textureless, non-Lambertian objects under unknown natural illumination. By bypassing solving the challenging joint estimation of reflectance and illumination but instead fully leveraging learned realistic prior on reflectance maps along with one on surface geometry, our method successfully recovers accurate 3D shapes from posed multi-view images without making any restricting assumptions (*e.g.*, known illumination). Experimental results including ablation studies, comparison with existing methods, and evaluations on real-world images clearly demonstrate the effectiveness of the proposed method. We believe our DeepShaRM can serve as practical means for passive 3D reconstruction in the wild.

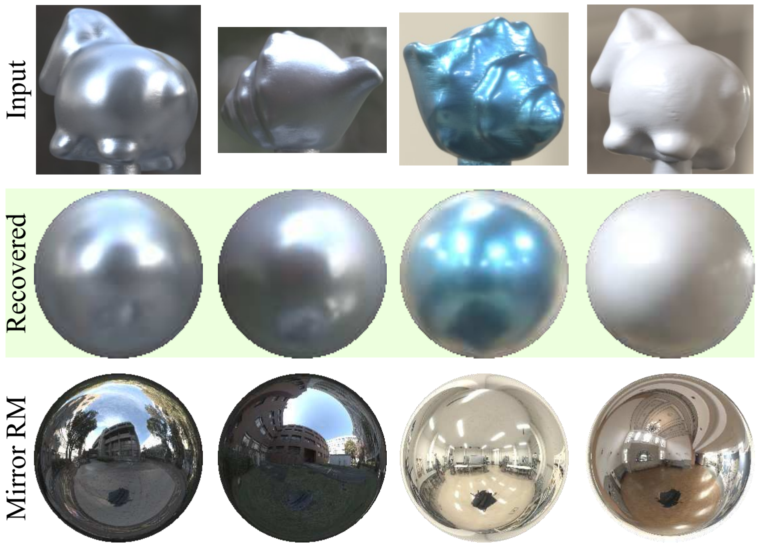


Figure 6.14: Recovered reflectance maps from real-world images and masks in the nLMVS-Real dataset. The reuslts are qualitatively consistent with Mirror RM, reflectance maps of mirror objects created from captured illumination maps. © 2024 IEEE [68]

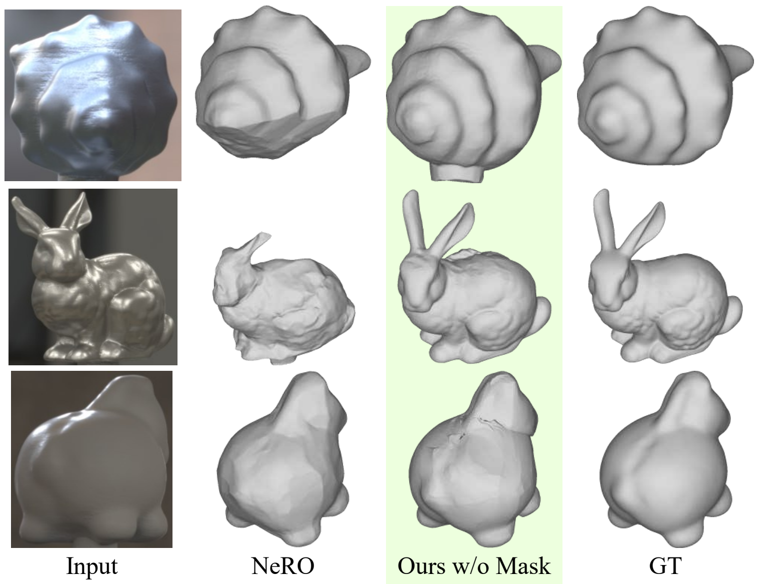


Figure 6.15: Geometry recovered from roughly ten-view images in the nLMVS-Real Dataset without using ground truth masks.

Object	Buildings-Metal-Horse	Court-Metal-Shell	Entrance-White-Horse	Labo-Blue-Shell	Labo-Metal-Bunny	Mean
IDR [73]	**0.37 %**	0.49 %	**0.38** %	0.40 %	0.95 %	0.52 %
NeuS [64]	3.14 %	2.70 %	0.60 %	1.51 %	1.00 %	1.79 %
PhySG [76]	0.70 %	1.01 %	0.69 %	0.57 %	0.90 %	0.78 %
NDRMC [21]	1.06 %	1.61 %	0.74 %	0.92 %	2.32 %	1.33 %
Ours (w/ mask)	0.42 %	**0.36 %**	0.41 %	**0.27** %	**0.33** %	**0.36** %
NeRO [38]	2.37 %	1.51 %	**0.62** %	**0.23 %**	1.51 %	1.25 %
Ours (w/o mask)	**0.38** %	**0.59 %**	1.00 %	0.25 %	**1.07** %	**0.66** %

(a) RMSE from recovered to ground truth (RMS1)

Object	Buildings-Metal-Horse	Court-Metal-Shell	Entrance-White-Horse	Labo-Blue-Shell	Labo-Metal-Bunny	Mean
IDR [73]	**0.36** %	0.49 %	0.41 %	**0.34** %	0.81 %	0.48 %
NeuS [64]	1.75 %	1.99 %	0.51 %	0.88 %	0.68 %	1.16 %
PhySG [76]	0.66 %	0.68 %	0.68 %	0.48 %	0.80 %	0.66 %
NDRMC [21]	0.76 %	1.11 %	0.46 %	0.68 %	0.99 %	0.80 %
Ours (w/ mask)	0.39 %	**0.36** %	**0.38 %**	0.34 %	**0.37 %**	**0.37 %**
NeRO [38]	2.61 %	1.68 %	**0.59** %	**0.24 %**	3.86 %	1.80 %
Ours (w/o mask)	**0.36** %	**0.62 %**	0.95 %	0.27 %	**1.05** %	**0.65** %

(b) RMSE from ground truth to recovered (RMS2)

Table 6.3: Geometry reconstruction accuracy (RMSE) on nLMVS-Real dataset.

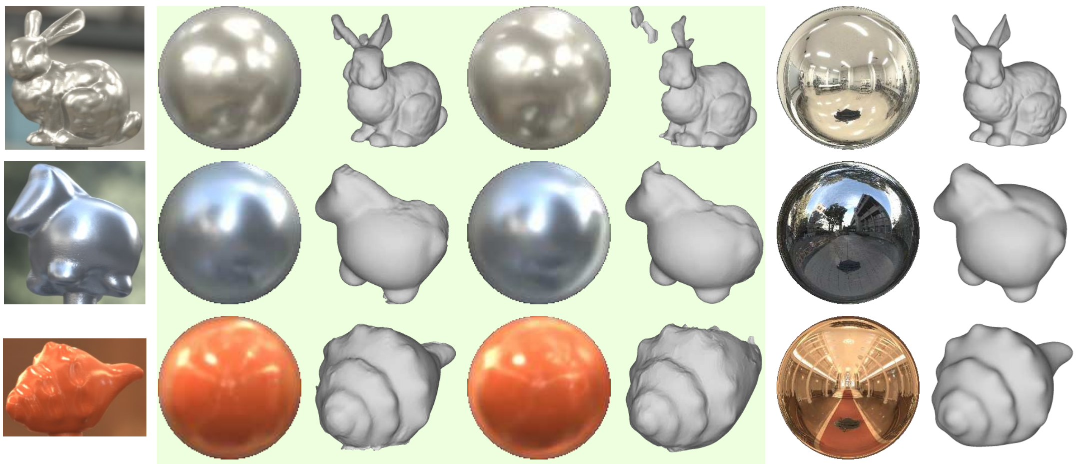


Figure 6.16: Recovered geometry and reflectance maps from five-view images in the nLMVS-Real dataset.

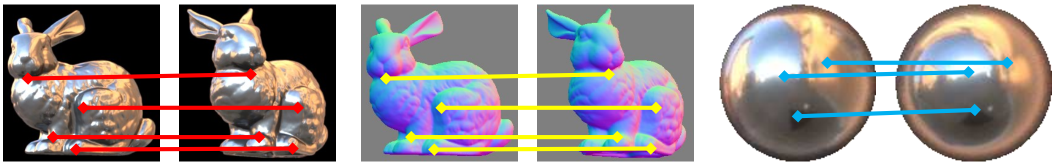
Chapter 7

Reflection Correspondences for Shape and Camera Pose Recovery

In the last two chapters, we focused on 3D reconstruction from posed multi-view images (images with known camera poses). Even if we do not know camera poses, we, as humans, can understand motions of the cameras and also the object 3D shape from two images of a textureless, non-Lambertian object. What can compute vision methods do for such unposed images of a textureless, non-Lambertian object?

Structure-from-motion methods handle unposed images by detecting and leveraging pixel correspondences, correspondences of pixels that are associated with the same surface point. Once we could detect pixel correspondences, we can jointly solve for the camera poses and 3D locations of sparse surface points that correspond to the pixel correspondences. Although it is difficult to directly detect pixel correspondences from images of a textureless, non-Lambertian object, by applying DeepShaRM to each input image separately, we can obtain a surface normal map for each view which becomes cues for the correspondence detection. Although the surface normals are estimated in each camera coordinate system, we can still extract view-independent features such as curvatures of surfaces for correspondence detection. The surface normal maps also provide sets of corresponding surface normals which become additional constraint for the camera pose estimation. We refer to this types of correspondences in normal maps as 3D correspondences.

Camera pose estimation from images of a textureless, non-Lambertian object is, however, still challenging even with the radiometrically recovred surface normals as the single-view estimation suffers from the fundamental ambiguity between geometry and reflectance maps. That is, as visualized in Fig. 7.2, there is a family of possible combinations of a normal map (geometry) and a reflectance map that are consistent with the input image, which is known as the (generalized) bas-relief ambiguity [3]. This makes the correspondence detection difficult. Besides, when we only have two-views and the cameras are orthographic (*i.e.*, the viewing direction is constant for each view), there is an ambiguity in the camera pose estimation which cannot be resolved even with correct pixel and 3D correspondences. Note that the orthographic assumption often holds true in real scenes. For instance, when the distance between the object and the cameras are much larger than the object size. The viewing direction from a camera to the object becomes almost constant. Note also that, even with more than two views, the camera pose estimation is still challenging as the joint estimation of 3D object shape (3D locations of sparse surface points) and the camera poses can easily falls into a local minima. These problems become critical especially when we need to estimate camera poses relative to a moving object. In



(a) Pixel Correspondences (b) 3D Correspondences (c) Reflection Correspondences

Figure 7.1: Three types of correspondences. (a) **Pixel correspondences** are classical correspondences that link pixels that correspond to the same point on the object surface. (b) If we have pixel correspondences and a surface normal map for each view, we can also obtain pairs of surface normals which we can leverage for the camera pose estimation. We refer to these correspondences as **3D correspondences**. (c) **Reflection correspondences** associate pixels in two views where the same object in the surrounding illumination environment is observed as specular reflection. We can achieve detection of reflection correspondences as detection of corresponding surface normal orientations in two-view reflectance maps.

such setting, we cannot leverage textures in backgrounds at all.

In this chapter, we introduce a novel joint camera pose and object 3D shape reconstruction method which only leverages appearance of a textureless object under static natural illumination. Our key idea is leveraging reflection correspondences, correspondences of pixels whose specular reflection correspond to the same point in the surrounding illumination. We show that, by detecting and leveraging this novel type of correspondences, we can resolve the bas-relief ambiguity and recover the relative camera poses and also the object 3D shape even from two-view images captured by orthographic cameras.

We formalize reflection correspondences and derive how we can exploit them along with conventional correspondences for obtaining a solution for the relative camera pose. We show that We can achieve the detection of reflection correspondences as detection of corresponding surface normals in reflectance maps. We train two neural mathchers for detection of reflection correspondences from reflectance maps and one of 3D correspondences from normal maps, respectively. We introduce a data augmentation method based on the generalized bas-relief transformation so that we can detect the correspondences robust to estimation errors caused by the bas-relief ambiguity. We introduce a RANSAC-based robust joint estimation framework for accurate joint object 3D shape and camera pose recovery which alternates between the camera pose estimation and multi-view geometry estimation based on DeepShaRM.

7.1 Three Types of Correspondences

In this chapter, we first detail the three types of correspondences and then explain how we can exploit them for joint shape and camera pose recovery from two-view images of a textureless, non-Lambertian object. Figure 7.1 provides visualization of the three types of correspondences. In this thesis, we refer to them as the following names:

- “Pixel correspondences”, for classical correspondences which link pixels in two view images that correspond to the same 3D point on the object surface.

- "3D correspondences", for correspondences between pixels in two-view normal maps that also be associated with the same surface point. Note that, the difference of the 3D correspondences from the pixel correspondecens is that 3D correspondences provide additional constraints regarding surface normals for the camera pose estimation.

- "Reflection correspondences", for correspondences between pixels whose specular reflection (specular highlights) corresponds to the same point in the surrounding illumination environment.

Note that, similar to the previous chapters, we assume orthographic projection as objects are usually distant from the cameras and we can regard the viewing directions as almost constant for each view. Note also that, as we see in the next sub-section, camera pose estimation is challenging especially under orthographic projection.

7.1.1 Pixel Correspondences

Let us assume that we can establish N_{IM} pixel correspondences $(u_1^i, v_1^i) \leftrightarrow (u_2^i, v_2^i)$ $(i = 1, 2, \ldots, N_{\mathrm{IM}})$. Under orthographic projection, there is a relationship between the pixel correspondences, the relative rotation R_{21}, and the relative translation $\mathbf{t}_{21} \equiv (t_x, t_y, t_z)^{\mathrm{T}}$

$$f'_{\mathrm{IM}} = \sum_i^{N_{\mathrm{IM}}} f_u^i + f_v^i = 0 \,, \tag{7.1}$$

with

$$f_u^i = (r_{11} u_2^i + r_{12} v_2^i + r_{13} z_2^i + t_x - u_1^i) \,, \tag{7.2}$$

$$f_v^i = (r_{21} u_2^i + r_{22} v_2^i + r_{23} z_2^i + t_y - v_1^i) \,, \tag{7.3}$$

where z_2^i is the depth of a surface point in the second view for the i-th correspondence. Harris [20] showed that, by minimizing f'_{IM} with respect to the depths z_2^i and the translation vector $\mathbf{t}_{21}$, we have an objective function for estimation of the relative rotation

$$f_{\mathrm{IM}} = \sum_i^{N_{\mathrm{IM}}} (t_\phi^i - t_\theta^i)^2 \,, \tag{7.4}$$

with

$$t_\phi^i = u_1^i \cos\phi + v_1^i \sin\phi \,, \tag{7.5}$$

$$t_\theta^i = u_2^i \cos\theta + v_2^i \sin\theta \,, \tag{7.6}$$

where ϕ and θ are two of three Euler angled (ϕ, η, θ) that correspond to the relative rotation R_{21} from the second view to the first view. Without loss of generality, we assume that these correspondences are shifted such that the average location (*e.g.*, $\frac{1}{N_{\mathrm{IM}}} \sum_i u_1^i$) is zero. We also assume the (z-x-z) sequence for the Euler angles

$$\mathrm{R}_{21} = \mathrm{R_z}[\phi] \mathrm{R_x}[\eta] \mathrm{R_z}[-\theta] \,, \tag{7.7}$$

where $\mathrm{R_z}[\cdot]$ and $\mathrm{R_x}[\cdot]$ are rotations around z and x axes, respectively. Please refer to Harris [20] for the derivation. Since Eq. (7.4) is independent of one of the Euler angles (η), it is not possible to fully recover the relative rotation matrix only from pixel correspondences between two views.

Distortions of the surface geometry Let us also consider how the ambiguity regarding η affects corresponding estimate of surface geometry, *i.e.*, the depths of the sparse surface points z_2^i. As the depths that minimize Eq. (7.1) should satisfy

$$\frac{\partial}{\partial z_2^i} f'_{\mathrm{IM}} = 0\,, \tag{7.8}$$

we have

$$z_2^i = -\frac{r_{13}(\tilde{u}^i + t_x) + r_{23}(\tilde{v}^i + t_y)}{r_{13}^2 + r_{23}^2}\,, \tag{7.9}$$

where

$$\tilde{u}^i = r_{11}u_2^i + r_{12}v_2^i - u_1^i\,, \tag{7.10}$$

$$\tilde{v}^i = r_{21}u_2^i + r_{22}v_2^i - v_1^i\,. \tag{7.11}$$

Relationships between the element of R_{21} and corresponding Euler angles (ϕ, η, θ) are

$$r_{11} = \cos\phi\cos\theta + \sin\phi\sin\theta\cos\eta\,, \tag{7.12}$$

$$r_{12} = \cos\phi\sin\theta - \sin\phi\cos\theta\cos\eta\,, \tag{7.13}$$

$$r_{13} = \sin\phi\sin\eta\,, \tag{7.14}$$

$$r_{21} = \sin\phi\cos\theta - \cos\phi\sin\theta\cos\eta\,, \tag{7.15}$$

$$r_{22} = \sin\phi\sin\theta + \cos\phi\cos\theta\cos\eta\,, \tag{7.16}$$

$$r_{23} = -\cos\phi\sin\eta\,, \tag{7.17}$$

$$r_{31} = -\sin\theta\sin\eta\,, \tag{7.18}$$

$$r_{32} = \cos\theta\sin\eta\,, \tag{7.19}$$

$$r_{33} = \cos\eta\,. \tag{7.20}$$

By substituting these for Eq. (7.9), we have

$$z_2^i = \frac{1}{\tan\eta} n_\theta^i - \frac{1}{\sin\eta}(n_\phi^i - n_{t,\phi})\,, \tag{7.21}$$

where

$$n_\theta^i = u_2^i \sin\theta - v_2^i \cos\theta\,, \tag{7.22}$$

$$n_\phi^i = u_1^i \sin\phi - v_1^i \cos\phi\,, \tag{7.23}$$

$$n_{t,\phi} = t_x \sin\phi - t_y \cos\phi\,. \tag{7.24}$$

Let us assume that $\tilde{\eta}$ is an inaccurate estimate of the unresolvable Euler angle and η is the corresponding ground truth. There are relationships between these quantities

$$\frac{1}{\sin\hat{\eta}} = \frac{\sin\eta}{\sin\hat{\eta}} \frac{1}{\sin\eta}\,, \tag{7.25}$$

$$\frac{1}{\tan\hat{\eta}} = \frac{\sin\eta}{\sin\hat{\eta}}\frac{1}{\tan\eta} + \left(\frac{\cos\hat{\eta}}{\cos\eta} - 1\right)\frac{\sin\eta}{\sin\hat{\eta}}\frac{1}{\tan\eta}\,. \tag{7.26}$$

By using Eqs. (7.21), (7.25) and (7.26), we can derive the relationship between inaccurate depth estimates $\hat{z}_2^i$ that correspond to $\hat{\eta}$ and the ground truth z_2^i

$$\hat{z}_2^i = \lambda_{\hat{\eta}} z_2^i + \mu_{\hat{\eta}} u_2^i + \nu_{\hat{\eta}} v_2^i + c_{\hat{\eta}}\,, \tag{7.27}$$

where

$$\lambda_{\hat{\eta}} = \frac{\sin\eta}{\sin\hat{\eta}}\,, \tag{7.28}$$

$$\mu_{\hat{\eta}} = \left(\frac{\cos\hat{\eta}}{\cos\eta} - 1\right)\frac{\sin\eta}{\sin\hat{\eta}}\frac{1}{\tan\eta}\sin\theta\,, \tag{7.29}$$

$$\nu_{\hat{\eta}} = -\left(\frac{\cos\hat{\eta}}{\cos\eta} - 1\right)\frac{\sin\eta}{\sin\hat{\eta}}\frac{1}{\tan\eta}\cos\theta\,, \tag{7.30}$$

$$c_{\hat{\eta}} = \frac{1}{\sin\hat{\eta}}\left(\hat{n}_{t,\phi} - n_{t,\phi}\right)\,, \tag{7.31}$$

and $\hat{n}_{t,\phi}$ is an estimate of $n_{t,\phi}$ that corresponds to $\hat{\eta}$. This relationship is identical to the generalized bas-relief (GBR) transformation [3]

$$d'[u_2^i, v_2^i] = \lambda_2 d[u_2^i, v_2^i] + \mu_2 u_2^i + \nu_2 v_2^i\,, \tag{7.32}$$

except for the global offset $c_{\hat{\eta}}$. Note that, we can also derive the relationship between depths in the first view z_1^i and η in a similar way. Due to this relationship between the ground truth and the wrong geometry estimate, as we see in the next sub-section, the ambiguity in the camera pose estimation is unresolvable even with 3D correspondences.

7.1.2 3D Correspondences

If we could recover accurate surface normal maps $\mathbf{N}_1$ and $\mathbf{N}_2$ for the input images, pixel correspondences $(u_1^i, v_1^i) \leftrightarrow (u_2^i, v_2^i)$ should also satisfy

$$\mathbf{N}_1[u_1^i, v_1^i] = \mathrm{R}_{21}\mathbf{N}_2[u_2^i, v_2^i]\,, \tag{7.33}$$

as surface normals at locations $\mathbf{N}_1[u_1^i, v_1^i]$ and $\mathbf{N}_2[u_2^i, v_2^i]$ should be identical up to the relative rotation R_{21}. Once we have these correspondences, we could recover the relative rotation by solving Eq. (7.33).

Unfortunately, as visualized in Fig. 7.2, the single-view estimation of surface normal and reflectance maps from the object appearance suffers from the generalized bas-relief ambiguity [3], the fundamental ambiguity between the two unknowns. Given a (ground truth) normal map $\mathbf{N}$ for an image I, any normal map $\mathbf{N}'$ with

$$\mathbf{N}'[u, v] = \mathrm{G}^{-\mathrm{T}}\mathbf{N}[u, v]\,, \tag{7.34}$$

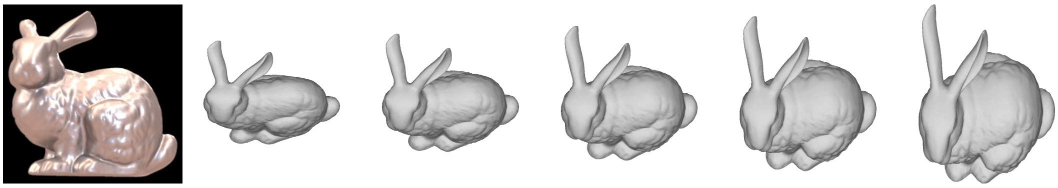


Figure 7.2: The bas-relief ambiguity. There is a family of possible 3D shapes that can explain single-view observation. For this, even if we recover surface normals for each view and establish correspondences on them for the camera pose estimation, the results are erroneous and affect the geometry estimation accuracy.

could result in the same image I under a different lighting, where G is the generalized bas-relief transformation [3]

$$\mathrm{G} \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \mu & \nu & \lambda \end{pmatrix} . \tag{7.35}$$

μ and ν can take any values and λ can take any positive value. Note that, while discussions about the bas-relief ambiguity (*e.g.*, ones in Belhumeur *et al.* [3]) usually assume directional lights, this holds true even for environmental illumination as we can regard it as a set of directional lights.

The bas-relief ambiguity changes the relationship in Eq. (7.33) to

$$\mathbf{N}_1[u_1^i, v_1^i] \propto \mathrm{G}_1^{-\mathrm{T}} \mathrm{R}_{21} \mathrm{G}_2^{\mathrm{T}} \mathbf{N}_2[u_2^i, v_2^i] , \tag{7.36}$$

where G_k $(k = 1, 2)$ is an unknown GBR transformation for each view that corresponds to the estimation errors of surface normal maps. If we have a sufficient number of correspondences, we can obtain a unique solution for the combined transformation

$$\mathrm{G}_{21} \equiv \mathrm{G}_1^{-\mathrm{T}} \mathrm{R}_{21} \mathrm{G}_2^{\mathrm{T}} . \tag{7.37}$$

There is, however, an unresolvable ambiguity in its decomposition into the three unknown matrices. Most important, this ambiguity corresponds to one regarding pixel correspondences. In other words, we cannot recover the relative camera pose even if we have correct pixel and 3D correspondences. For any wrong estimate of the Euler angle $\hat{\eta}$, there are is a GBR transformation

$$\hat{\mathrm{G}}_2 = \mathrm{G}_2 \mathrm{G}_{\hat{\eta},2}^{-1} , \tag{7.38}$$

that transforms surface normals of the inaccurate surface geometry that corresponds to $\hat{\eta}$ to ones in the estimated normal map in the second view. G_2 is a transformation from the ground-truth geometry to the recovered normal map and $\mathrm{G}_{\hat{\eta},2}$ is one from the ground truth to the inaccurate geometry

$$\mathrm{G}_{\hat{\eta},2} = \begin{pmatrix} 1 & 0 & o \\ 0 & 1 & 0 \\ \mu_{\hat{\eta}} & \nu_{\hat{\eta}} & \lambda_{\hat{\eta}} \end{pmatrix} . \tag{7.39}$$

We can also derive the relationship between $\hat{\eta}$ and the GBR transformation of the first view $\hat{\mathrm{G}}_1$ in a similar way. For this, pixel and 3D correspondences are insufficient for the camera pose estimation when we only have two views.

7.1.3 Reflection Correspondences

Let us assume we have correspondences between image locations $(u_1^i, v_1^i) \leftrightarrow (u_2^i, v_2^i)$ where incident rays come from the same direction. In other words we have correspondences of the surroundings reflected on the object surface. If we, for now, ignore the generalized bas-relief ambiguity, each reflection correspondences gives us a constraint of the form

$$\boldsymbol{\omega}_\mathbf{r}(\mathbf{N}_1[u_1^i, v_1^i]) \propto \mathrm{R}_{21}\boldsymbol{\omega}_\mathbf{r}(\mathbf{N}_2[u_2^i, v_2^i])\,, \tag{7.40}$$

where function $\boldsymbol{\omega}_\mathbf{r}(\mathbf{n})$ returns the reflection direction of the viewing direction $\boldsymbol{\omega}_o$ on the surface with normal $\mathbf{n}$

$$\boldsymbol{\omega}_\mathbf{r}(\mathbf{n}) = -\boldsymbol{\omega}_o + 2\frac{\boldsymbol{\omega}_o \cdot \mathbf{n}}{\mathbf{n} \cdot \mathbf{n}}\mathbf{n}\,. \tag{7.41}$$

Again, from a single-view observation, we can recover surface normals only up to the bas-relief ambiguity. We need to introduce the two GBR transformations in Eq. (7.40)

$$\mathbf{G_1^T}\boldsymbol{\omega}_\mathbf{r}(\mathrm{G}_2^\mathrm{T}\mathbf{N}_1[u_1^i, v_1^i]) \propto \mathrm{R}_{21}\boldsymbol{\omega}_\mathbf{r}(\mathbf{N}_2[u_2^i, v_2^i])\,, \tag{7.42}$$

Each reflection correspondences thus gives us a new type of constraints for the estimation of the relative rotation R_{21}.

Detection of reflection correspondences directly from images is, however, extremely challenging as surface reflection depends not only on the surrounding illumination but also on the surface geometry. As we assume that we have (erroneous) normal maps for the input images, we can also assume that we have corresponding reflectance map estimates. Note that, in practice, we jointly recover surface normal and reflectance maps for each view using DeepShaRM. Since detection of reflection correspondences is equivalent to detection of corresponding surface normals in two-view reflectance maps, by recovering reflectance maps, we can detect reflection correspondences regardless of the surface geometry. Note that the reflectance map estimate $E'(\mathbf{n})$ recovered from the erroneous surface normal map estimate also suffer from the bas-relief ambiguity

$$E'(\mathbf{n}) = E(\mathrm{G}^\mathrm{T}\mathbf{n})\,, \tag{7.43}$$

where $E(\mathbf{n})$ is the ground truth. Hence we still need to use Eq. (7.42) for these reflection correspondences.

7.1.4 Relative Rotation from Correspondences

From Eqs. (7.4), (7.36) and (7.42), we derive the objective function for the estimation of the relative rotation R_{21}

$$f = f_\mathrm{IM} + f_\mathrm{NM}^{(12)} + f_\mathrm{NM}^{(21)} + f_\mathrm{RM}^{(12)} + f_\mathrm{RM}^{(21)}\,, \tag{7.44}$$

where

$$f_{\mathrm{NM}}^{jk} = \sum_{i}^{N_{\mathrm{NM}}} \left\| \mathbf{N}_j[u_j^i, v_j^i] - \mathrm{Norm}\left(\mathrm{G}_{kj}\mathbf{N}_k[u_k^i, v_k^i]\right) \right\|^2 , \tag{7.45}$$

$$f_{\mathrm{RM}}^{jk} = \sum_{i}^{N_{\mathrm{RM}}} \left\| \mathbf{N}_j[u_j^i, v_j^i] - \Omega^{(jk)}\left(\mathrm{G}_{kj}\mathbf{N}_k[u_k^i, v_k^i]\right) \right\|^2 . \tag{7.46}$$

$\mathrm{Norm}\left(\cdot\right)$ is the vector normalization operator and $\Omega^{(jk)}(\mathbf{n})$ transforms a surface normal in the j-th view reflectance map to the surface normal in the k-th view

$$\Omega^{(jk)}(\mathbf{n}) = \mathrm{Norm}\left(\mathrm{G_k}^{-\mathrm{T}}\boldsymbol{\omega}_{\mathbf{r}}^{-1}\left(\mathrm{R}_{jk}\boldsymbol{\omega}_{\mathbf{r}}\left(\mathrm{G}_j{}^{\mathrm{T}}\mathbf{n}\right)\right)\right) . \tag{7.47}$$

Note that, in Eqs. (7.45) and (7.46), we can switch the role of the two views.

Given 3 sets of N_{IM} pixel correspondences, N_{NM} 3D correspondences, and N_{RM} reflection correspondences, we compute the relative rotation R_{21} by minimizing the objective function f in Eq. (7.44) for the three Euler angles θ, ϕ, and η, and two sets of parameters of the GBR transformations for both images μ_1, ν_1, λ_1, μ_2, ν_2, and λ_2 under constraints that λ_1 and λ_2 are positive. We ensure these constraints by optimizing logarithms of λ_1 and λ_2 instead of directly optimizing them.

A naive approach to solve this optimization problem is using an off-the-shelf solver for nonlinear optimization problems such as ones in Scipy [63]. Such a naive optimization, however, easily falls into a local minima as the objective function, especially f_{RM}^{jk}, includes complex nonlinear operation. For this, we derive a two-step algorithm that first recover the compound transformation G_{21} using the pixel and the 3D correspondences and then resolve the remaining ambiguity regarding η using the reflection correspondences. Specifically, we first obtain initial estimates of unknown parameters by minimizing the sum of Eqs. (7.4) and (7.45) using an off-the-shelf solver [63]. This gives us a unique solution for ϕ, θ, and the compound transformation matrix G_{21}.

In the second step, given the reflection correspondences, ϕ, θ, and G_{21}, we solve for the remaining unknown parameters, *i.e.*, η and parameters of the bas-relief transformations. Although the decomposition of G_{21} into the three matrices R_{21}, G_1, and G_2 is ambiguous, if η is given in addition to θ and ϕ, *i.e.*, if R_{21} is determined, we can solve for the parameters of the GBR transformations analytically. From the definitions of G_{21}, G_1, G_2, and R_{21}, we have

$$\mathrm{G}_{21} = \begin{pmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} \\ \gamma_{21} & \gamma_{22} & \gamma_{23} \\ \gamma_{31} & \gamma_{32} & \gamma_{33} \end{pmatrix} , \tag{7.48}$$

$$\gamma_{11} = r_{11} - r_{31}\frac{\mu_1}{\lambda_1}\,, \tag{7.49}$$
$$\gamma_{12} = r_{12} - r_{32}\frac{\mu_1}{\lambda_1}\,, \tag{7.50}$$
$$\gamma_{13} = r_{11}\mu_2 + r_{12}\nu_2 + r_{13}\lambda_2 - \mu_1\gamma_{33}\,, \tag{7.51}$$
$$\gamma_{21} = r_{21} - r_{31}\frac{\nu_1}{\lambda_1}\,, \tag{7.52}$$
$$\gamma_{22} = r_{22} - r_{32}\frac{\nu_1}{\lambda_1}\,, \tag{7.53}$$
$$\gamma_{23} = r_{21}\mu_2 + r_{22}\nu_2 + r_{23}\lambda_2 - \nu_1\gamma_{33}\,, \tag{7.54}$$
$$\gamma_{31} = \frac{r_{31}}{\lambda_1}\,, \tag{7.55}$$
$$\gamma_{32} = \frac{r_{32}}{\lambda_1}\,, \tag{7.56}$$
$$\gamma_{33} = \frac{1}{\lambda_1}(r_{31}\mu_2 + r_{32}\nu_2 + r_{33}\lambda_2)\,, \tag{7.57}$$

where r_{ij} is the (i,j) element of R_{21}. From Eqs. (7.55) and (7.56), we have

$$\lambda_1^2 = \frac{r_{31}^2 + r_{32}^2}{\gamma_{31}^2 + \gamma_{32}^2}\,. \tag{7.58}$$

Since λ_1 is positive, by taking the square root of both sides, we have a solution for λ_1

$$\lambda_1 = \sqrt{\frac{r_{31}^2 + r_{32}^2}{\gamma_{31}^2 + \gamma_{32}^2}}\,. \tag{7.59}$$

As there is a relationship about determinants of the matrices

$$\det(\mathrm{G}_{21}) = \det\left(\mathrm{G}_1{}^{-\mathrm{T}}\right)\det(\mathrm{R}_{21})\det\left(\mathrm{G}_2{}^{\mathrm{T}}\right) = \frac{\lambda_2}{\lambda_1}\,, \tag{7.60}$$

we also have a solution for λ_2

$$\lambda_2 = \det(\mathrm{G}_{21})\,\lambda_1\,. \tag{7.61}$$

As Eqs. (7.49) to (7.54) and (7.57) are linear equations for the remaining unknown parameters, we can analytically solve these equations in least-squares sense. Note that, we empirically found that these linear equations have a solution only when the sign of the given η and one of the corresponding ground truth are identical. Based on these, in the second step, we sample discretized possible solutions of η in 360 angles, solve for the other parameters analytically, and check if the obtained parameters are consistent with the reflection correspondences. Please see the next section for what metric we use to evaluate the consistency in practice.

Intuition on the required number of correspondences In practice, the numbers of correspondences required for the camera pose estimation is important as we can use the information for implementing a robust estimation method with RANSAC [15].

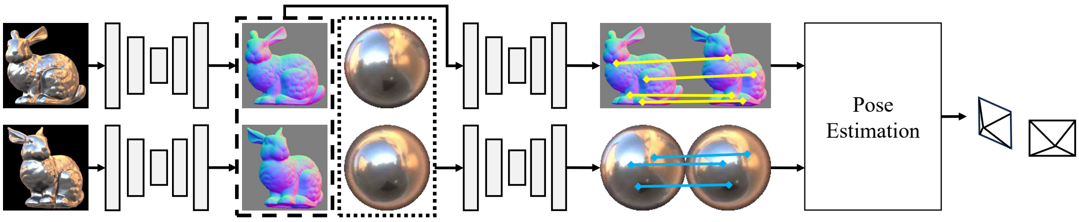


Figure 7.3: Given two-view images of a textureless, non-Lambertian object, we first recover normal and reflectance maps for the input images by applying DeepShaRM to each image separately. We then detect correspondences in the normal and reflectance maps using neural matchers, and recover the camera pose from the correspondences.

Although it is difficult to derive a rigorous proof about the numbers, we can build intuition about how many correspondences are required for each view pair. We first consider the estimation of the compound transformation G_{21} from correspondences in normal maps. Each correspondence gives us 1 constraint from a pixel correspondence and 2 from a 3D correspondence. Note that, the 3D correspondence gives only two constraints as it is normalized. The number of unknown parameters except for the unresolvable one (η) is 8. Thus the expected number of required correspondences in normal maps is 3. One additional reflection correspondence is also required for resolving the remaining ambiguity about η. We indeed experimentally validated that three correspondences in normal maps and one reflection correspondence are sufficient for a direct optimization of the all unknown parameters (although it often falls into a local minima and is unusable in practice). We. however, empirically found that the two-step estimation algorithm requires four correspondences in normal maps for the first stage, *i.e.*, the estimation of the combined transformation G_{21}. This would be because the nonlinear optimization of the all parameters always requires four correspondences regardless of whether the ambiguity about η remains or not.

7.2 Camera Pose Estimation from Two Images

Given the estimation algorithm with reflection correspondences, we can recover the relative camera pose from only two images of a textureless, non-Lambertian object. Figure 7.3 depicts how we recover the camera pose between two views. To recover the normal maps $\mathbf{N}_1$ and $\mathbf{N}_2$ and corresponding reflectance maps for the images, we first apply DeepShaRM to each of the input images separately. We use the recovered normal and reflectance maps for detection of 3D and reflection correspondences.

To obtain 3D correspondences in the normal maps, we train a deep network to extract pixel-wise, view-independent features that we use to match locations across the normal maps. Figure 7.4 depicts how we train the feature extraction network. We train the network by contrastive learning. We use pairs of synthetic normals maps of two views from the nLMVS-Synth dataset as training data. We feed the normal maps to the feature extraction network separately and obtain corresponding feature maps $\mathbf{F}_1$ and $\mathbf{F}_2$. Given ground truth correspondences $(u_1^i, v_1^i) \leftrightarrow (u_2^i, v_2^i)$, we impose the

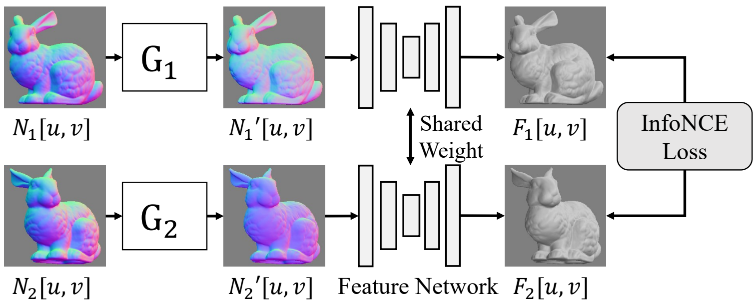


Figure 7.4: We train a deep neural network to extract view-independent, per-pixel features for the correspondence detection from normal maps. We achieve this training by contrastive learning with the InfoNCE loss [61]. We also introduce a data augmentation method based on the generalized bas-relief (GBR) transformation [3] so that we can detect correspondences even from normal maps that suffer from the bas-relief ambiguity.

InfoNCE loss [61]

$$L = \sum_i \sum_j -\gamma_{ij} \log \left(\frac{\exp(c_{ij})}{\sum_{i'} \sum_{j'} \exp(c_{i'j'})} \right) , \tag{7.62}$$

where c_{ij} is the cosine similarity of feature vectors $\mathbf{F_1}[u_1^i, v_1^i]$ and $\mathbf{F_2}[u_2^j, v_2^j]$

$$c_{ij} = \mathrm{Norm}\left(\mathbf{F_1}[u_1^i, v_1^i]\right) \cdot \mathbf{F_2}[u_2^j, v_2^j] , \tag{7.63}$$

and δ_{ij} is the Kronecker delta. By this, we ensure that features of pixels that correspond to the same surface point become similar.

The training with synthetic normal maps is, however, insufficient for 3D correspondence detection in the estimated normal maps as they are distorted by the GBR transformation. We overcome this with data augmentation. We randomly sample parameters of the GBR transformation and transform the input normal maps according to Eq. (7.34). We use the transformed normal maps instead of the original ones so that the network can learn to extract features robuts to estimation errors caused by the bas-relif ambiguity.

At inference time, using the extracted feature maps, we detect correspondences by brute-force matching and then filer them with the ratio test by Lowe [42]. For any location on the object in the first image, we look for the best and second best location in the second images in terms of the cosine similarity of the features. If the ratio of the similarity of the second best location to one of the best location is lower than a threshold, we use the best location as the matched location. This gives us both pixel and 3D correspondences. We also use another deep feature extraction network and the brute-force matching algorithm to detect reflection correspondences from reflectance maps.

Once we obtain the three types of correspondences, we recover the relative rotation using the proposed two-step algorithm. For robust estimation, we perform the first step with RANSAC [15], *i.e.*, we estimate different candidates of the combined transformation G_{21} from randomly sampled minimal sets of pixel and 3D correspondences. We then select one of them according to the number of inlier correspondences, *i.e.*, ones that are consistent with the estimated parameters. According to

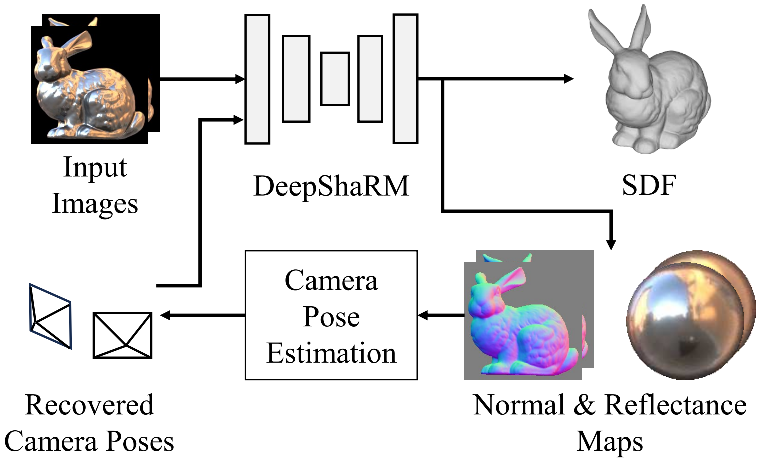


Figure 7.5: Once we recover the camera poses, we can recover the 3D object shape and also refine the camera pose estimate by alternating between multi-view geometry and reflectance map recovery by DeepShaRM and the proposed camera pose estimation method.

an empirical analysis in Sec. 7.5.1, we use four pixel and four 3D correspondences from the normal map as the minimal set for this first step. In the second step, for each possible decomposition of G_{21}, we check the number of inlier reflection correspondences that are consistent with the decomposed parameters and select parameters that maximized the number of the inliers.

Translation Estimation Once R_{21} is determined, by leveraging the pixel correspondences, we can also solve for the relative translation between the two views except for offsets regarding the two viewing directions [20]. That is, we can estimate the relative translation vector $\mathbf{t}_{21} = (t_x, t_y, t_z)^{\mathrm{T}}$ using a constraint

$$r_{23}t_x - r_{13}t_y = -\sum_i (r_{23}\tilde{u}^i - r_{13}\tilde{v}^i)\,, \tag{7.64}$$

where r_{jk} is (j,k) element of R_{21}. $\tilde{u}^i$ and $\tilde{v}^i$ are identical to ones in Eqs. (7.10) and (7.11).

7.3 Joint Shape and Camera Pose Recovery

As depicted in Fig. 7.5, we can further improve the camera pose estimation results by alternating between estimating the relative camera pose from normal and reflectance map estimates and updating the maps using DeepShaRM, a multi-view geometry and reflectance map estimation framework. As a byproduct, DeepShaRM also outputs an accurate object 3D shape represented by a 3D grid of a signed distance function.

7.4 Implementation of Single-View DeepShaRM

As we explained in Sec. 7.2, at first, we apply DeepShaRM to each image separately to obtain normal and reflectance map estimates. Although DeepShaRM is originally designed for multi-view 3D reconstruction, we empirically found that it can work even with the single-view input. To reduce computational burden, instead of using a 3D grid of a signed distance function, we represent the surface geometry as a 2D grid of surface depths. As the single-view estimation suffers from the bas-relief ambiguity, during the joint optimization of surface geometry and a reflectance map, we impose a regularization loss

$$L_{\text{reg}} = \left\{ \begin{pmatrix} 0 \\ 0 \\ l \end{pmatrix} - \frac{1}{|P|} \sum_{u,v \in P} \mathbf{N}[u,v] \right\}^2 , \tag{7.65}$$

where $\mathbf{N}[u,v]$ is a normal map extracted from the depth grid, P is a set of pixels inside the object, and $|P|$ is the number of such pixels. As we can assume that the surface is roughly fronto-parallel, we enforce the x and y components of the average surface normal orientation to be zero. We also encourage its z component to be identical to its average value in training data l. As the value of the z component corresponds to flatness of the surface (*i.e.*, higher values mean that almost all surface normals are facing the camera), this loss ensures that the surface is not extremely flat or sharp. For the rest, we use the method described in Chapter 6 as is.

7.5 Experiments

We focus on our experiments on answering the following key questions. How many correspondences are required for the camera pose estimation? Can the novel feature extraction network establish correspondences robust to distortions caused by the bas-relief ambiguity? How the proposed method work compared to existing camera pose estimation methods? To answer these questions, we validate the following points.

- Empirical evaluation shows that a set of four correspondences from normal maps and one correspondences from reflectance maps is the minimal set for the two-step algorithm.

- The proposed components (the data augmentation and the joint shape and camera pose estimation) are essential for accurate camera pose reconstruction from a sparse set (2) of images.

- Our method can work on two-view real-world images of textureless, non-Lambertian objects, which are extremely challenging to existing methods.

7.5.1 The Number of Correspondences

We first conduct empirical evaluation for the number of correspondences required for the two-step estimation method with the three types of correspondences. We created 10000 sets of synthetic correspondences by randomly sampling surface normals, relative rotation matrices, and parameters of the GBR transformations. Using this synthetic data, we checked how the numbers of correspondences used for the two-step algorithm (Sec. 7.2) affects the estimation accuracy. In this experiment, we optimized all the unknown parameters with all available correspondences using an

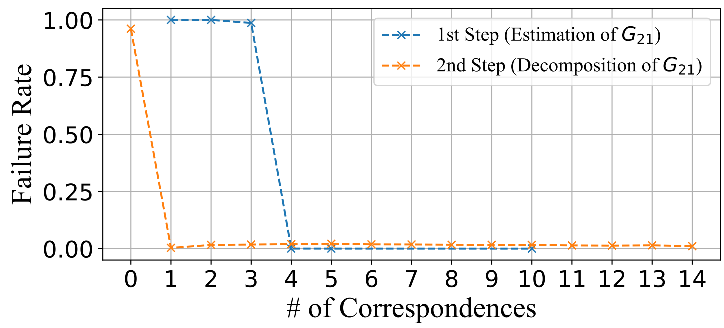


Figure 7.6: Empirical analysis of the number of correspondences required for the camera pose estimation. Please see the text for details.

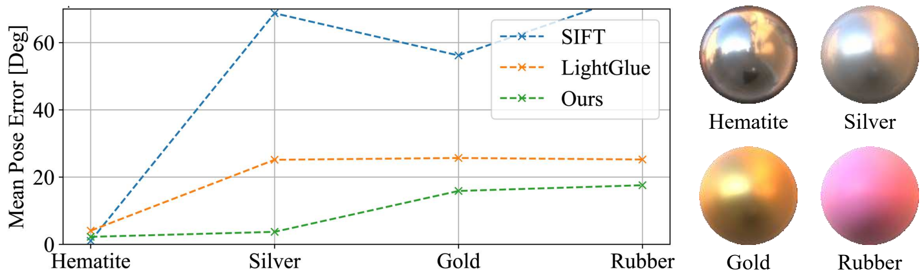


Figure 7.7: Accuracy of the detected reflection correspondences. In this evaluation, we used two-view ground truth reflectance maps as inputs and evaluated mean errors of the relative rotation recovered from the detected reflection correspondences.

off-the-shelf solve [63] as the two-step algorithm cannot be applied to cases where reflection correspondences are not available. In the following results, we exclude cases where the objective function did not converge to zero as it means that the optimization algorithm did not converge correctly.

The blue line in Fig. 7.6 shows the relationship between the number of correspondences in normal maps (pixel and 3D correspondences) and rate of failure cases where the error (Frobenius norm) between the estimated combined transformation matrix G_{21} and the ground truth is higher than 0.01. To simulate the first step of the algorithm, we did not use reflection correspondences in this experiment. The results show that four correspondence are required for the estimation of G_{21}. As discussed in Sec. 7.1.4, this would be because four correspondences are always required regardless of whether the ambiguity about η remains or not. The orange line in Fig. 7.6 shows the relationship between the number of reflection correspondences and rate of failure cases where the error between the estimated camera pose and the ground truth is higher than 0.1 degree. In this experiment, we used four pixel and four 3D correspondences in addition to the reflection correspondences. The results show that one reflection correspondence resolve the ambiguity in the camera pose estimation. Note that, the leftmost point in the orange line shows that, without reflection correspondences, we cannot decompose the combined transformation G_{21}.

	Pose Error (↓)
w/o Data Augm.	13.4 deg
w/o Joint	7.1 deg
Ours	**6.1 deg**

Table 7.1: Camera pose estimation accuracy on images in the nLMVS-Synth dataset. The results show the effectiveness of each of the proposed components.

7.5.2 Accuracy of Reflection Correspondences

We then evaluate the accuracy of the learning-based detection of reflection correspondences. We tested this in a simplified setting where two-view ground truth reflectance maps are given, *i.e.*, we do not need to consider the bas-relief ambiguity and can recover the relative rotation between the views using Eq. (7.40). We computed the mean error of the recovered relative rotation. We also applied traditional correspondence detection methods (SIFT [42] and LightGlue [37]) to the (projected 2D) reflectance maps and compared their accuracy with ours.

Figure 7.7 shows quantitative accuracy for each material. While the traditional methods do not generalize to rough surfaces at all, our method works well even with the "Silver" material. The results show that the necessity of the neural matcher specialized for the reflection correspondences.

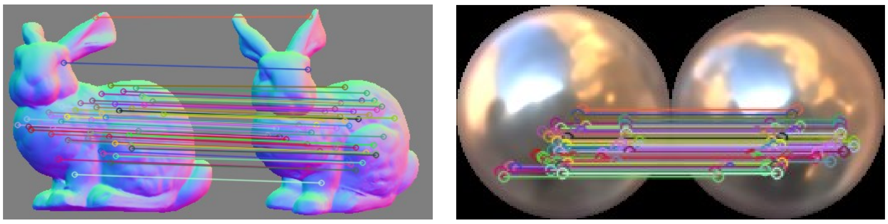
The results also show that, even with our method, detection of reflection correspondences on very rough surfaces (*e.g.*, the "Rubber" material) is still challenging. Such surfaces are extremely challenging to both our method and the existing shape and camera pose estimation methods and remain for future study.

7.5.3 Shape and Camera Pose Estimation

We then evaluate accuracy of the joint camera pose and object shape estimation framework on synthetic images in the nLMVS-Synth dataset. We train the feature extraction networks in our method using the training set of the nLMVS-Synth dataset and test our method on the test set. To evaluate the effectiveness of each component in our method, we compare our camera pose estimation accuracy (the average pose estimation error) with those by its variants, "w/o Data Augm." and "w/o Joint". "w/o Data Augm." uses the feature extraction networks trained without the data augmentation based on the GBR transformations. "w/o Joint" is our method without the alternating estimation which recoveres camera poses only from the per-view initial estimates of normal and reflectance maps.

Table 7.1 shows the average camera pose estimation errors. The results show that the joint estimation improves the estimation accuracy and the data augmentation method is essential for the joint estimation of the relative camera pose and the parameters of the bas-relief transformation. Figure 7.8 shows inlier correspondences detected by our method. The results are qualitatively plausible, *i.e.*, the correspondences link locations that correspond to the same surface point or the same object in the surrounding environment.

Figure 7.9 shows how the initial estimates of surface normal and reflectance maps are improved by the joint estimation of geometry and camera poses. Even when the initial estimates are erroneous, our method successfully resolves the bas-relief



(a) 3D Correspondences (b) Reflection Correspondences

Figure 7.8: Detected 3D and reflection correspondences from images in the nLMVS-Synth dataset. The correspondences link pixels that are semantically correct.

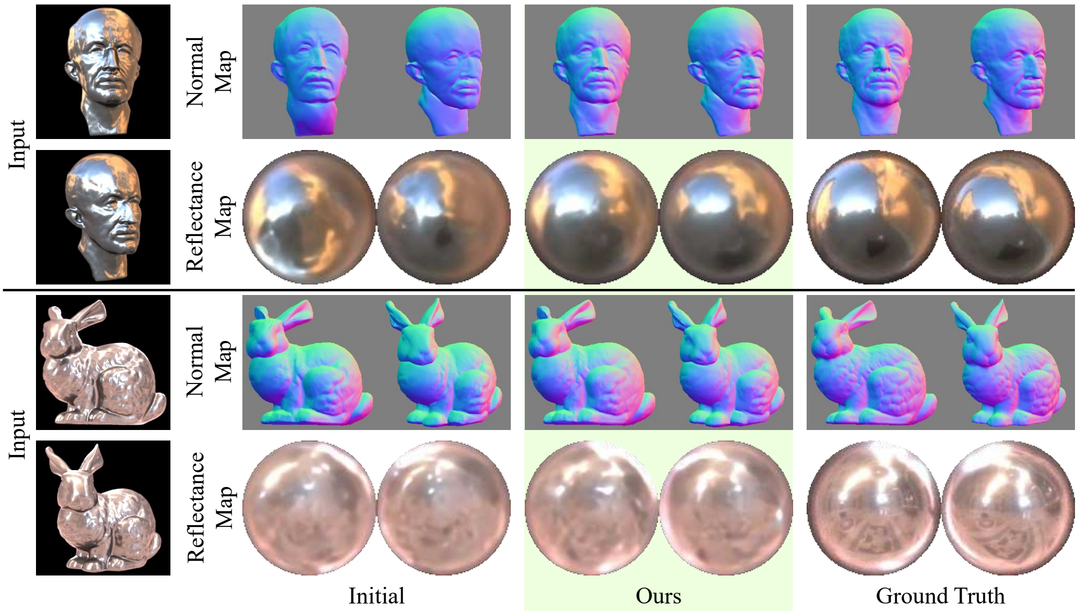


Figure 7.9: Surface normal and reflectance maps recovered from two synthetic images in the nLMVS-Synth dataset. The bas-relief ambiguity is successfully resolved and the recovered maps are consistent with ground truth.

ambiguity and recovers plausible surface normal and reflectance maps along with the relative camera pose. As shown in Fig. 7.10, surface geometry is also successfully recovered by our method.

7.5.4 Evaluation on Real Data

We also qualitatively and quantitatively evaluate our method on real-world two-view images from the nLMVS-Real dataset. We compared our results with ones of COLMAP [57] and SAMURAI [5]. We found that COLMAP completely fails on our inputs, *i.e.*, two-view images that covers only the target object. We used 11 view uncropped images that capture not only the target object but als textured object aroung the targes (*i.e.*, ChArUco boards) sa inputs to COLMAP. As SAMURAI requires coarse initial camera poses, we use the same camear pose as the initial value for all input views.

Table 7.2 and Fig. 7.11 show quantitative and qualitative results. Note that COMLAP completely failed on the "Horse" object. Note also that, as SAMURAI failed

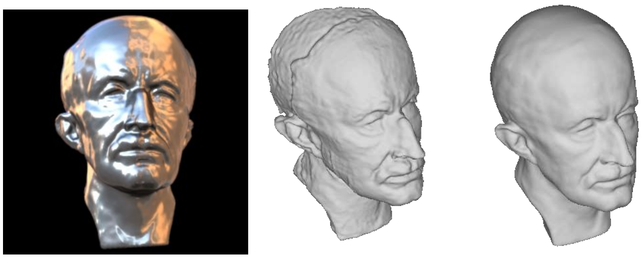


Figure 7.10: Surface geometry recovered by the joint estimation.

	Planck	Horse	Bunny
COLMAP [57] (*)	15.7 deg	N/A	24.5 deg
SAMURAI [5]	5.6 deg	24.3 deg	30.5 deg
w/o RM	3.8 deg	20.2 deg	11.9 deg
Ours	**1.9 deg**	**12.5 deg**	**7.2 deg**

Table 7.2: Camera pose estimation accuracy on real-world images in the nLMVS-Real dataset. (*) COLMAP [57] uses 11 view uncropped images as inputs. Please see the text for details.

to extract a 3D mesh model from their volumetric geometry representation, we instead show a rendered normal map in the first view for SAMURAI. While existing methods fail on the extremely challenging inputs, our method successfully recovers plausible camera poses, surface geometry, and reflectance maps.

Effectiveness of Reflection Correspondences We also compared our method with our method without reflection correspondences ("w/o RM"), a variant of our method that only leverages pixel and 3D correspondences. To obtain a unique solution without using reflection correspondences, we ignore the bas-relief ambiguity and recover the relative camera pose based on Eq. (7.33). For detection of 3D correspondences, we use the same RANSAC-based algorithm as one in the first step of our two-step method.

Table 7.2 and Fig. 7.12 show quantitative and qualitative results. Note that the joint shape and camera pose recovery without reflection correspondences completely failed on the "Horse" object. In Tab. 7.2, for the "Horse" obejct, we report the accuracy of our method "w/o RM" without the joint estimation. Without reflection correspondences, the recovered camera poses can be wrong and the recovered geometry can be distorted due to the bas-relief ambiguity.

7.6 Summary

In this chapter, we introduced a method that exploits new type of correspondences, reflection correspondences, for joint object 3D shape, reflectance maps, and camera pose recovery even from two-view images of textureless, non-Lambertian objects without using any background textures, which remains extremely challenging to existing methods. By leveraging the reflection correspondences for the camera pose estimation and integrating it with the joint shape and reflectance recovery for posed

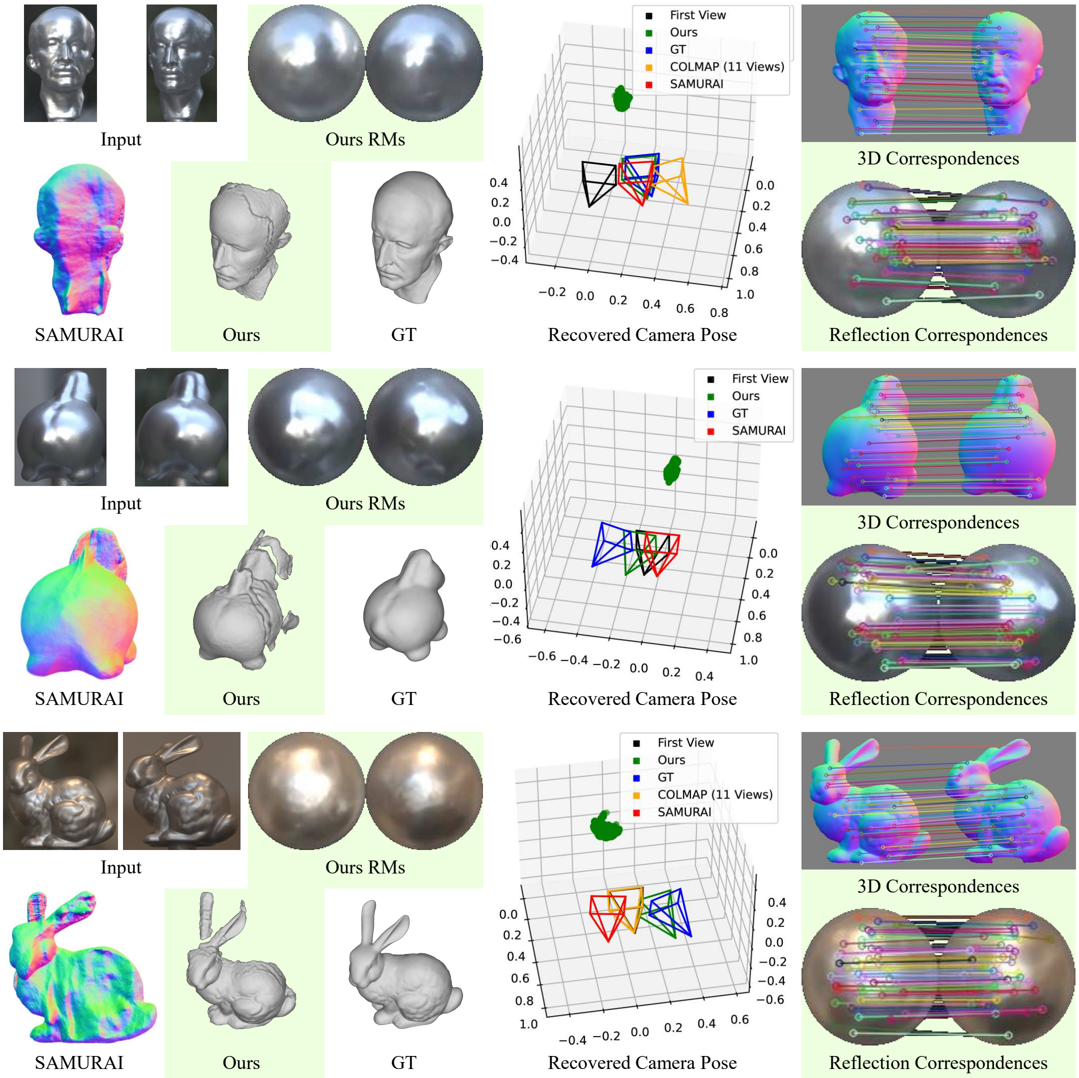


Figure 7.11: Reflectance maps (RMs), surface geometry, relative camera poses, and correspondences recovered from two-view real-world images. Our results are qualitatively plausible.

images (Chapter 6), our method successfully handles unposed images of textureless, non-Lambertian objects. The experimental analysis qualitatively and quantitatively shows the effective of the use of the reflection correspondences, the joint shape and camera pose estimation framework, and the data augmentation based on the generalized bas-relief (GBR) transformation. We believe the reflection correspondences enable not only object shape and camera pose recovery in challenging settings but also can play important roles in different applications such as camera calibration and object pose estimation.

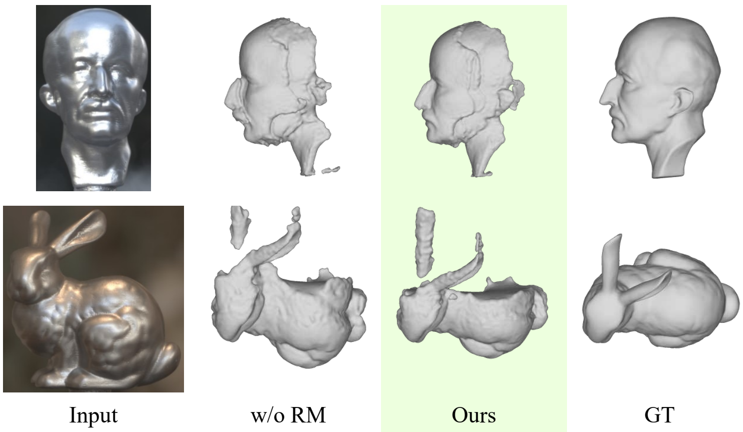


Figure 7.12: Effectiveness of the use of reflection correspondences. Thanks to the use of reflection correspondences, our results are not distorted due to the bas-relief ambiguity,

Chapter 8

Conclusion

In this dissertation, we introduced two novel datasets and three novel methods for image-based 3D reconstruction of a textureless, non-Lambertian objects from a sparse set (2 at minimum) of multi-view images taken under unknown natural illumination.

First, as illustrated in Chapter 4, we introduced novel synthetic and real-world datasets for training and evaluation of non-Lambertian 3D reconstruction methods. The synthetic image dataset, nLMVS-Synth, provides 26850 training images of 2685 objects and 4320 test images of 216 objects with corresponding ground truth BRDF data, illumination maps, and surface geometry. These are sufficient for training of deep neural networks and thorough qualitative and quantitative evaluations. The real object dataset, nLMVS-Real, consists of 2569 multi-view high dynamic range images of 120 combinations of 5 shapes, 4 paints, and 6 illumination environments. We provide ground truth camera poses and aligned ground truth 3D mesh models of the objects by data curation and careful calibration, which enable quantitative evaluation on the real-world images. We also provide illumination maps of the surrounding environments which are also useful for evaluation of inverse rendering methods. We believe our two novel datasets, especially the nLMVS-Real, can serve as a useful platform for not only non-Lambertian 3D reconstruction but also other inverse rendering research such as joint estimation of reflectance and illumination.

In Chapter 5, we introduced *nLMVS-Net*, a deep multi-view geometry estimation method for posed multi-view images of a textureless, non-Lambertian object taken under known but natural illumination. The proposed single-view deep shape-from-shading network effectively leverages radiometric likelihoods of surface normals computed from reflectance maps and successfully estimates surface normals as well-defined pixel-wise probability densities, which become tractable cues for not only the non-Lambertian multi-view stereo, but also the joint shape and reflectance map estimation (DeepShaRM in Chapter 6) and detection of pixel correspondences for the joint shape, camera pose, and reflectance map estimation (Chapter 7). Undoubtedly, the shape-from-shading network plays important roles in our non-Lambertian 3D reconstruction methods. The proposed cost volume filtering network with the estimated probability densities and the joint shape and reflectance estimation framework with a neural reflectance estimation method [11] enable plausible geometry and reflectance estimation for textureless, non-Lambertian objects under known but natural illumination. Although the geometry recovered by the cost volume filtering is view-dependent (*i.e.*, depths and surface normals estimated for each view are not strictly consistent with ones of the other views) and relatively poor compared to one by DeepShaRM. The cost volume filtering network is, however, superior in terms of inference time (*i.e.*, it does not require costly iterative nonlinear optimization) and can be useful for applications that require inference speed such as autonomous driving. Although the assumption of known illumination is somewhat restrictive, the explicit estimation of surface reflectance would be useful for XR applications, *i.e.*, we can synthesize realistic appearance for objects whose surrounding illumination is virtually changed.

In Chapter 6, we introduced *DeepShaRM*, a joint shape and reflectance map estimation framework for posed multi-view images of a textureless, non-Lambertian object captured under unknown natural lighting. The proposed reflectance map estimation network recovers plausible reflectance maps from a sparse set of observations (pixel values and surface normals) by leveraging realistic prior on reflectance maps learned from the nLMVS-Synth dataset. The image feature filtering layer and the explicit estimation of the pixel-wise confidence scores inside the network enable us to handle objects with complex surface geometry which causes deviations from the reflectance maps (*e.g.*, shadows and interreflections). The proposed joint estimation framework with the shape-from-shading network, the reflectance map estimation networks, and the SDF optimization enables us to recover plausible surface geometry and reflectance maps even from a very sparse (*e.g.*, 2) set of posed multi-view images without requiring any additional information (*e.g.*, one of the surrounding illumination). As DeepShaRM estimates camera-view reflectance maps for each view, *i.e.*, it does not assume a static illumination environment, we can apply DeepShaRM to images taken under real scenes whose surrounding illumination environments dynamically change. We believe DeepShaRM can serve as a practical means for real-world applications where we can calibrate camera poses in advance or from background textures.

In Chapter 7, we proposed to leverage *Reflection Correspondences* for a joint estimation of object 3D shape, camera poses, and reflectance maps that only requires two-view images of a textureless, non-Lambertian object under static but unknown natural illumination. By leveraging reflection correspondences, *i.e.*, correspondences of the surrounding environment reflected on the surface (ones in reflectance maps), we can resolve the bas-relief ambiguity [3] even with the two-view observations. The carefully designed two-step relative rotation estimation algorithm successfully handles the different kinds of correspondences without falling into a local minima solution. The learned neural matchers, the robust pose estimation method based on RANSAC, and the joint estimation framework enable plausible reconstruction from the sparse observations without using background textures, which remains extremely challenging to existing methods. The data augmentation method for the neural matchers based on the generalized bas-relief transformation is essential for accurate estimation. We believe reflection correspondences open new applications of the object appearance. For instance, they would be useful especially when the target textureless, non-Lambertian object (*e.g.*, car) is moving and the background textures cannot be used for the relative camera pose estimation.

8.1 Contributions

By introducing the non-Lambertian 3D reconstruction method(s) that can be applied to real-world objects with different materials (*e.g.*, metal, plastic, and ceramic), we believe our method opens a new avenue of applications of image-based 3D reconstruction in the wild. Specifically, we realized the followings which remain extremely challenging to existing methods.

Non-Lambertian 3D reconstruction from sparse posed multi-view images taken under unknown natural illumination By integrating the geometry estimation network of nLMVS-Net with the joint estimation framework of DeepShaRM, we realized dense surface reconstruction of textureless, non-Lambertian surfaces from sparse (2 or more) posed multi-view images taken under unknown natural illumination. As we recover camera-view reflectance maps for each view separately and do not exploit constraints that assume cross-view consistency of reflectance maps for the estimation from posed images, our method can recover geometry even from images taken under dynamic illumination environment (Fig. 6.11). Thanks to these advantages, it would be possible to apply the method to real-world applications where we cannot assume textures and non-Lambertian reflectance (*e.g.*, reconstruction of cars) and static illumination (*e.g.*, reconstruction under outdoor scenes).

Non-Lambertian 3D reconstruction from sparse unposed multi-view images captured under unknown static natural illumination By integrating the three key method, we realized geometry reconstruction of textureless, non-Lambertian surfaces from sparse (*e.g.*, 2) posed multi-view images taken under static but unknown natural illumination. Our overall method does not require any additional input and leverages only the appearance of the target object. This would be useful for applications where there is no textures even in backgrounds of the target objects. For instance, by using our method, it would be possible to recover geometry of table ware put on a table with a uniform color, which remains extremely challenging to existing methods.

8.2 Towards Full Inverse Rendering

We mainly focused on geometry estimation and, except for the nLMVS-Net that assumes known illumination, we did not decompose the recovered reflectance maps into its radiometric roots, *i.e.*, reflectance and illumination. Image-based full inverse rendering, *i.e.*, joint estimation of surface geometry, surface reflectance, and surrounding illumination from only images, still remains for future study and would be helpful for analysis of the scene. We believe our geometry and reflectance map estimation methods can bootstrap research of the challenging full inverse rendering in the wild. That is, using our method, we can recover an initial estimate of surface geometry only from input images. Also, from reflectance maps recovered by our method, we can obtain initial estimates of reflectance and illumination using existing inverse rendering methods [39, 18, 11]. Once we obtain good initial estimates for all unknown physical quantities, we can jointly refine them by nonlinear optimization based on the rendering equation.

8.3 Future Directions

In this dissertation, we focused on non-Lambertian 3D reconstruction of a textureless, homogeneous object. We also assumed almost orthographic cameras and distant illumination, and only considered direct lighting (*i.e.*, we ignored shadows and interreflections). Although these assumptions are reasonable to a certain degree (*e.g.*, geometry reconstruction is challenging especially for homogeneous surfaces) and essential for making a steady step towards non-Lambertian 3D reconstruction in the wild, they sometimes do not hold true in real-world scenes. Therefore, there are also several possible future directions for realizing real-world applications of non-Lambertian 3D reconstruction.

Spatially Varying Reflectance While we assumed completely homogeneous objects, real-world objects are usually composed of partially homogeneous regions. Thus one possible future direction would be extension to reconstruction for such partially-homogeneous objects. One possible approach to achieve this would be integrating our methods with a material segmentation method (*e.g.*, [36, 7]) and 3D reconstruction for textured surfaces (*i.e.*, multi-view stereo). We can apply our methods to each of the homogeneous regions separately based on results of the material segmentation method. As changes of materials can act like textures, we would also be able to leverage a standard multi-view stereo method for recovering 3D locations of surface points on the borders between the homogeneous regions. How to integrate the three kinds of approaches and jointly refine the segmentation results, the geometry estimate, and the reflectance map estimates would be a key to accurate reconstruction.

Global Illumination Effects The definition of the reflectance map ignores global illumination effects, *i.e.*, our methods cannot recover accurate geometry for surfaces with shadows and interreflections. This may be resolved by conducting a full inverse rendering that considers the global illumination effects. Another idea to tackle this problem would be modeling such spatially varying changes and jointly estimating its parameters with the other unknown physical quantities. The key to achieving this would be ensuring the consistency between the spatial changes of reflectance maps and the surface geometry estimate.

Perspective Projection Our methods assume orthographic projection, *i.e.*, a constant viewing direction for each view. When the distances between the object and the cameras are not large, we need to consider pixel-wise changes of reflectance maps caused by perspective projection. One possible idea to overcome this is modeling the pixel-wise changes based on a physics-based reflectance model. Reflectance maps for Lambertian surfaces are independent of the viewing directions, while view-dependent changes for purely specular objects can be (approximately) computed according to the relationship between the incident, surface normal, and viewing directions. Therefore, if we could decompose the reflectance maps into Lambertian and specular components, we can approximately compute the pixel-wise changes of the reflectance maps when computing the radiometric likelihoods of surface normals from them.

Efficient Estimation All of our proposed methods require costly iterative optimizations at inference which take hours. Note that, although the geometry estimation networks in the nLMVS-Net do not require such an iterative optimization, the neural reflectance estimation method inside the nLMVS-Net requires a nonlinear optimization. This would not be acceptable in real applications that require high inference speed such as autonomous driving. Although our overall methods cannot be directly applied to such tasks, the proposed sub-networks, *i.e.*, the shape-from-shading network, the cost volume filtering network, and the reflectance map estimation network, do not require such an iterative optimization and can serve as means for realizing the efficient estimation. For instance, an efficient geometry and reflectance map estimation may be realized by integrating the nLMVS-Net with the reflectance map estimation network inside the DeepShaRM.

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